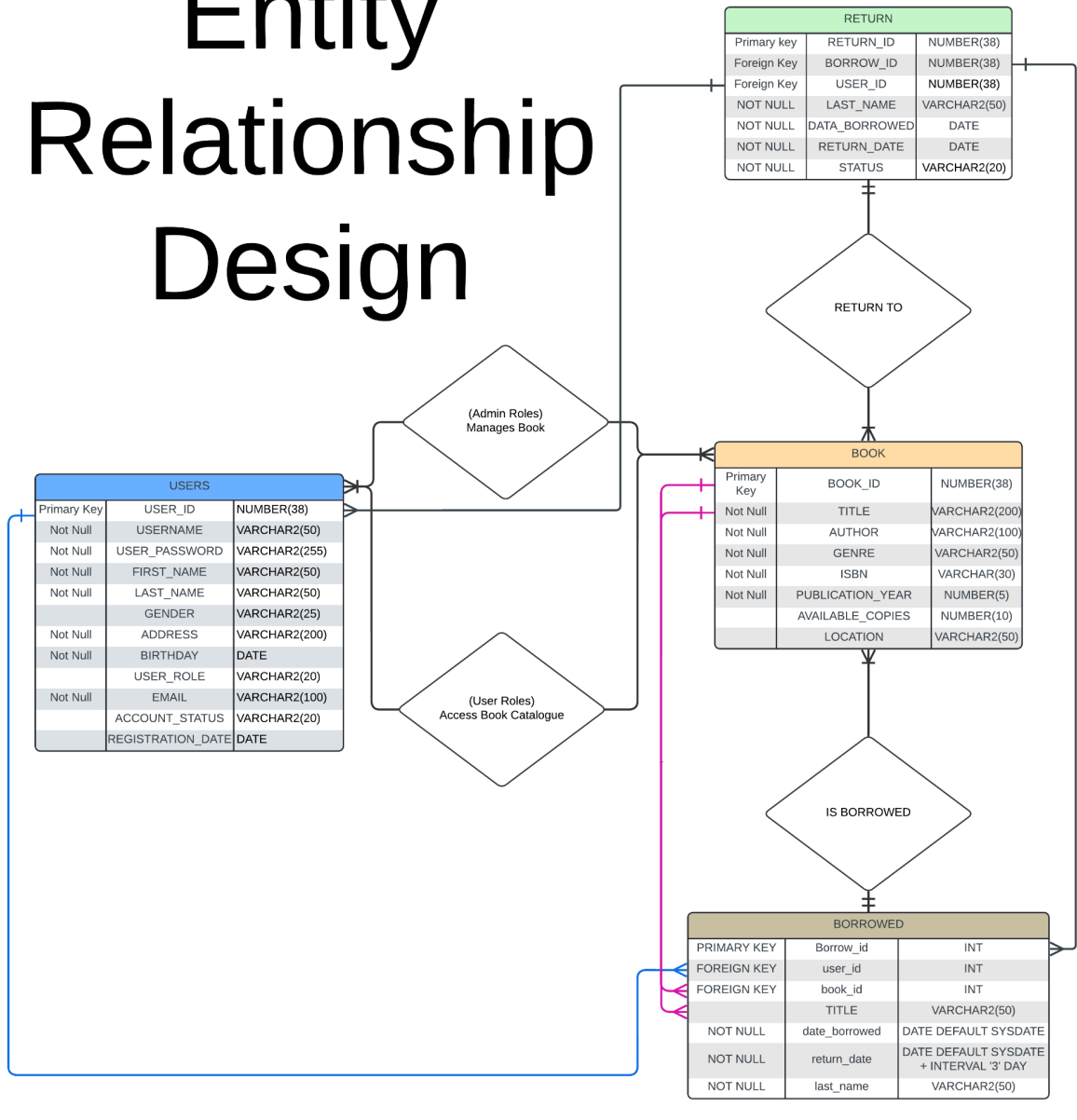
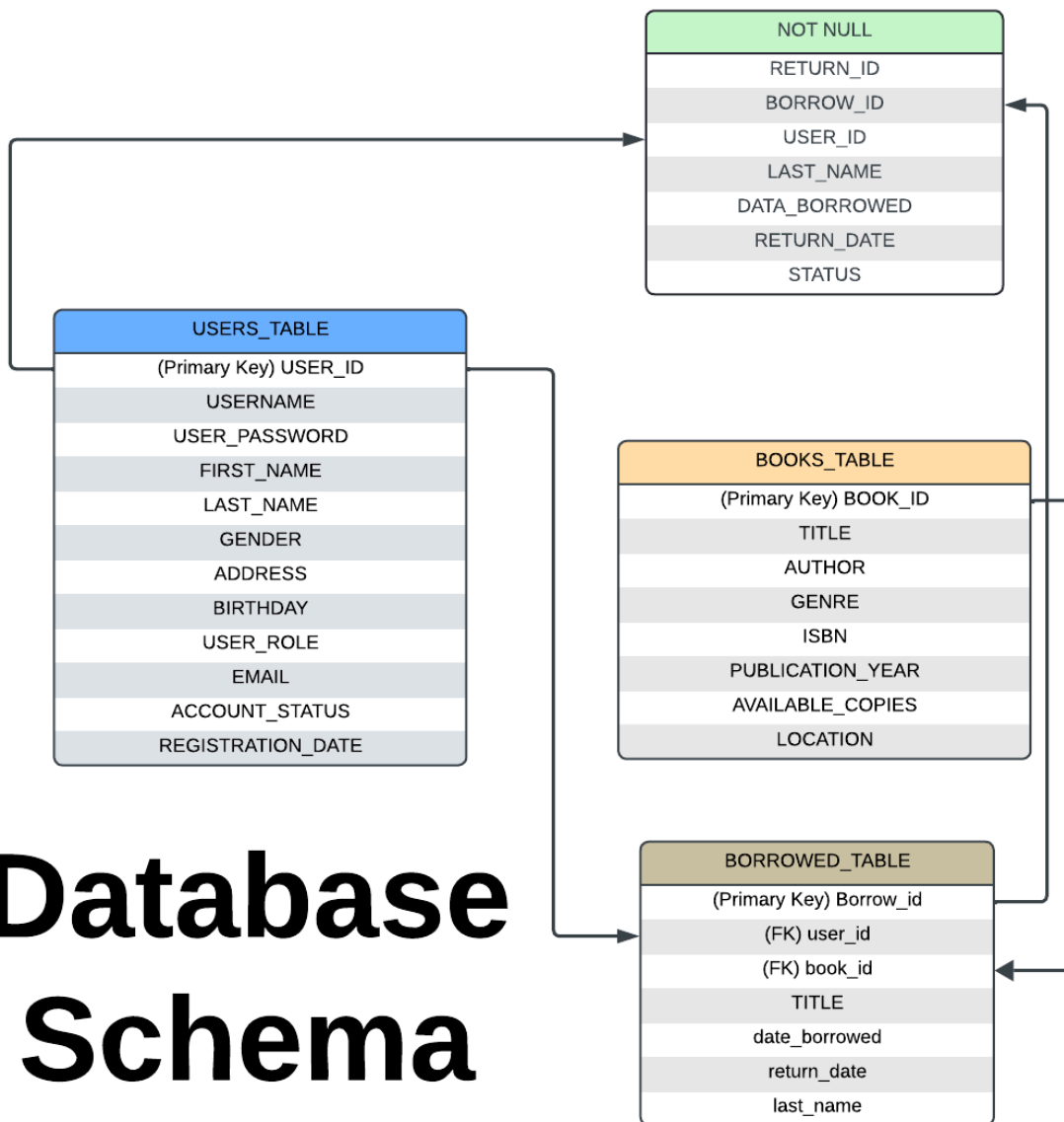


Entity Relationship Diagram

Entity Relationship Design



Database Design/Schema



Database Schema

The Database Schema Diagram shows the tables and their relationship to each other. It also shows the connection of one primary key to another as well, using the arrows to point their connection and describes which are the Primary key and foreign key of each succeeding table.

I. Screenshots of Login and Maintenance modules with Source Codes

Screen No: 1

Screen Name: LOGIN FORM

Description: This form allows authorized users only. Access rights of the users are also verified.



Object Name: BTNOK

Trigger Level: Item-Level

Trigger Name: When-Button-Pressed

```
SELECT USERNAME, PASSWORD
INTO :LOG.UN, :LOG.PW
FROM PASS1
WHERE :LOG.UN=USERNAME
AND :LOG.PW=PASSWORD;
CLEAR_RECORD;
GO_BLOCK('MENU');
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        IF SHOW_ALERT('OKS')=ALERT_BUTTON1 THEN
            EXIT_FORM;
        END IF;

Exit_Form;
```

Screen No: 2

Screen Name: SIGNUP FORM



The logo for RATAS PODEROSAS LIBRARY features a stylized open book with a plant growing from it, set against a pink background. The text "RATAS PODEROSAS LIBRARY" is written below the icon.

User Id

First Name

Last Name

Gender

Birthday

Address

Contact Number

Username

Email

Password

Screen No: 3

Screen Name: USER MAINTENANCE (ADMIN)

Users Maintenance

User Id

Username Password

User Role

First Name Last Name

Gender Birthday

Address

Email Contact Number

Account Status

Register a User **Save Account** **Edit Account**

Delete Account **Suspend Account** **Cancel Action**

< << >> >|

Screen No: 4

Screen Name: BOOK MAINTENANCE (ADMIN)

WINDOW1



**RATAS
PODEROSAS
LIBRARY**

Users Maintenance

Book Maintenance

Borrowed Books

View Penalties

Logout

Books Maintenance

Book Id

Title

Author

Genre

ISBN

Year

Avail. Copies

Location

ADD RECORD

EDIT RECORD

SAVE RECORD

DELETE RECORD

CANCEL ACTION

< << >> >|

Screen No: 5

Screen Name: BORROWED BOOKS (ADMIN)

Borrow Id	Book Id	Title	User Id	Date Borrowed	Last Name	Status

II. Generated Different Reports

a. Tabular

Report Wizard

Choose the report style that most closely resembles the report you would like to create. You may also include a title.

Title: List of Books

☒ Tabular ☐ Group Left ☐ Group Above

☐ Matrix ☐ Matrix with Group

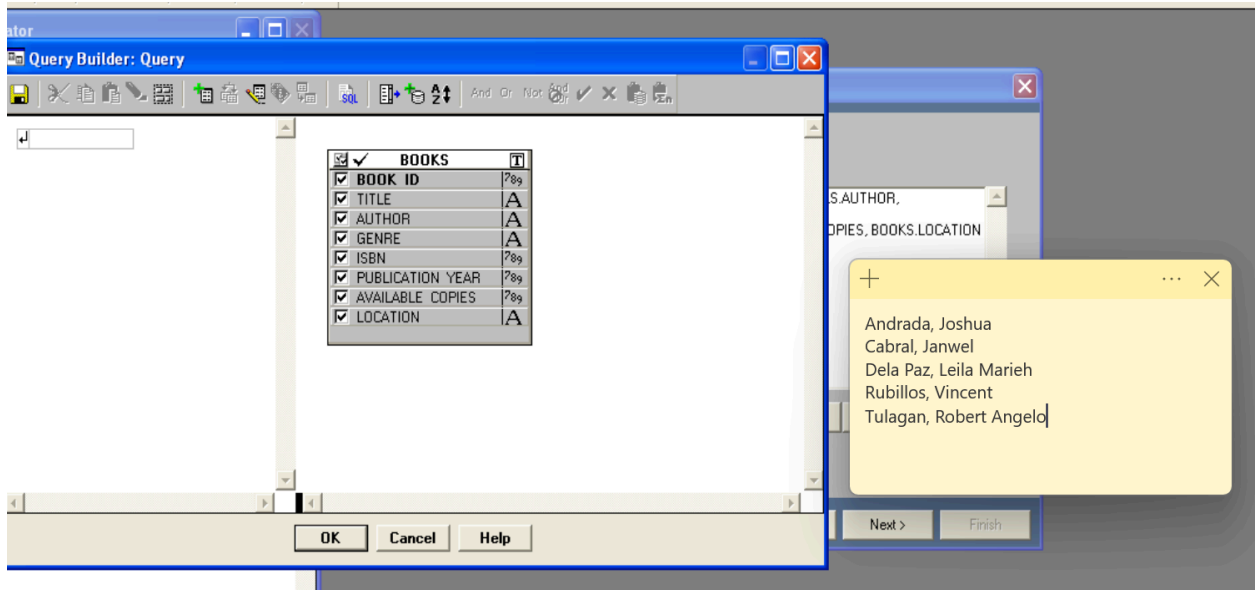
☐ Form ☐ Form Letter

Cancel Help < Back Next > Finish

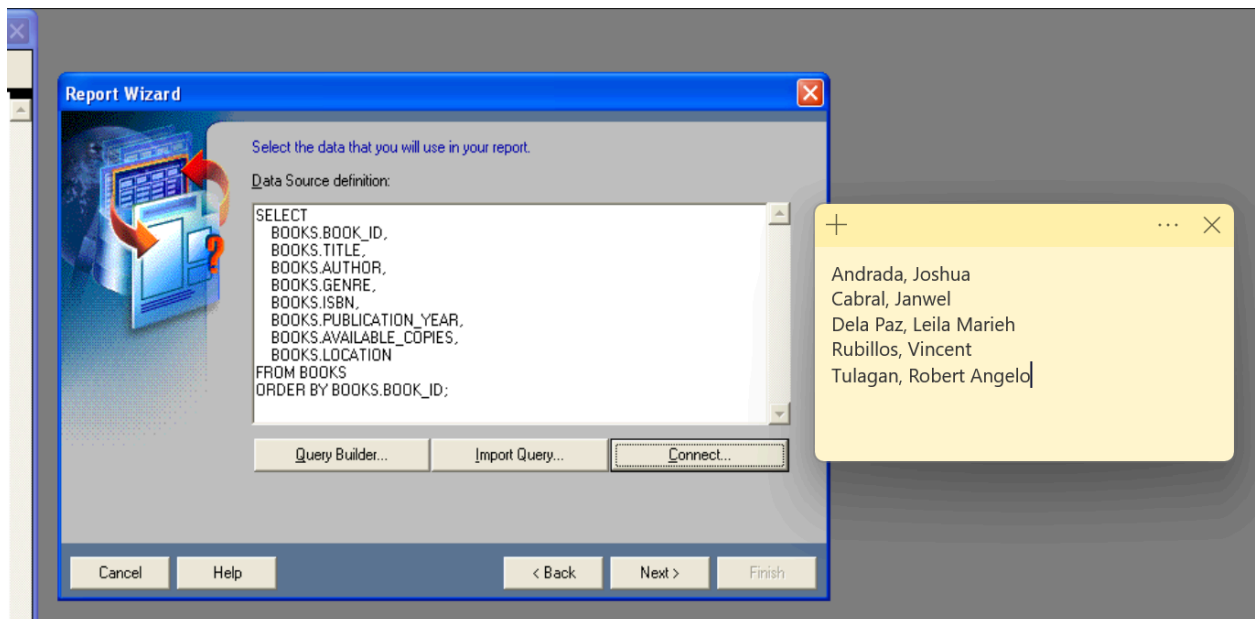
Preview:

- Andrada, Joshua
- Cabral, Janwel
- Dela Paz, Leila Marieh
- Rubillos, Vincent
- Tulagan, Robert Angelo

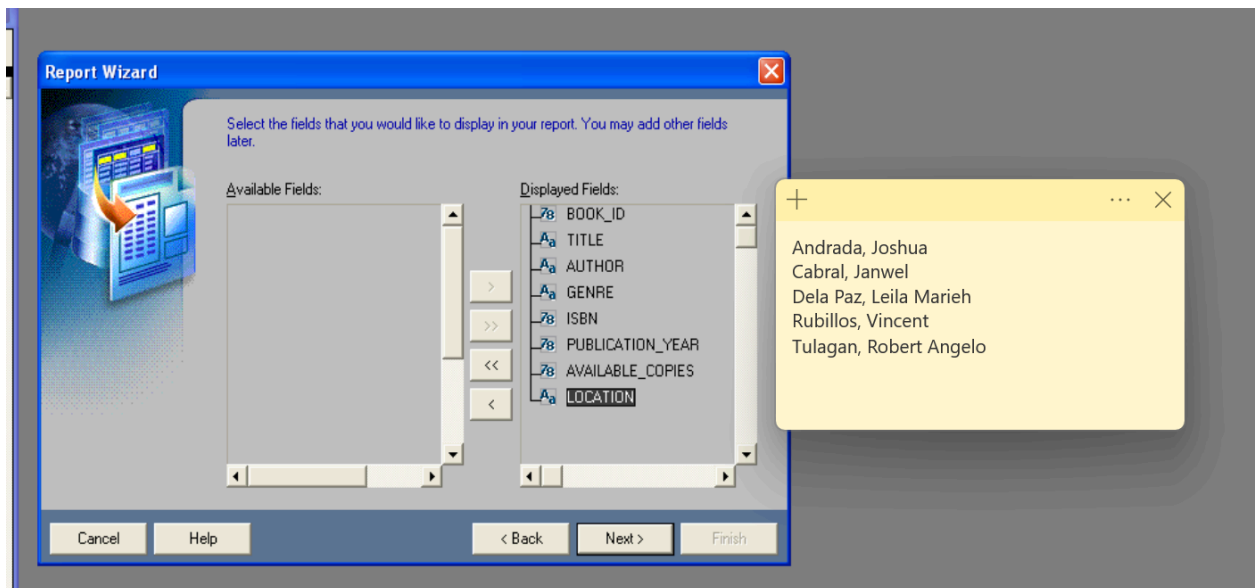
Use the tabular and named the title “List of Books”



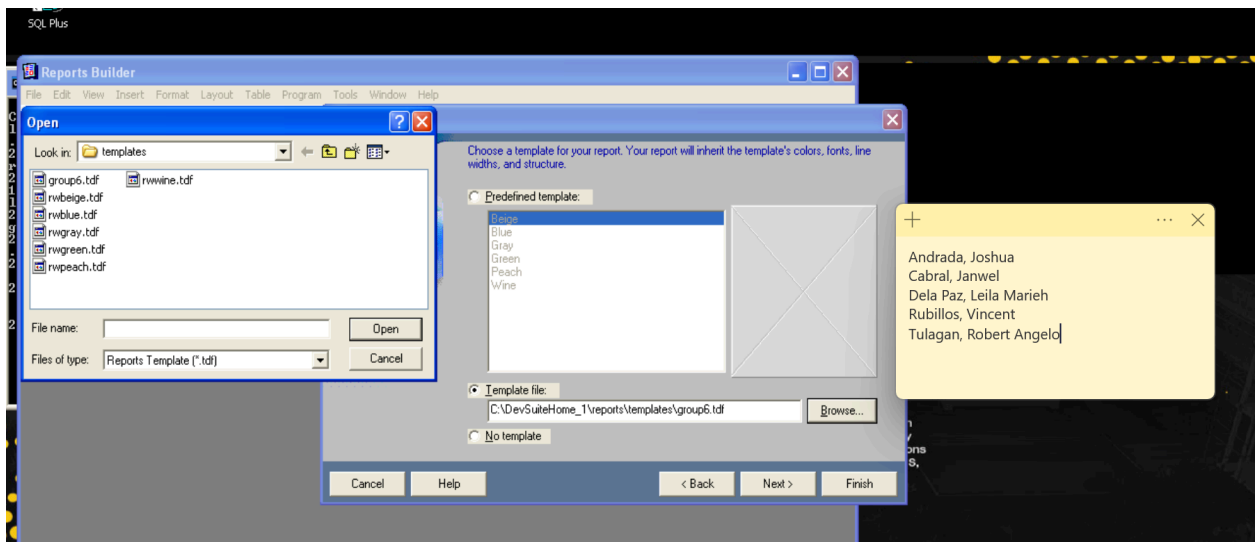
In the query builder, choose books and choose all that is available.



In the query changed the orders by book_id.



Display all fields



Choose the template we created for our own design.

 RATAS PODEROSAS LIBRARY RATAS PODEROSAS List of Books 938 Aurora Blvd. Cubao, Quezon City November 27, 2024							
Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location
1	Letters to Milena	Franz Kafka	Memoir	*****	1952	2	Aisle 4, Shelf B, Row 1 (Main Library)
2	1984	George Orwell	Fiction	*****	1949	4	Aisle 1, Shelf C, Row 1 (Special Collection s)
3	To Kill a Mockingbird	Harper Lee	Fiction	*****	1960	5	Aisle 3, Shelf C, Row 4 (Special Collection s)
4	Brave New World	Aldous Huxley	Science Fiction	*****	1932	3	Aisle 4, Shelf B

+ ... X

Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

7	Moby Dick	Herman Melville	Adventure	*****	1851	2	Wing) Aisle 1, Shelf E, Row 1 (Special Collection s)
8	The Catcher in the Rye	J.D. Salinger	Fiction	*****	1951	4	Aisle 3, Shelf C, Row 4 (Main Library)
9	The Hobbit	J.R.R. Tolkien	Fantasy	*****	1937	5	Aisle 1, Shelf D, Row 1 (East Wing)
10	The Fellowship of the Ring	J.R.R. Tolkien	Fantasy	*****	1954	7	Aisle 4, Shelf B, Row 4

Page 1 of 9

+ ... X

Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

17	The Old Man and the Sea	Ernest Hemingway	Fiction	*****	1952	8	Shelf E, Row 1 (West Wing) Aisle 1, Shelf D, Row 2 (East Wing)
18	Lolita	Vladimir Nabokov	Fiction	*****	1955	3	Aisle 5, Shelf D, Row 1 (Special Collection s)
19	Beloved	Toni Morrison	Historical Fiction	*****	1987	5	Aisle 3, Shelf D, Row 5 (West Wing)

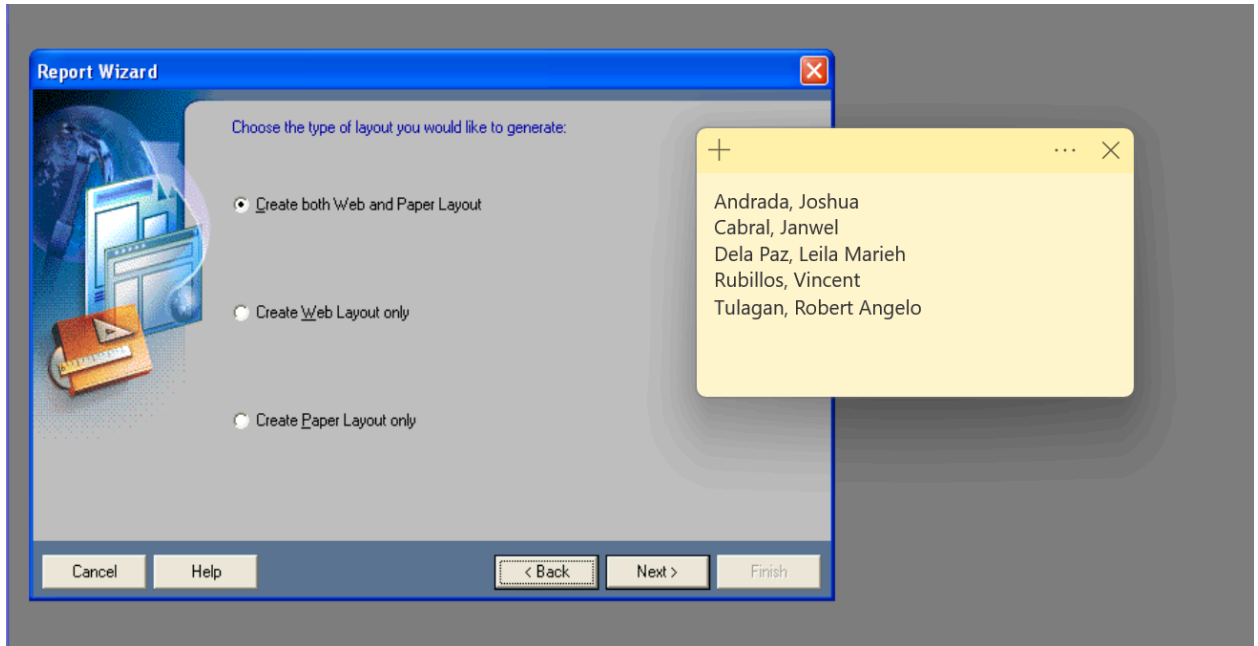
Page 2 of 9

+ ... X

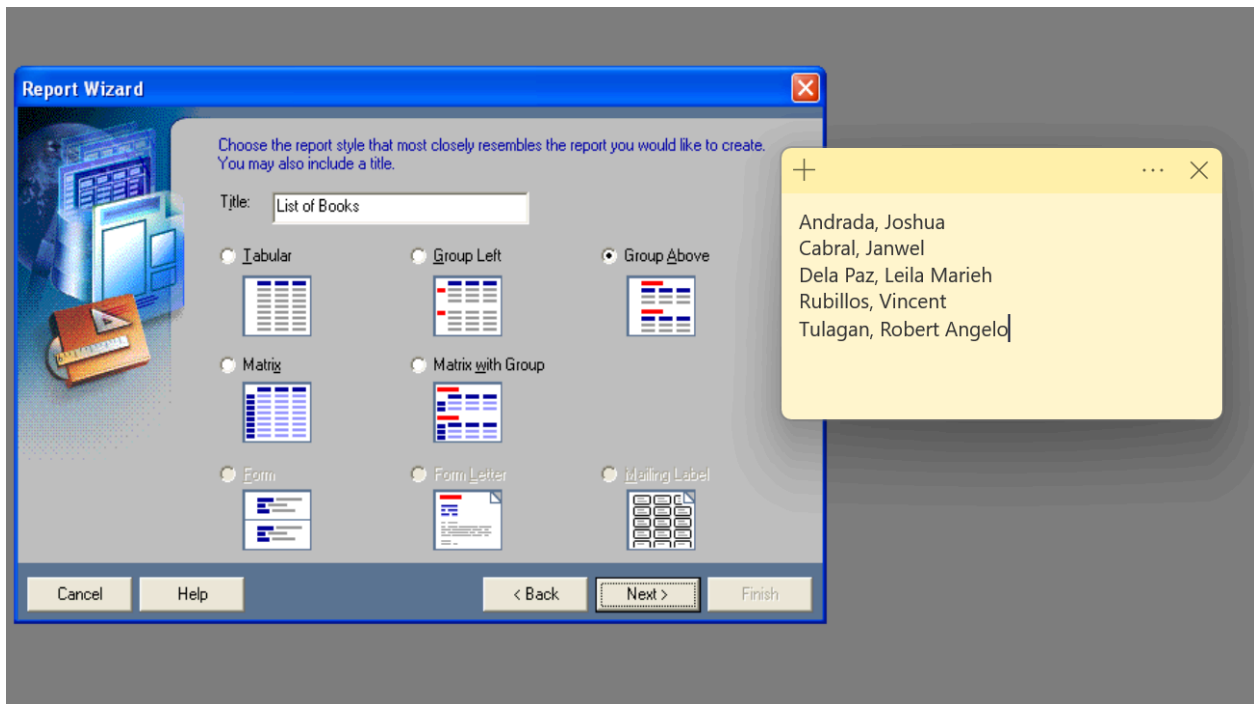
Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

b. Group Above

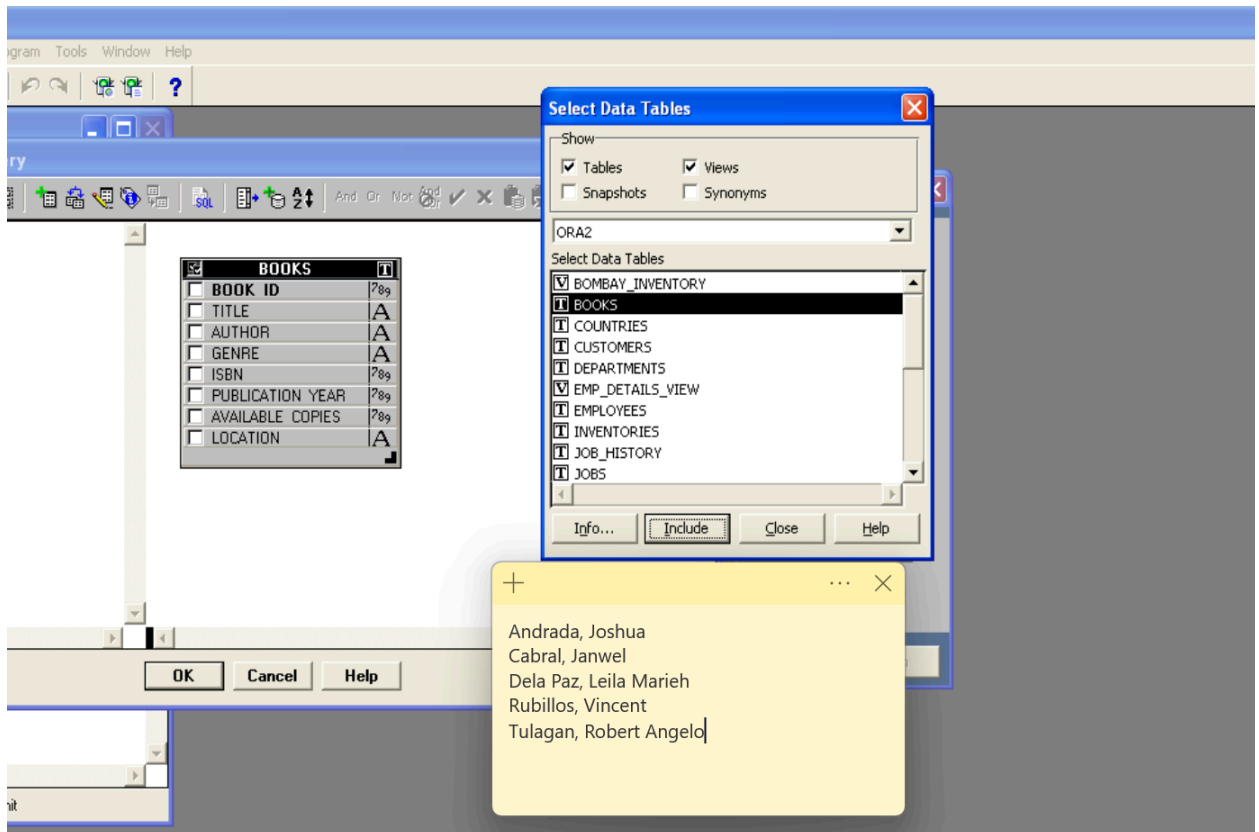
Group Above



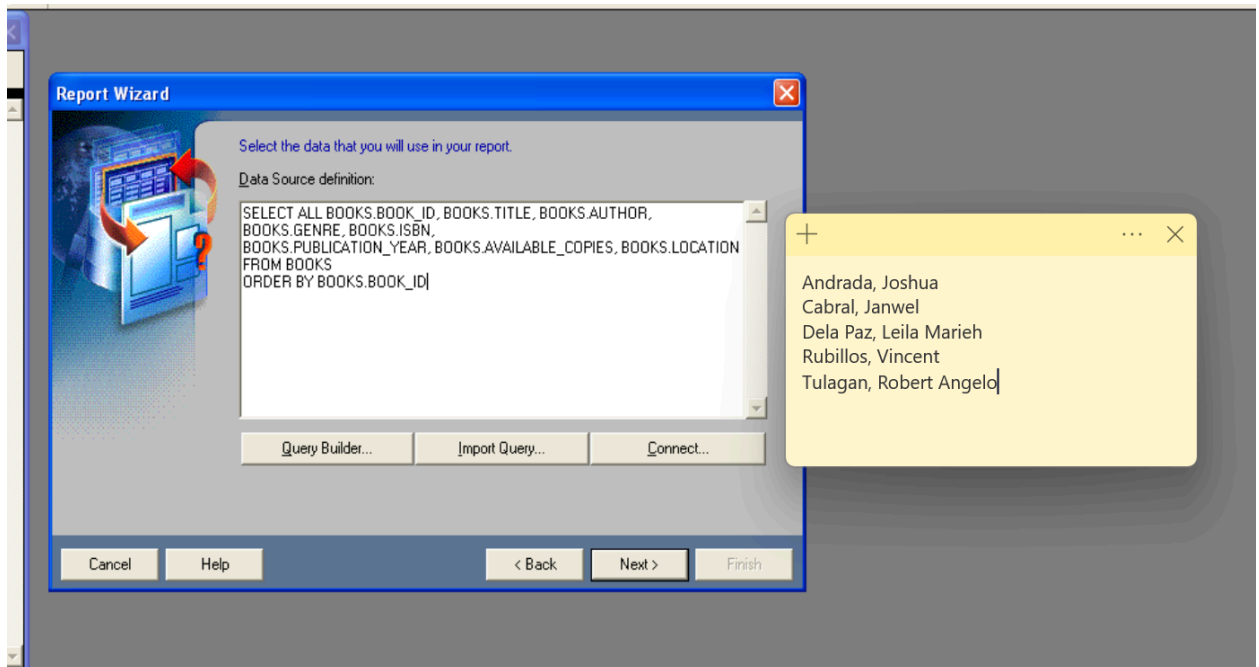
In report wizard create both web and paper layout



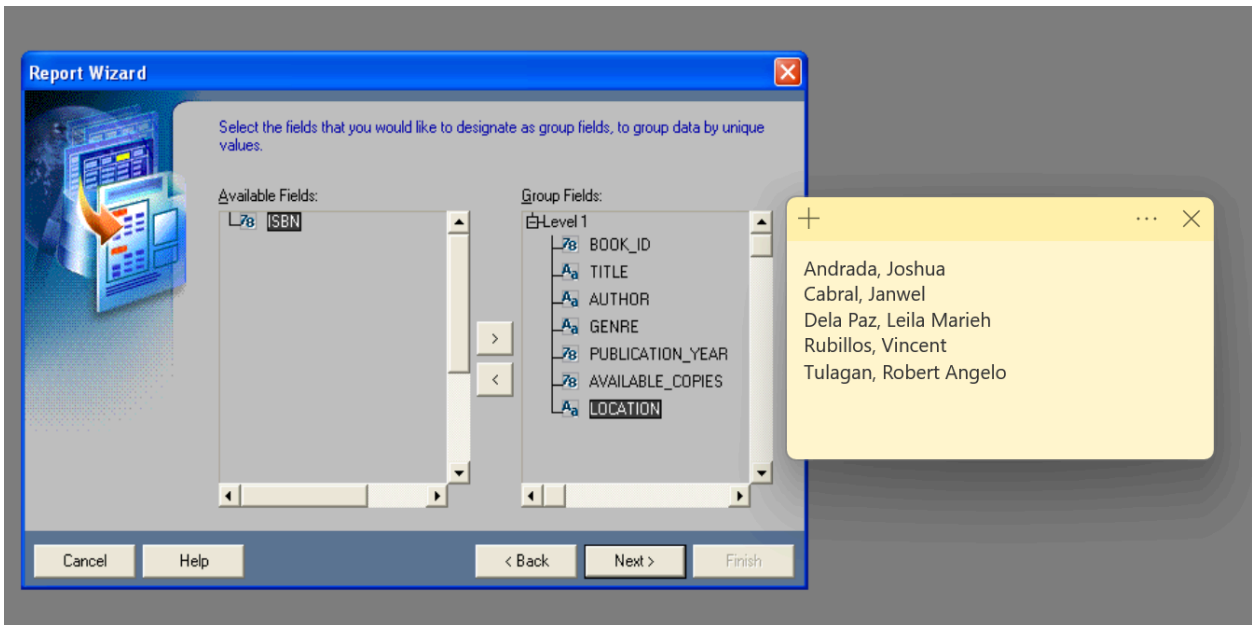
Name the title List of Books and select the Group Above option



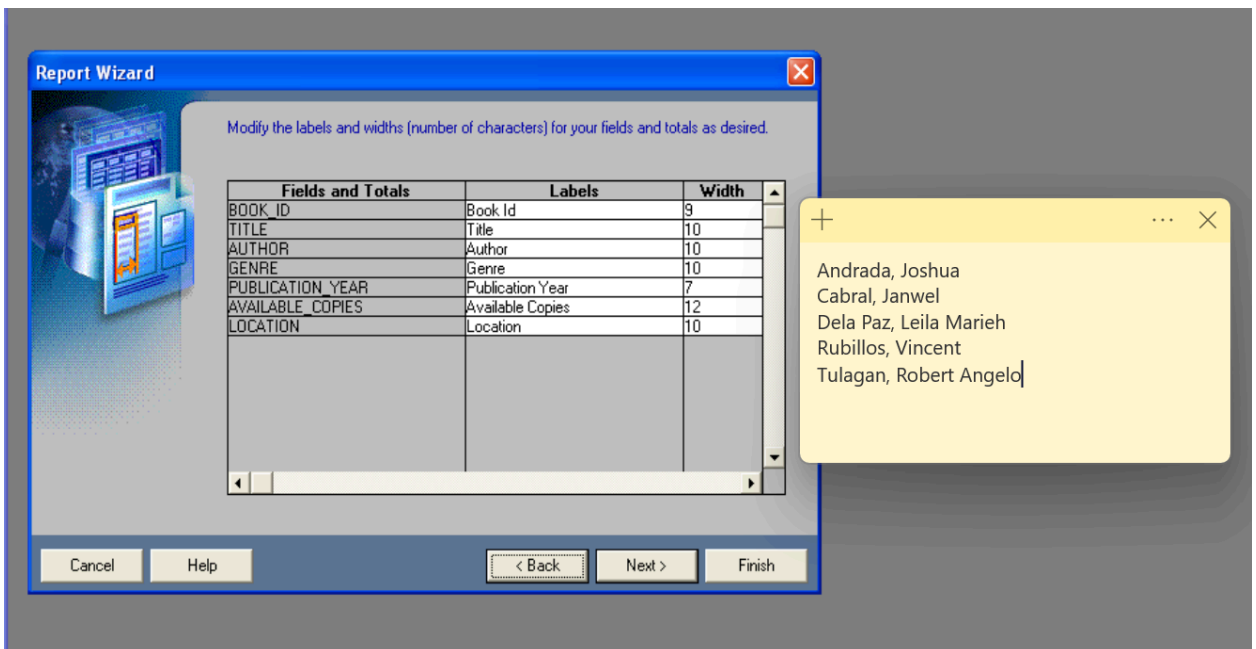
Choose the books table and select all that is available

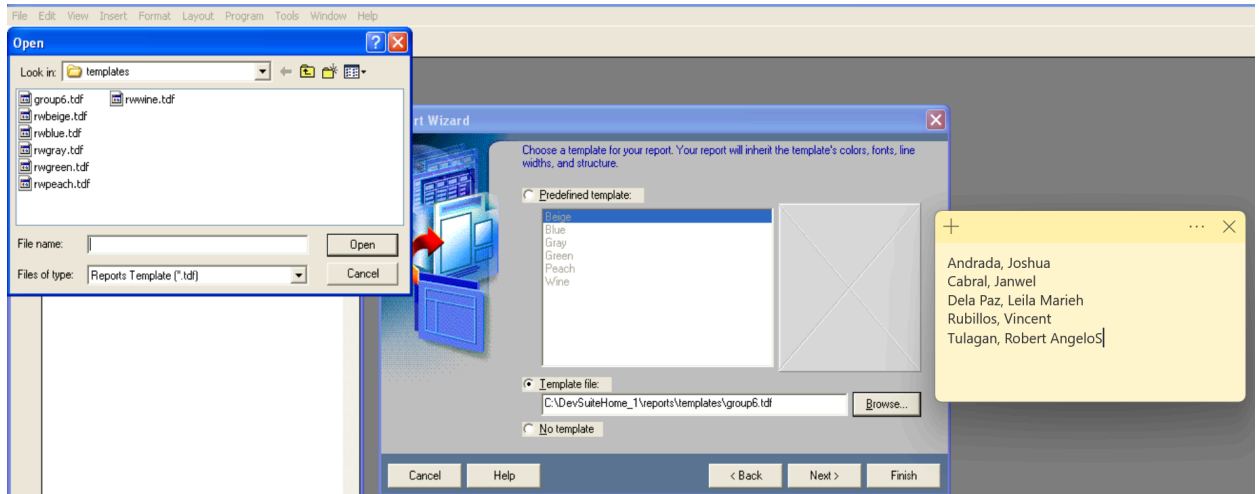


In the query add the order by book_id to change the order of the items by their book id

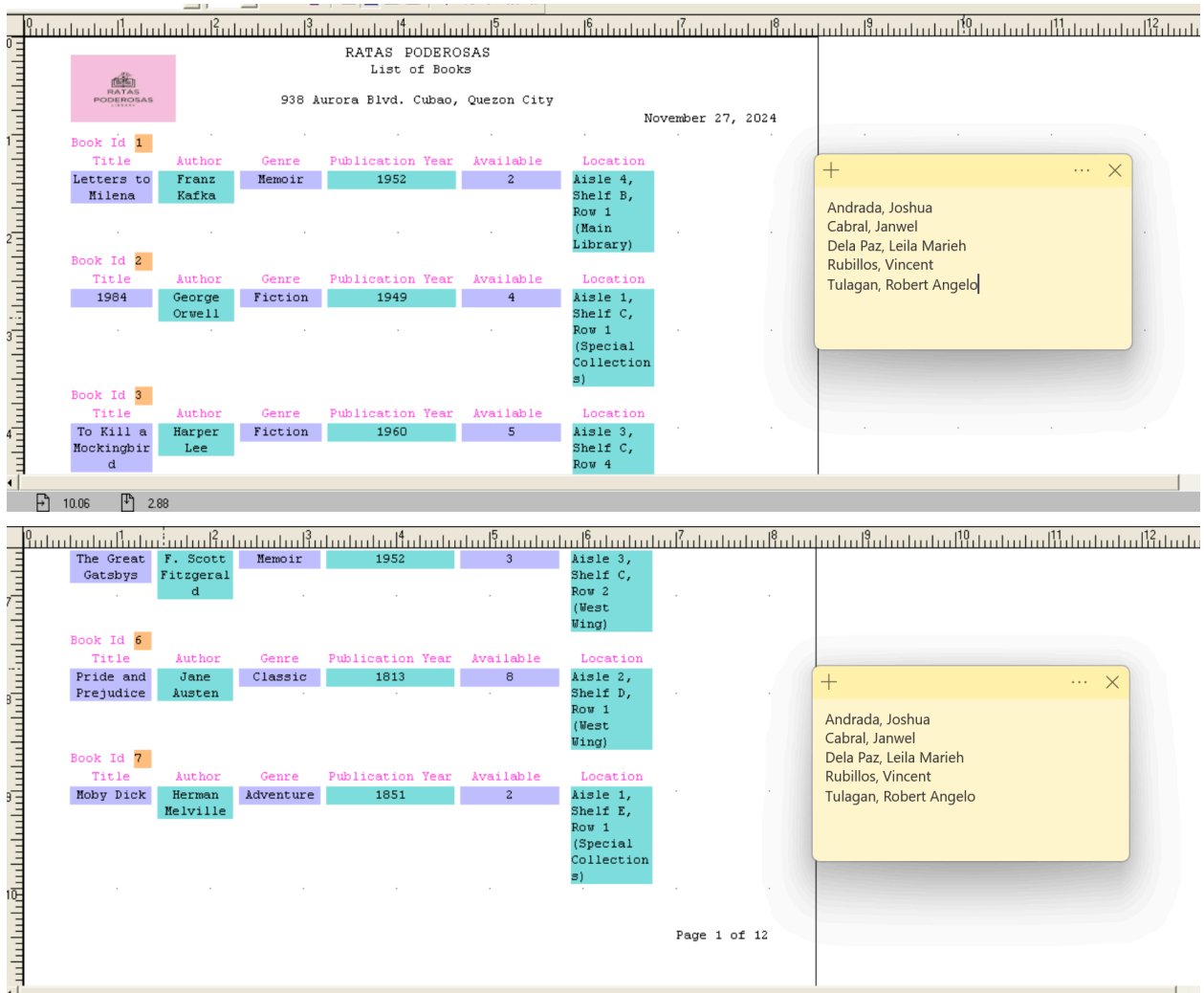


Include all except the ISBN



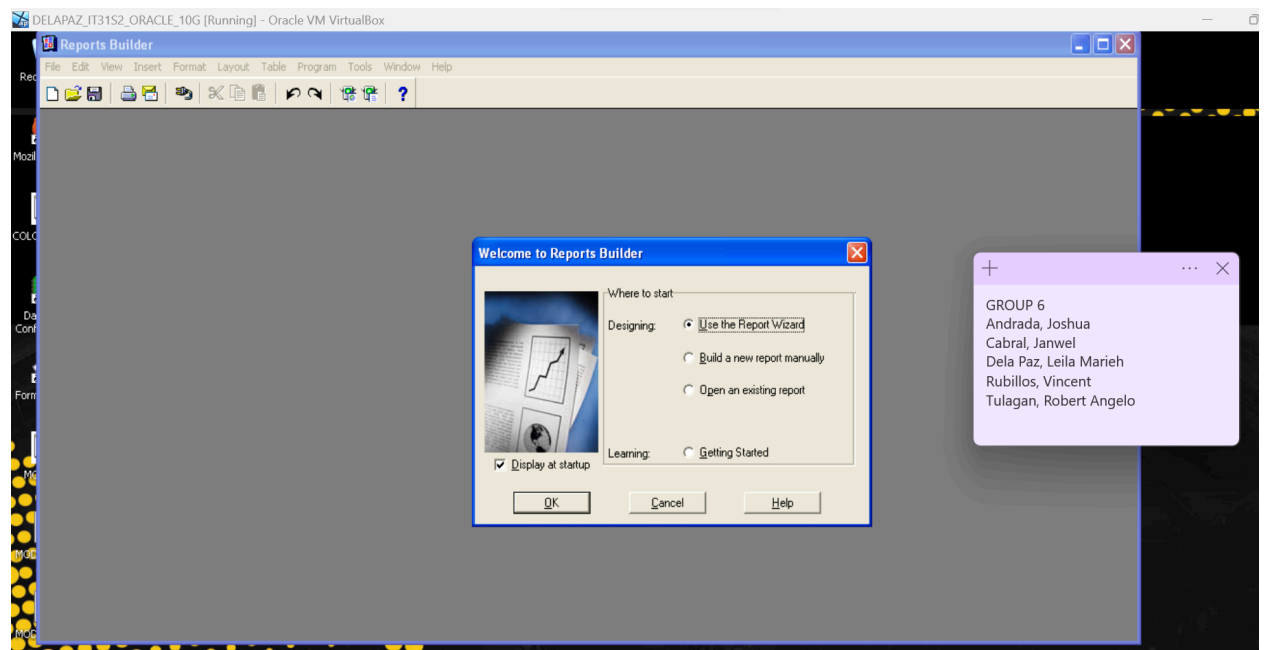


Use the template that the group created for original design.

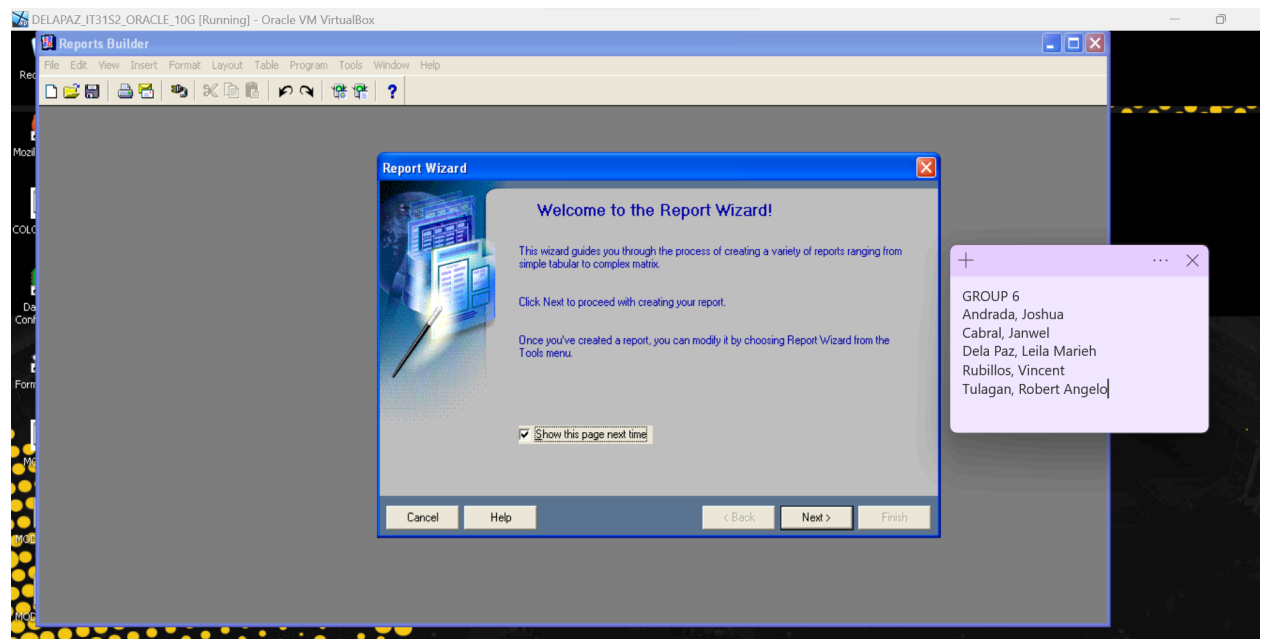


Compile and finish. (Page 1 of 12)

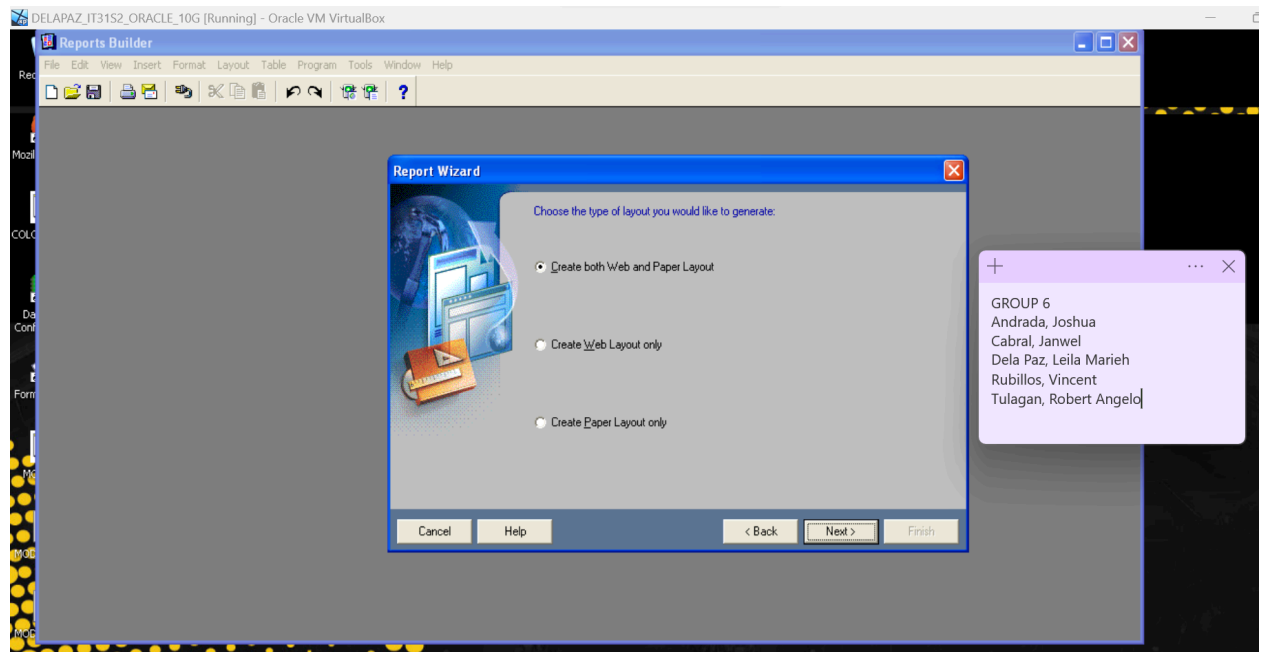
c. Group Left



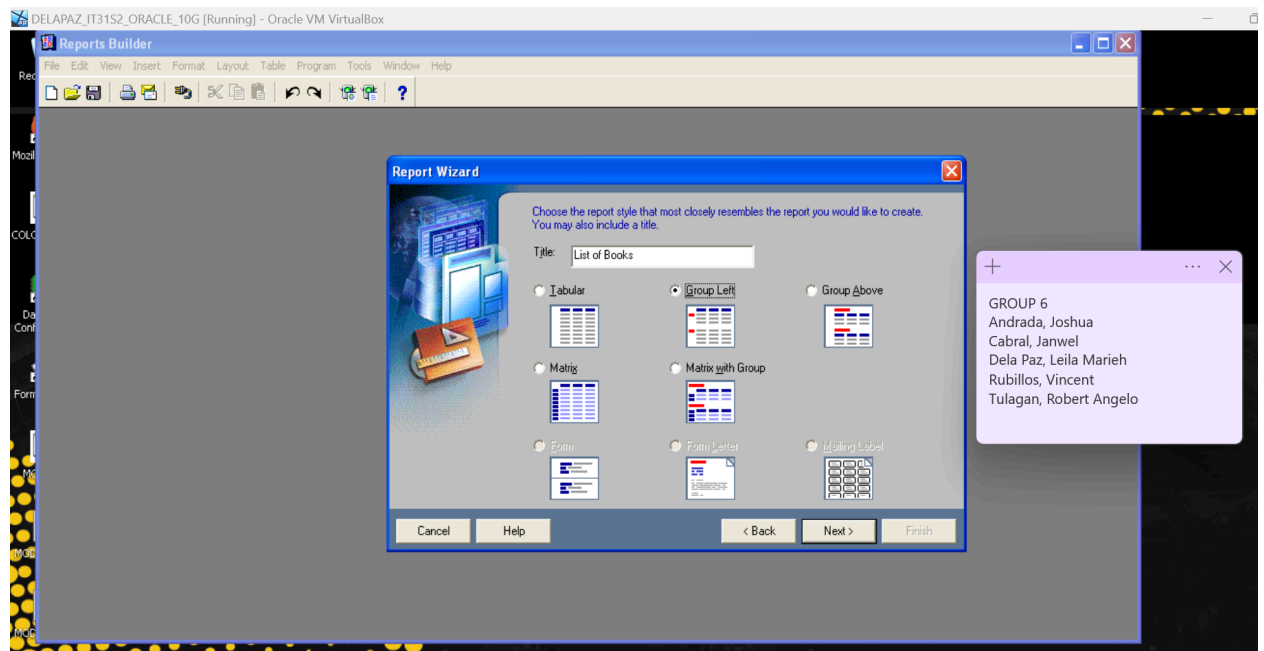
Choose the "Use the Report Wizard" option and click Ok.



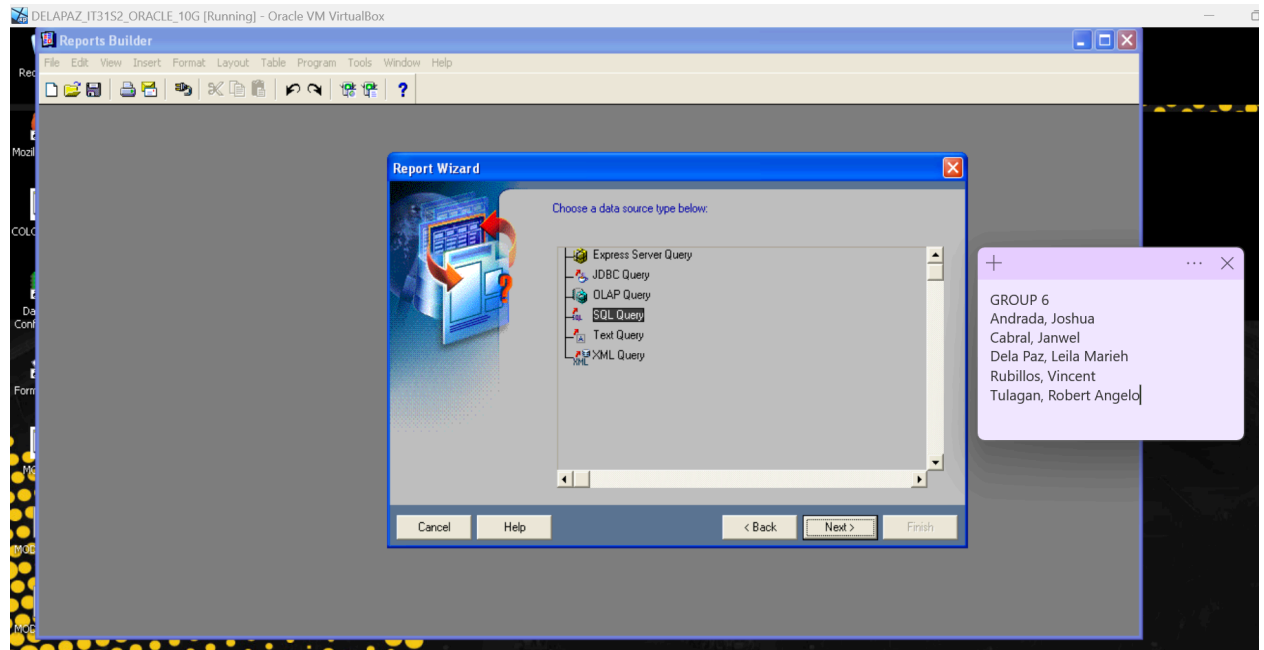
After clicking ok, this will show and just click the Next button.



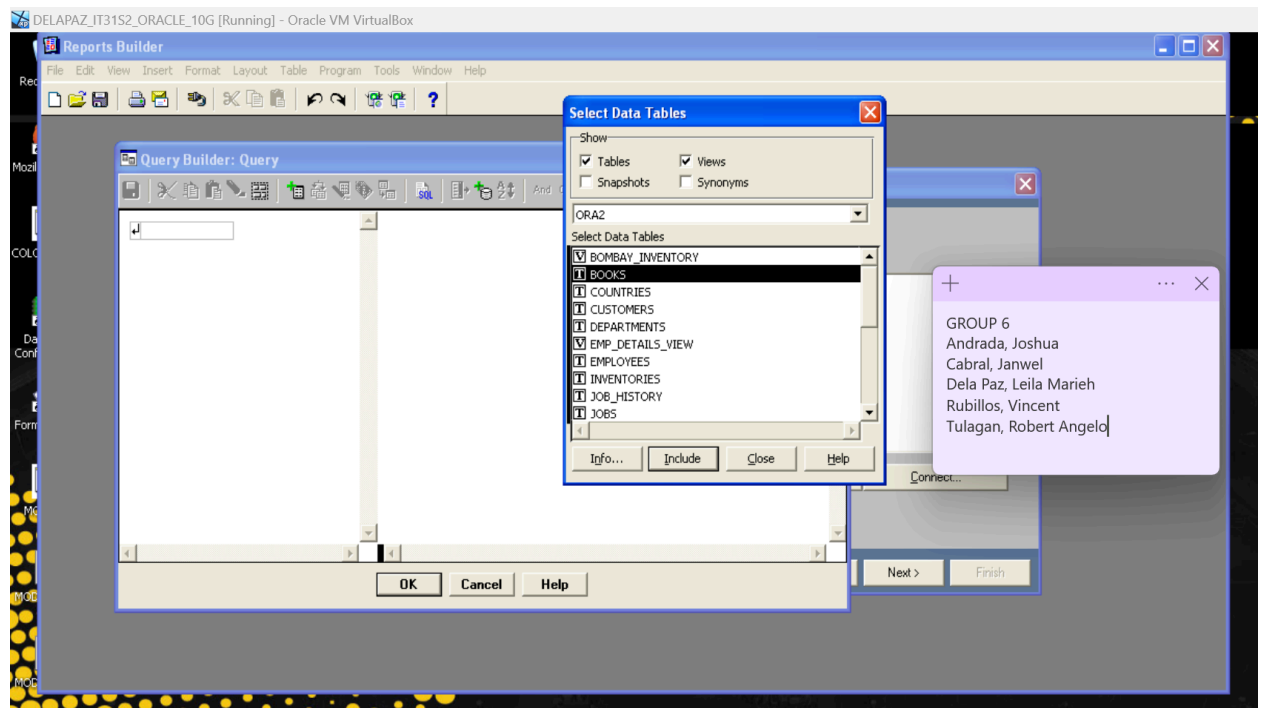
I selected the “Create both Web and Paper Layout” option then clicked the Next button.



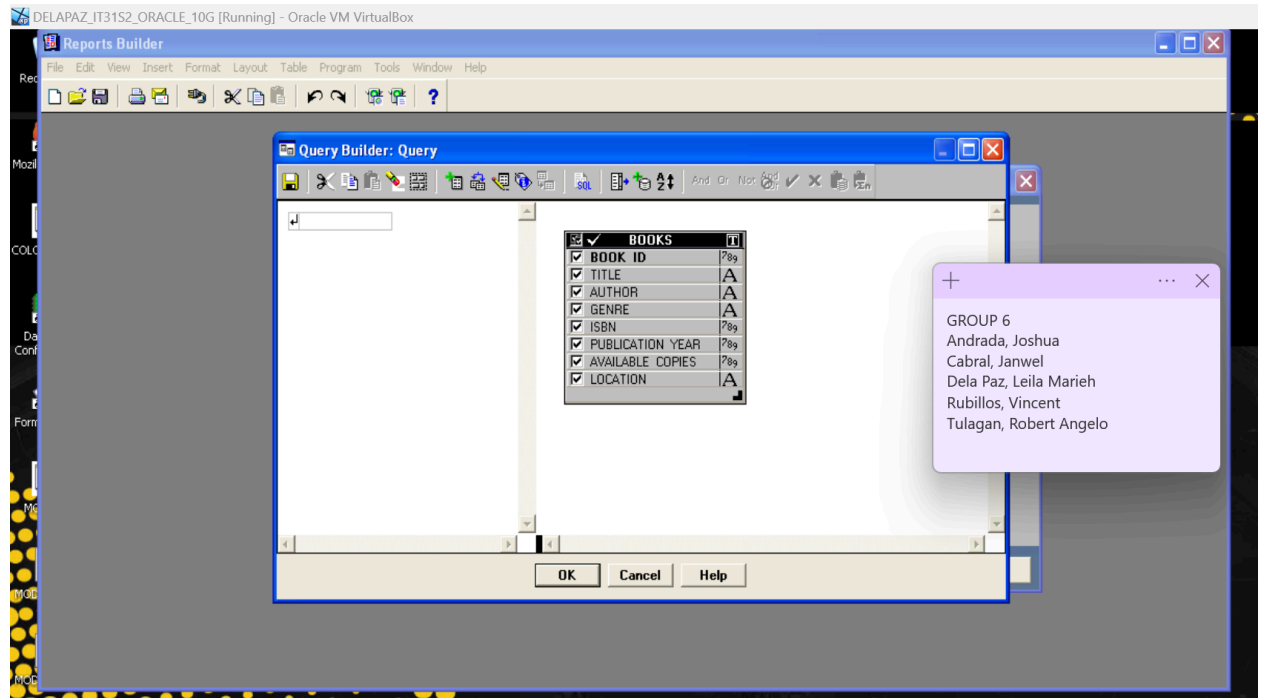
I input the title “List of Books” first then select the Group Left layout.



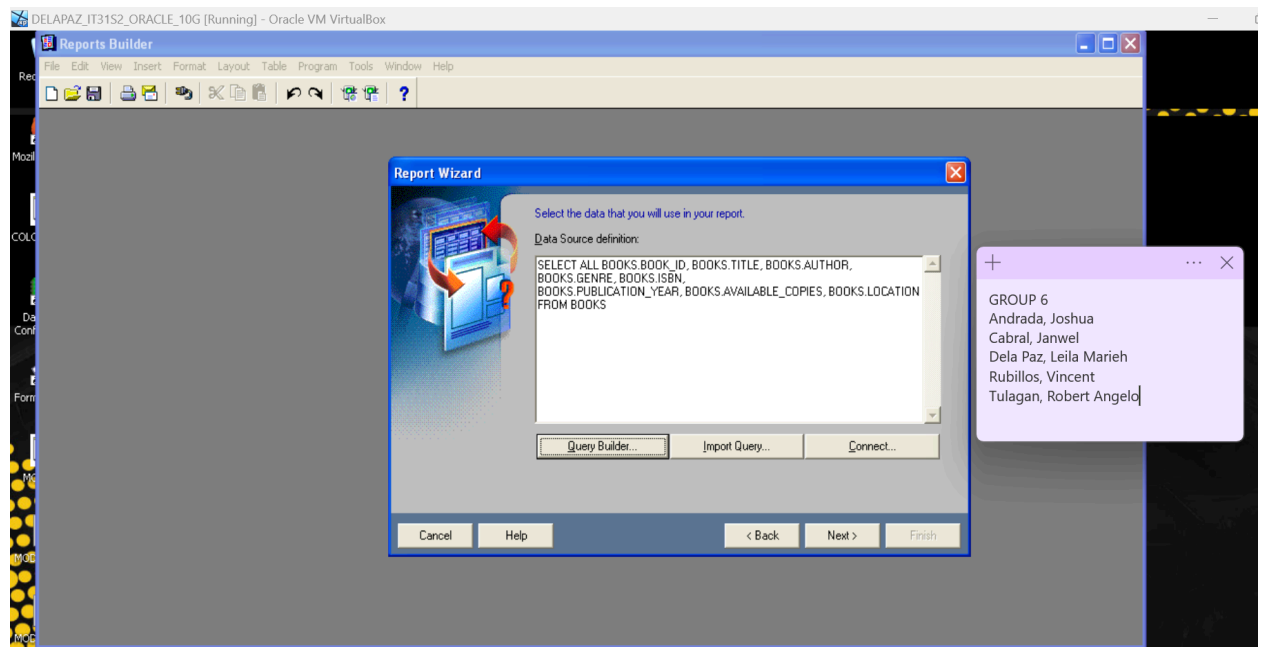
I selected the SQL query.



I included the Books table for the Group Left report.

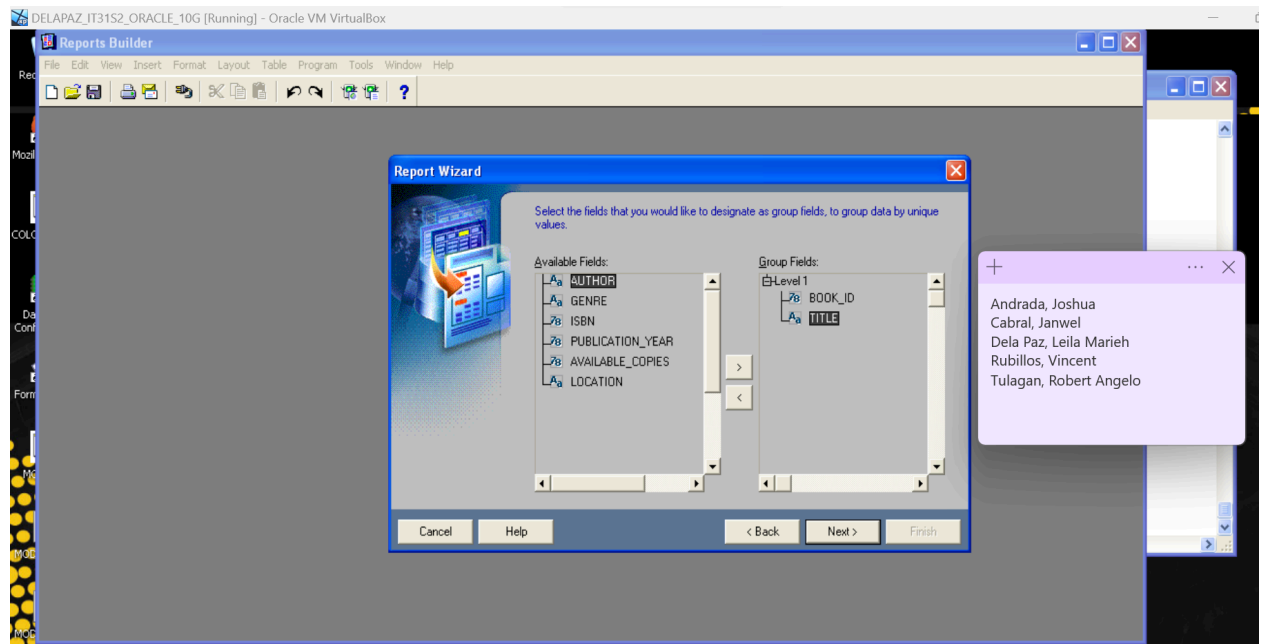


I selected all the checkboxes to include everything in the report.

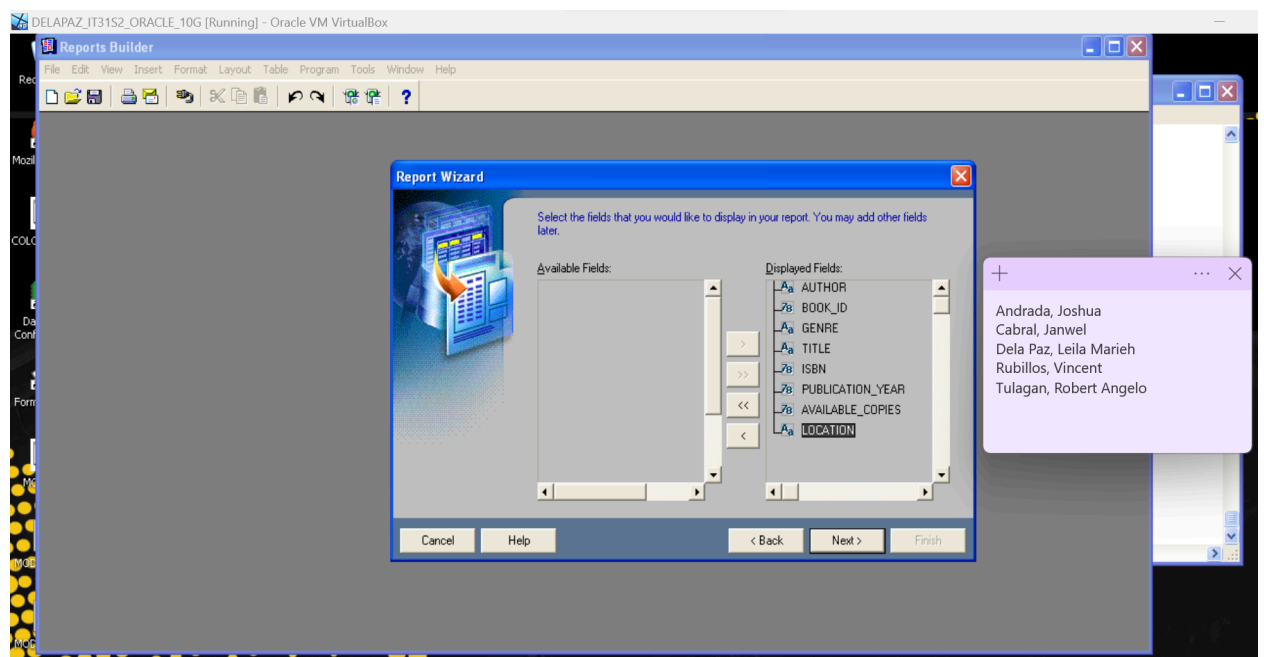


After selecting all the columns, I checked the SQL query if everything is included as it generated itself automatically.

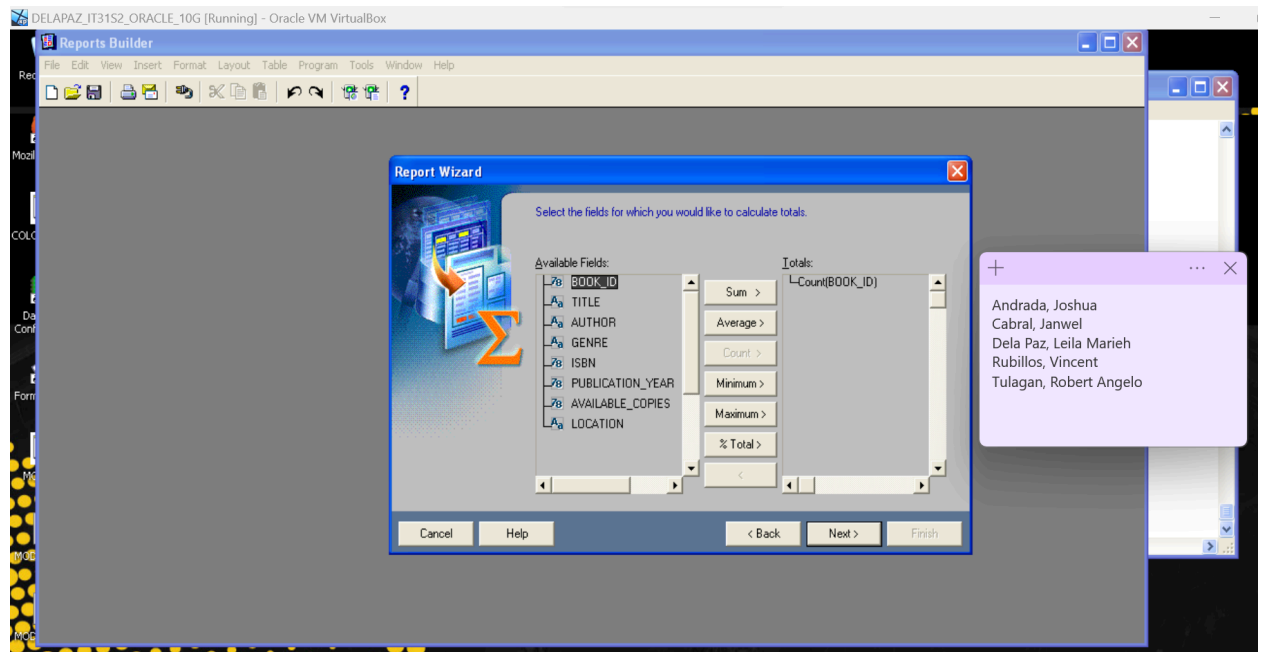
d.



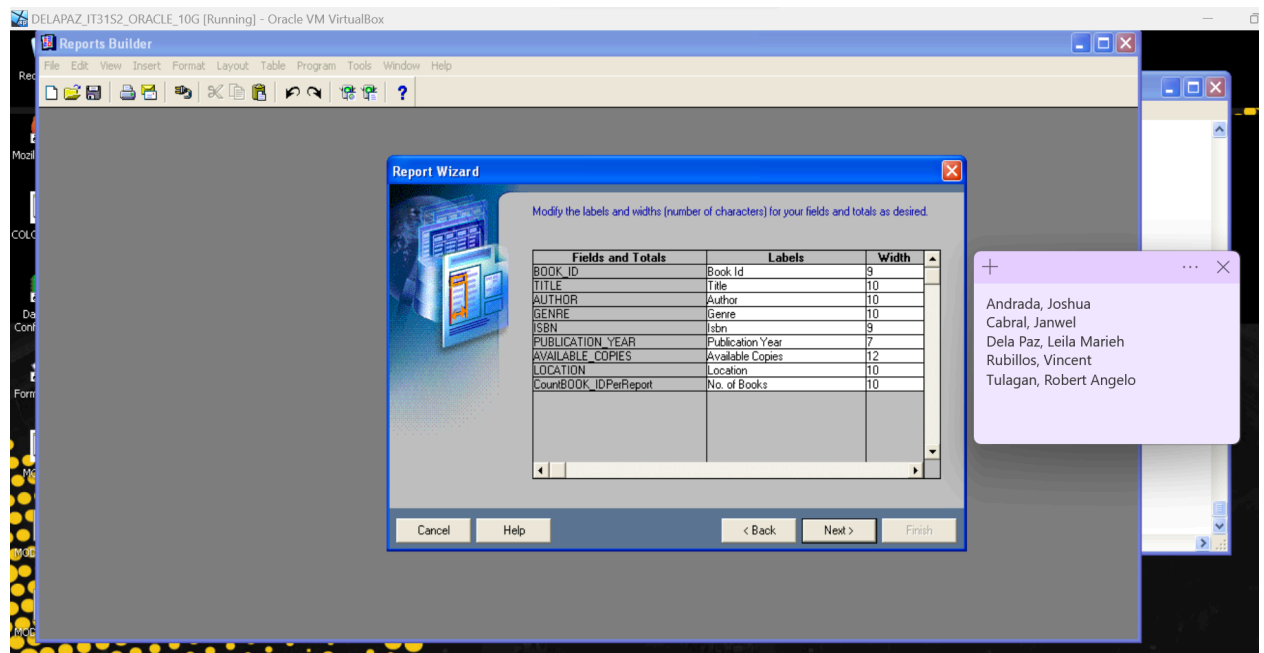
I selected the Book_ID and the title to group the data by unique values then click the Next button.



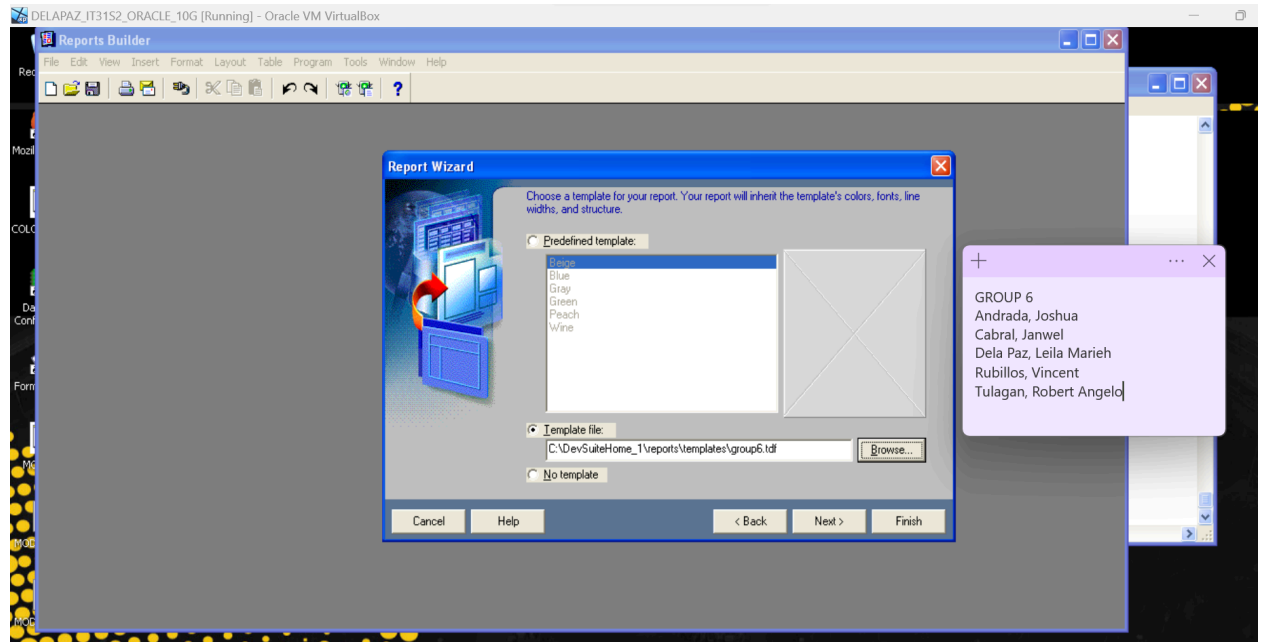
I added everything to display on my report.



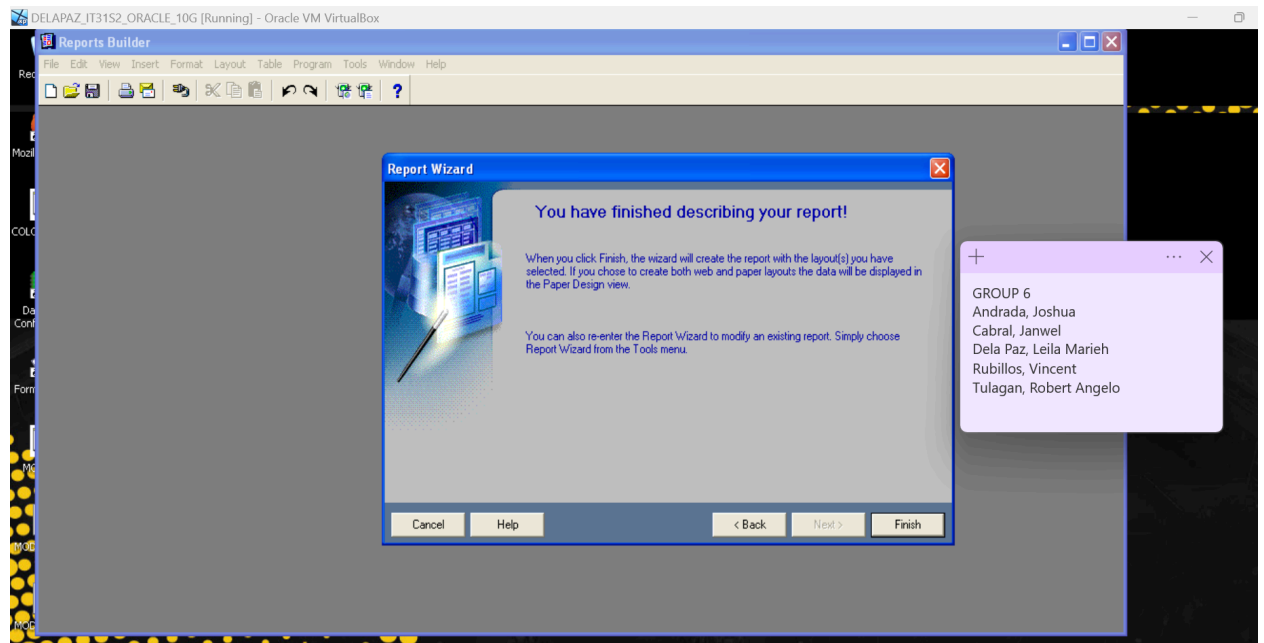
I added Count for the book ID to determine the number of books and to include in the report.



I didn't modify every label except for the Count in Book_id and changed it into "No. of Books".



After that, I chose the template for the report and I used the template that I made.



Click the Finish button to generate the Group left report.

DELAPAZ_IT31S2_ORACLE_10G [Running] - Oracle VM VirtualBox

Reports Builder - [GroupLeft: Report Editor - Paper Design]

File Edit View Insert Format Layout Program Tools Window Help

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Page: 1

RATAS PODEROSA
List of Books
938 Aurora Blvd. Cubao, Quezon City
November 29, 2024

Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location
1	Letters to Milena	Franz Kafka	Memoir	*****	1952	2	Aisle 4, Shelf B, Row 1 (Main Library)
2	1984	George Orwell	Fiction	*****	1949	4	Aisle 1, Shelf C, Row 1 (Special Collections)
3	To Kill a Mockingbird	Harper Lee	Fiction	*****	1960	5	Aisle 3, Shelf C, Row 4 (Special Collections)
4	Brave New World	Aldous Huxley	Science Fiction	*****	1932	3	Aisle 4, Shelf D, Row 5 (Main Library)
5	The Great Gatsby	F. Scott Fitzgerald	Memoir	*****	1952	3	Aisle 3, Shelf C, Row 2 (West Wing)
6	Pride and Prejudice	Jane Austen	Classic	*****	1813	8	Aisle 2, Shelf

GROUP 6
Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

1x 9.34 4.13

This is the final Group Left report style that I added some colors and changed the font style of the text. I also fixed some positions and added a margin to the name of the column.

DELAPAZ_IT31S2_ORACLE_10G [Running] - Oracle VM VirtualBox

Reports Builder - [GroupLeft: Report Editor - Paper Design]

File Edit View Insert Format Layout Program Tools Window Help

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Page: 1

7	Moby Dick	Herman Melville	Adventure	*****	1851	2	Aisle 1, Shelf E, Row 1 (Special Collections)
8	The Catcher in the Rye	J.D. Salinger	Fiction	*****	1951	4	Aisle 3, Shelf C, Row 4 (Main Library)
9	The Hobbit	J.R.R. Tolkien	Fantasy	*****	1937	5	Aisle 1, Shelf D, Row 1 (East Wing)
10	The Fellowship of the Ring	J.R.R. Tolkien	Fantasy	*****	1954	7	Aisle 4, Shelf B, Row 4 (Special Collections)
11	Fahrenheit 451	Ray Bradbury	Science Fiction	*****	1953	3	Aisle 2, Shelf D, Row 4 (West Wing)
12	Slaughterhouse-Five	Kurt Vonnegut	Fiction	*****	1969	6	Aisle 2, Shelf C, Row 5 (Special Collections)
13	Crime and Punishment	Fyodor Dostoevsky	Classic	*****	1866	4	Aisle 1, Shelf D, Row 4 (Special Collections)
14	War and Peace	Leo Tolstoy	Historical	*****	1869	2	Aisle 4, Shelf

GROUP 6
Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

1x 9.38 8.44

This is the second part of the final Group Left report style.

DELAPAZ_IT31S2_ORACLE_10G [Running] - Oracle VM VirtualBox

Reports Builder - [GroupLeft: Report Editor - Paper Design]

File Edit View Insert Format Layout Program Tools Window Help

Page: 1

Courier New (Western)

	Fellowship of the Ring						B, Row 4 (Special Collections)	
11	Fahrenheit 451	Ray Bradbury	Science Fiction	*****	1953	3	Aisle 2, Shelf D, Row 4 (West Wing)	
12	Slaughterhouse-Five	Kurt Vonnegut	Fiction	*****	1969	6	Aisle 2, Shelf C, Row 5 (Special Collections)	
13	Crime and Punishment	Fyodor Dostoevsky	Classic	*****	1866	4	Aisle 1, Shelf D, Row 4 (Special Collections)	
14	War and Peace	Leo Tolstoy	Historical Fiction	*****	1869	2	Aisle 4, Shelf E, Row 2 (West Wing)	
15	One Hundred Years of Solitude	Gabriel Garcia Marquez	Fiction	*****	1967	4	Aisle 2, Shelf C, Row 3 (West Wing)	

Page 1 of 6

10.44 10.75

+

...

×

GROUP 6

Andrada, Joshua

Cabral, Janwel

Dela Paz, Leila Marieh

Rubillos, Vincent

Tulagan, Robert Angelo

This is the last part of the final Group Left report style.

DELAPAZ_IT31S2_ORACLE_10G [Running] - Oracle VM VirtualBox

Reports Builder - [GroupLeft: Report Editor - Paper Design]

File Edit View Insert Format Layout Program Tools Window Help

Page: 6

Courier New (Western)

RATAS PODEROSA							
List of Books							
938 Aurora Blvd. Cubao, Quezon City							
November 29, 2024							
Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location
77	Anna Karenina	Jane Austen	Classic	*****	1813	6	A, Row 5 (West Wing)
78	Pride and Prejudice	Jane Austen	Classic	*****	1815	4	Aisle 2, Shelf A, Row 1 (Special Collections)
79	Emma	Jane Austen	Classic	*****	1815	4	Aisle 4, Shelf D, Row 2 (Special Collections)

No. of Books: 79

+

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×

GROUP 6

Andrada, Joshua

Cabral, Janwel

Dela Paz, Leila Marieh

Rubillos, Vincent

Tulagan, Robert Angelo

In the last page, you can see the total number of books.

e. Matrix

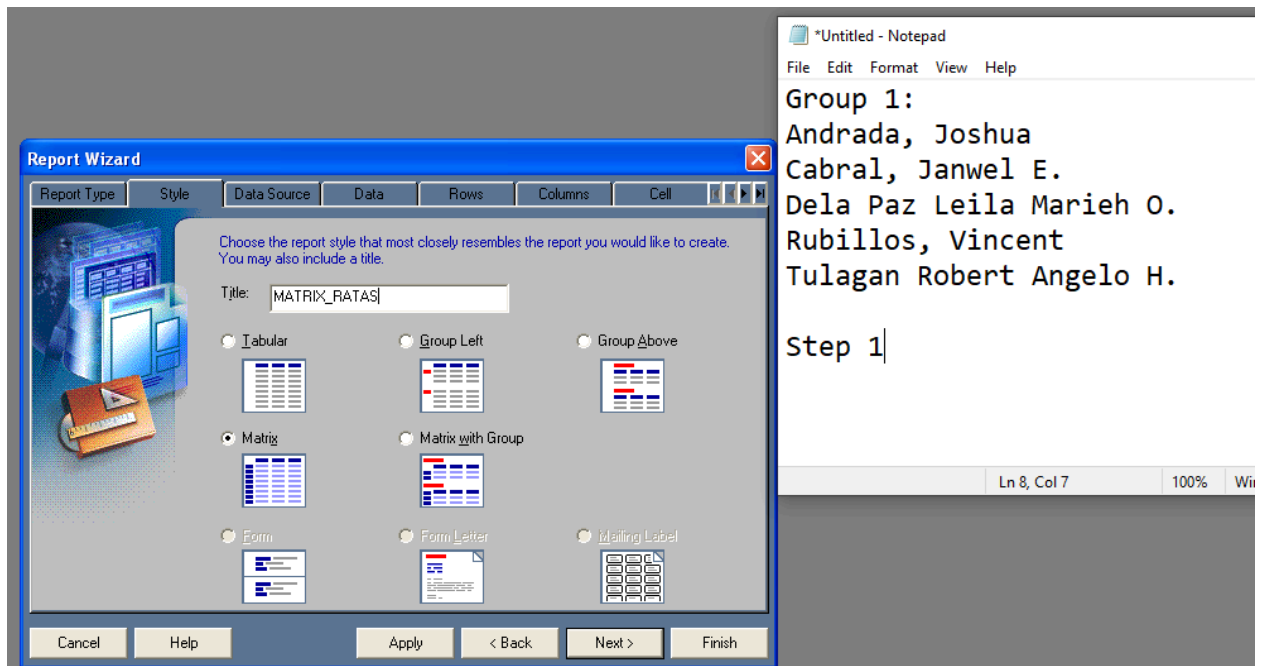


Figure 6,1 Group Matrix style

Utilizing the report wizard and selecting the Group Matrix as our report style format

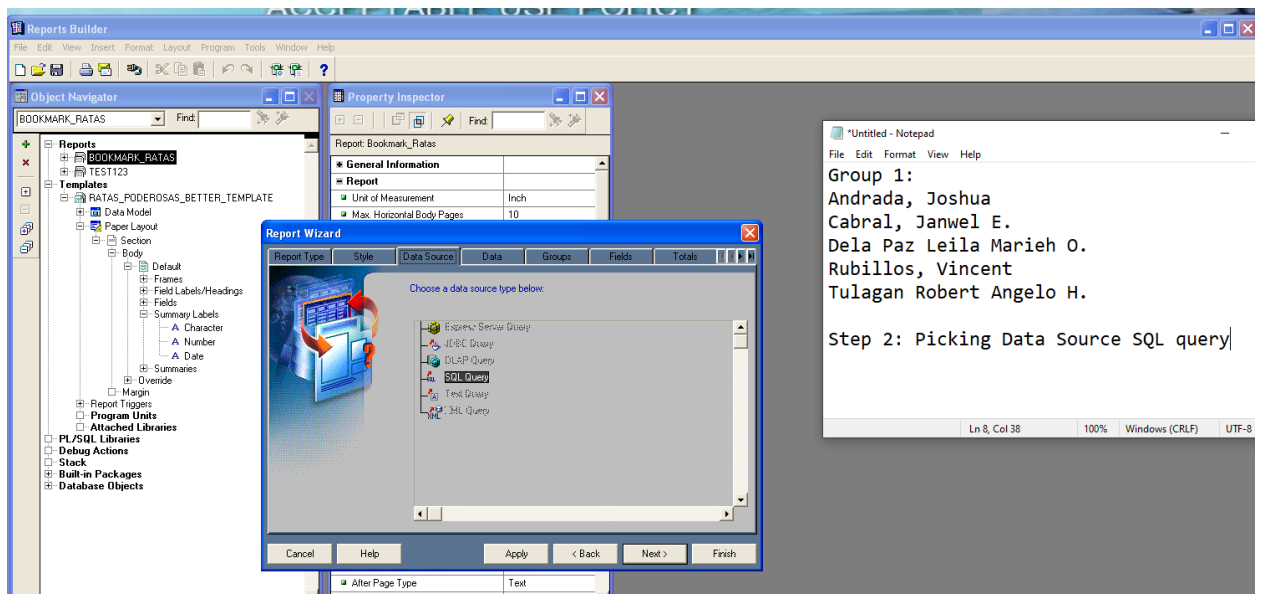


Figure 6.2 Data source selection

Using the SQL query as our data source to manipulate our data in our reports.

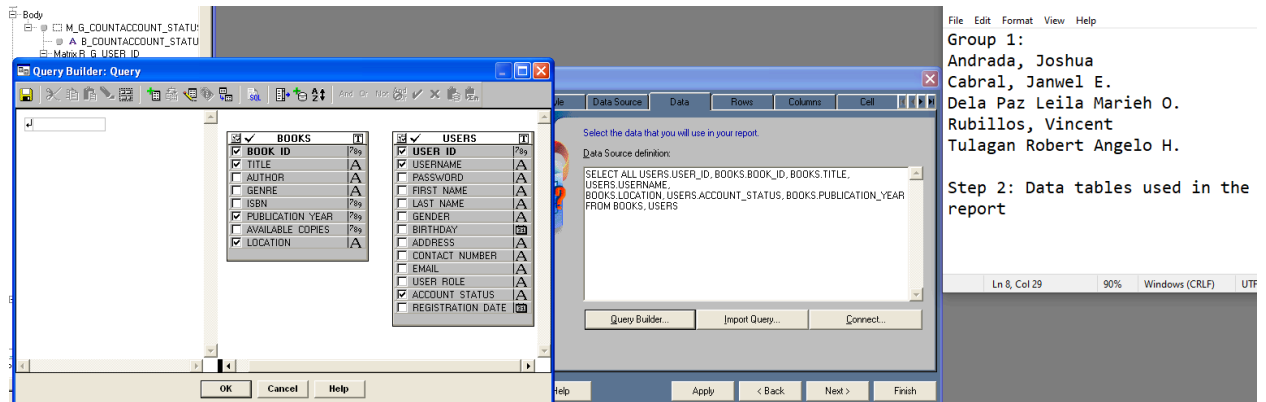


Figure 6.3 Data used in the report

We utilized Query builder and selected Users and Books table to represent as our data report for this sample

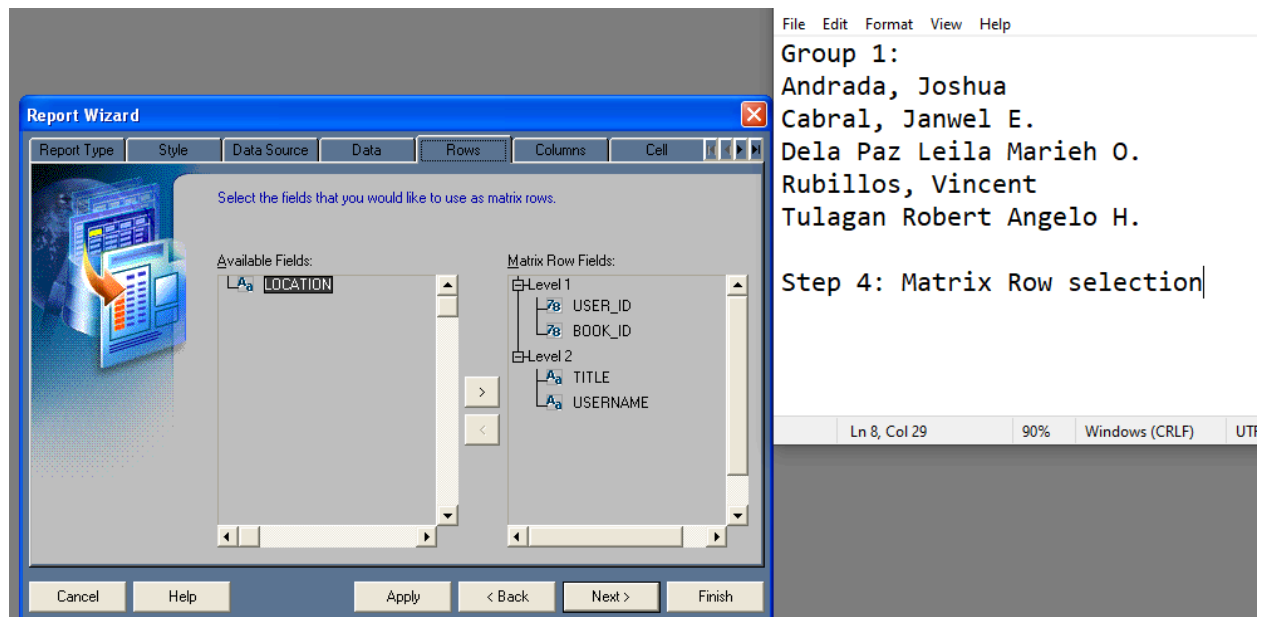


Figure 6.4 Matrix Rows

We made book_id and User_Id as our first level, and Title and Username as the second level row.

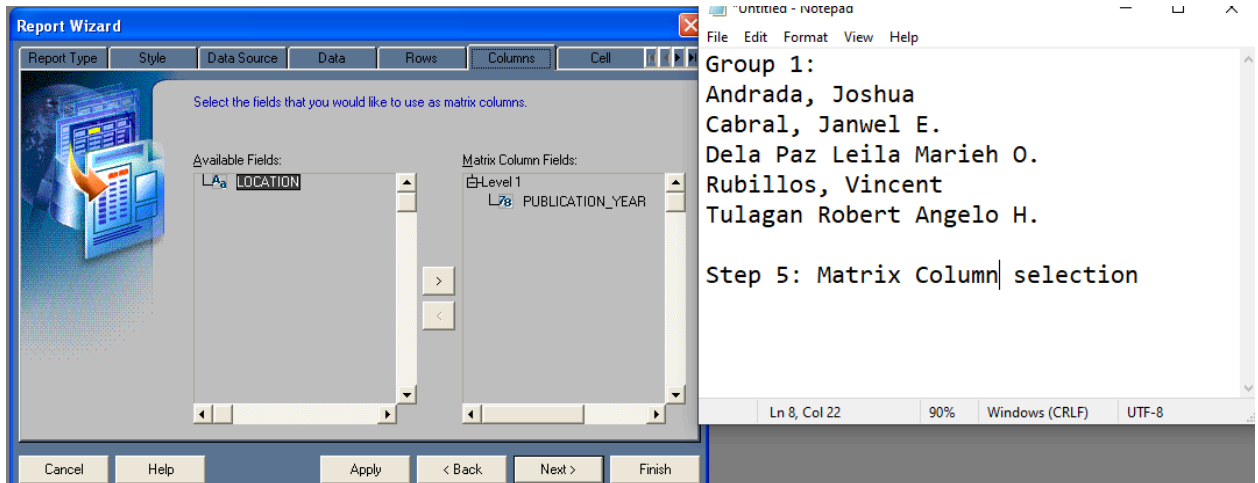


Figure 6.5 Matrix Columns

Next, we chose Publication_Year column to act as our matrix Column

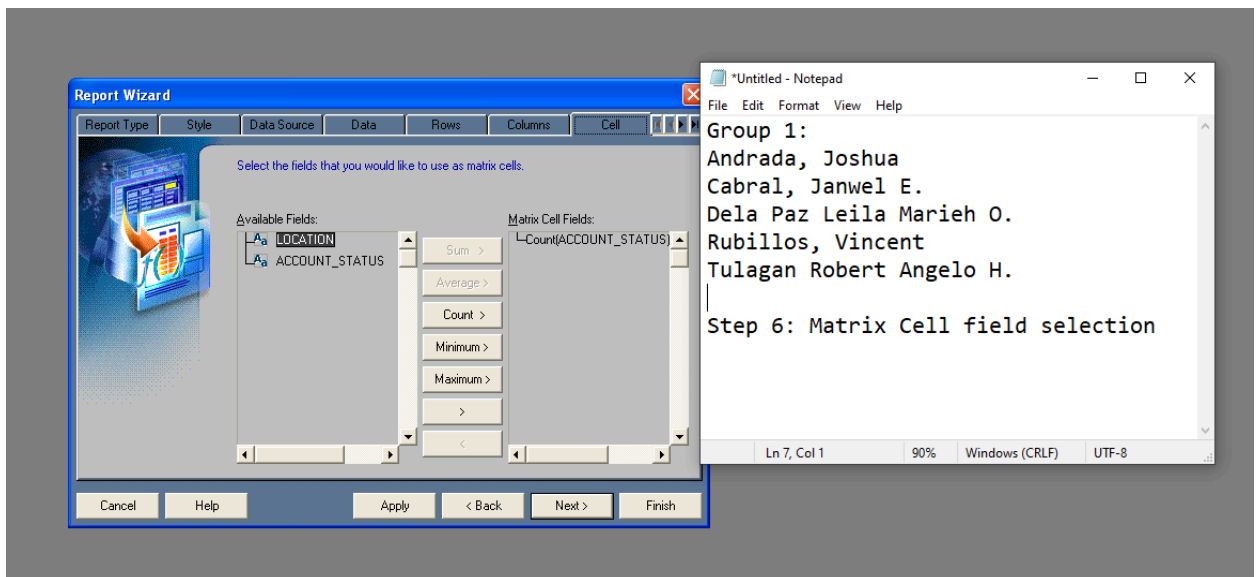


Figure 6.6 Matrix Cell Selection

We decided to use our table column "Account_Status" as the Matrix Cell field, while applying count operations on the selected table column..

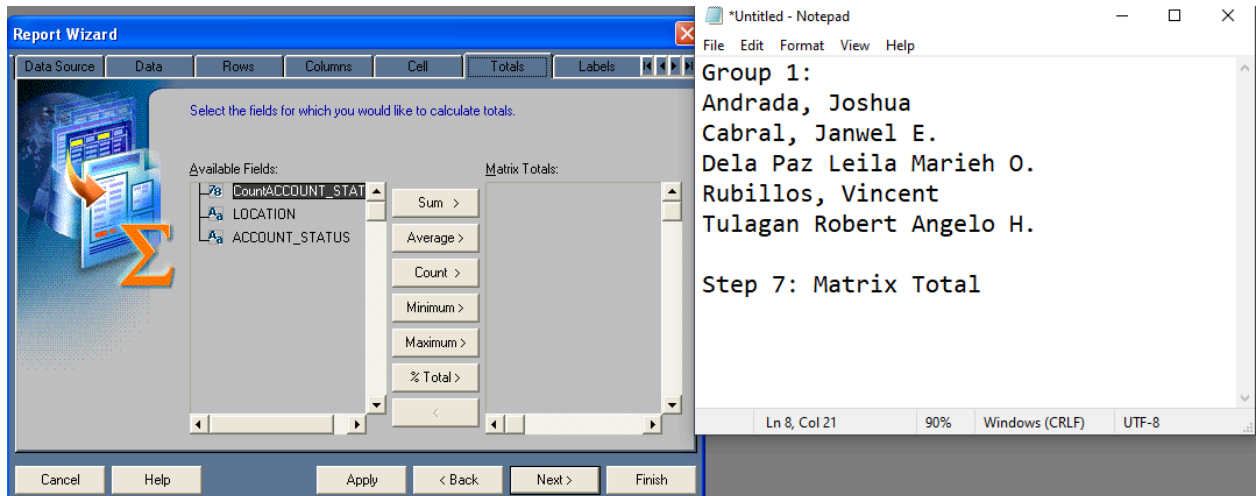


Figure 6.7 Matrix Total section

On the matrix total section, we chose to forgo and leaving it empty as it isn't needed for this sample.

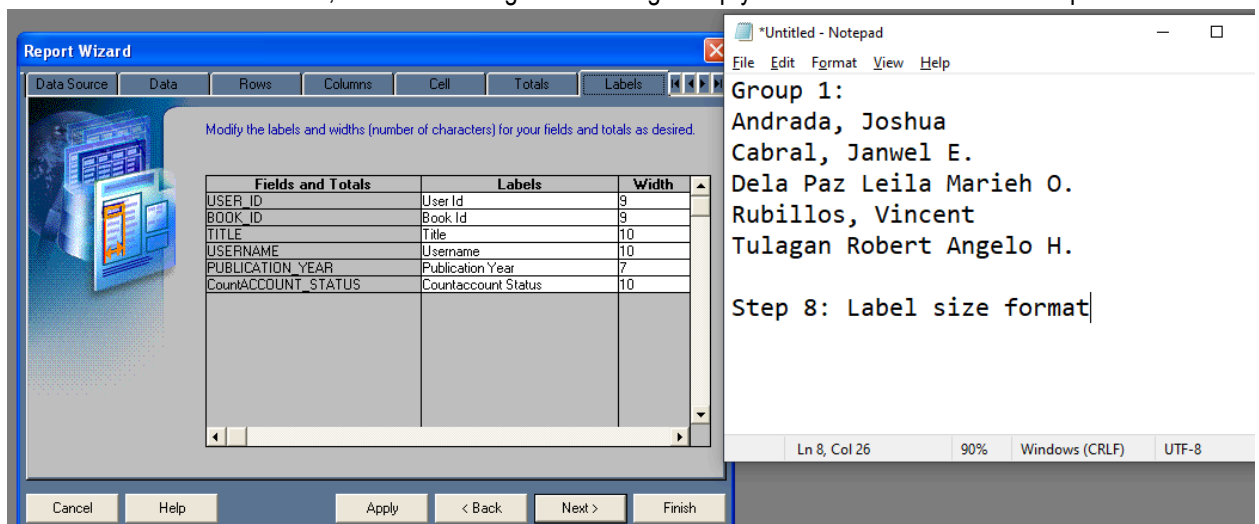


Figure 6.8 Label size format

our table data contents label format width size and labels are slightly changed to organize data sizes evenly.

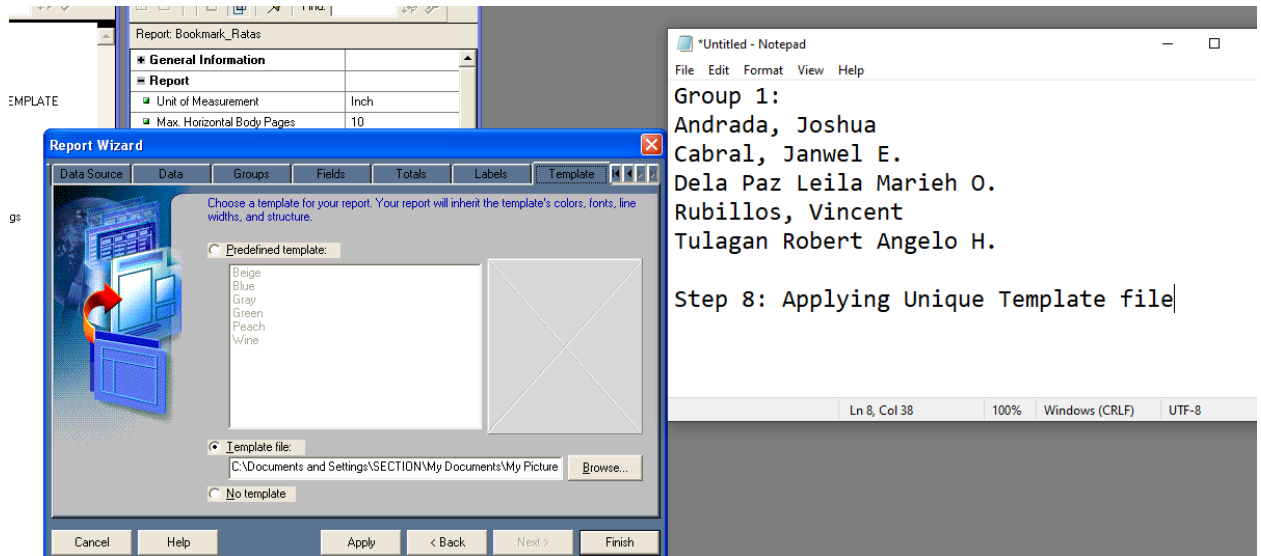


Figure 6.9 Applying the unique template format
We utilized our custom made template that projects our company logo design and color scheme.

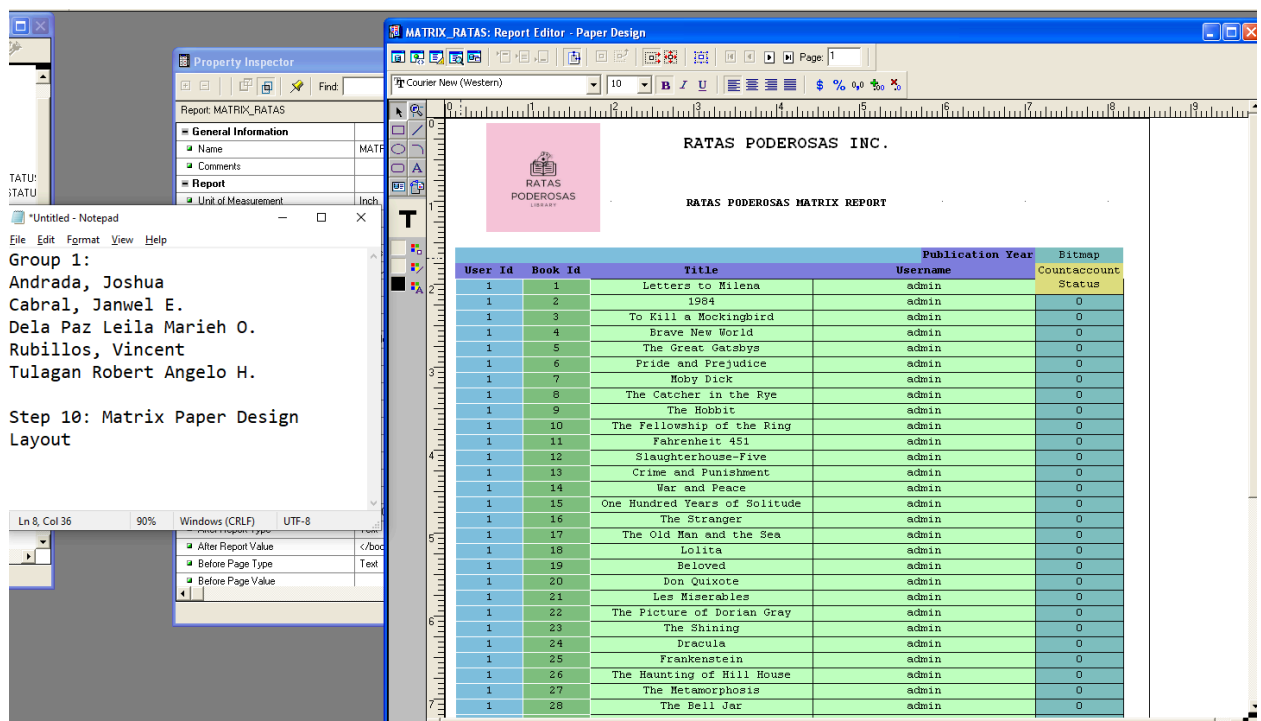
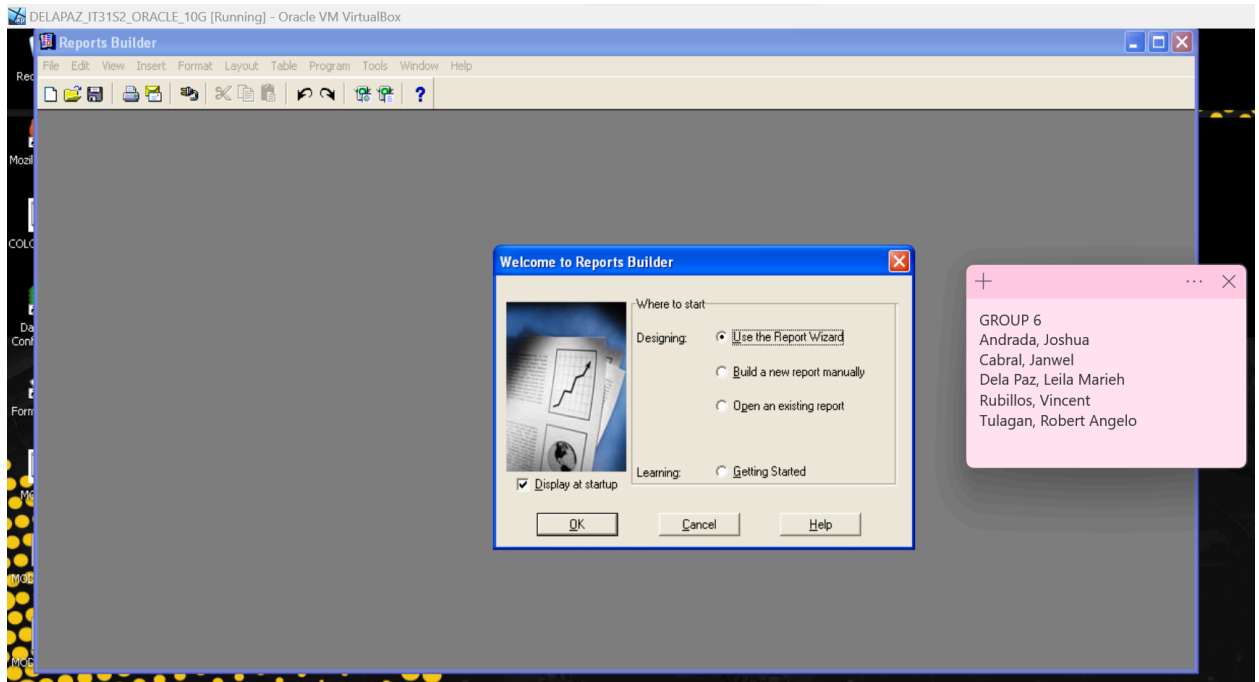
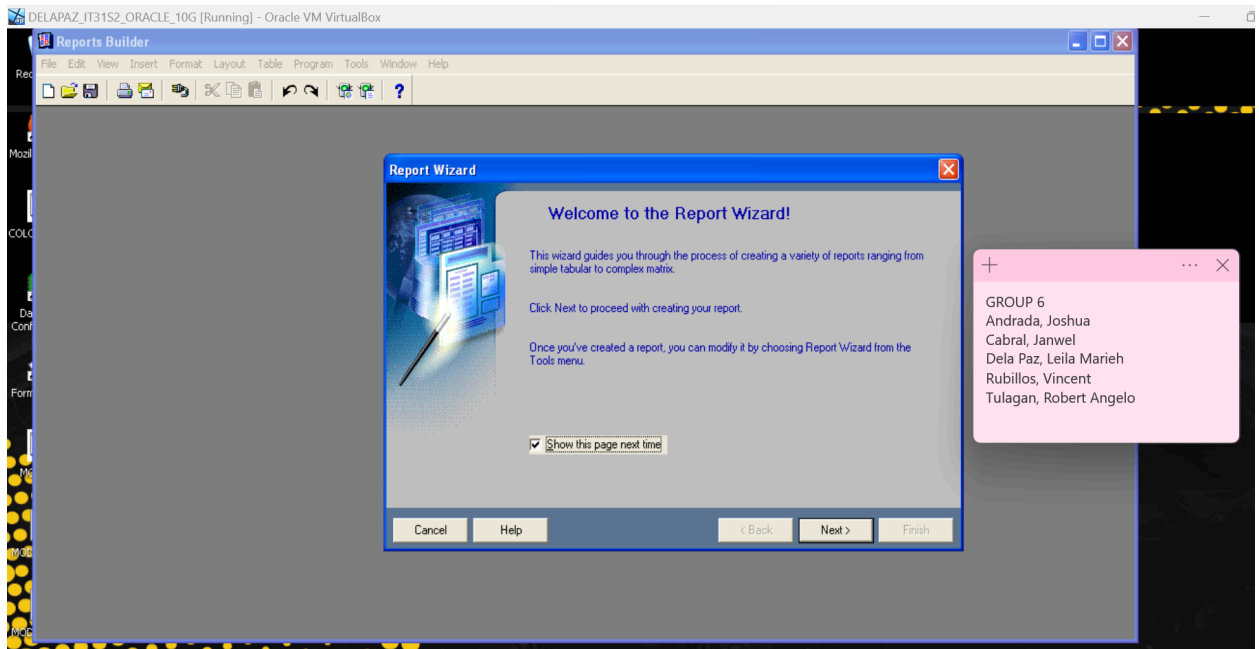


Figure 6.9 Paper design layout
Once applying and finishing the steps and minor tweak, this is the final layout of the matrix report of RATAS PODEROSAS. .

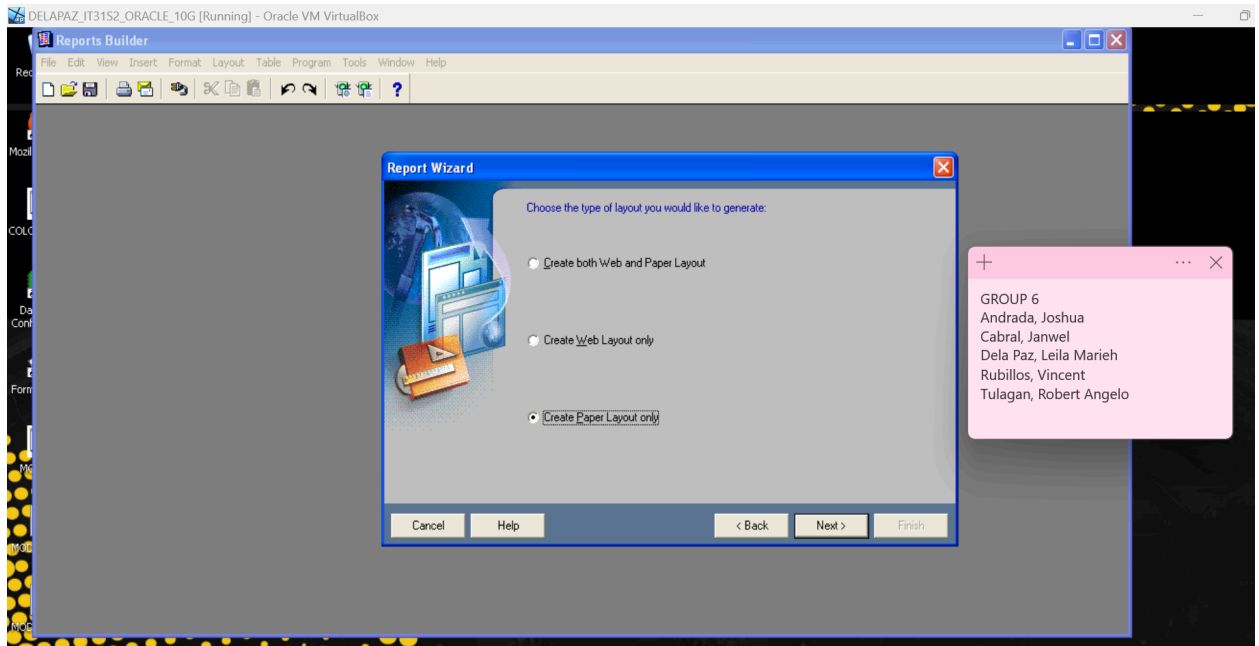
f. Mailing Label and Mailing Letter



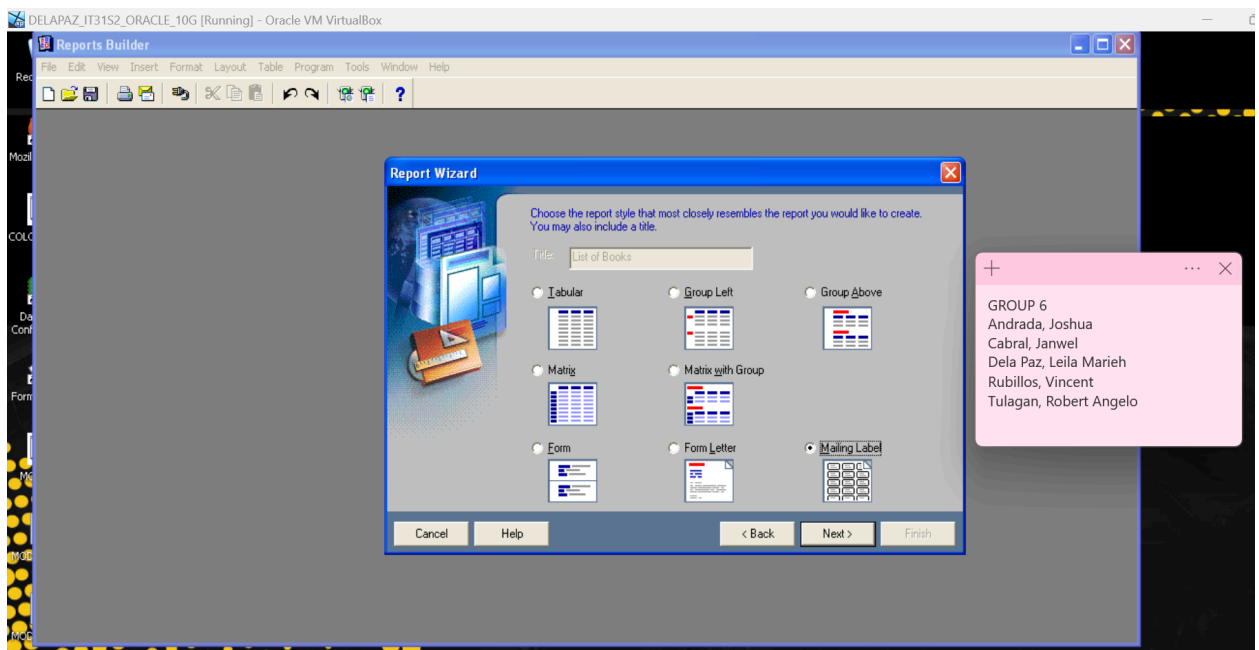
Choose the "Use the Report Wizard" option and click Ok.



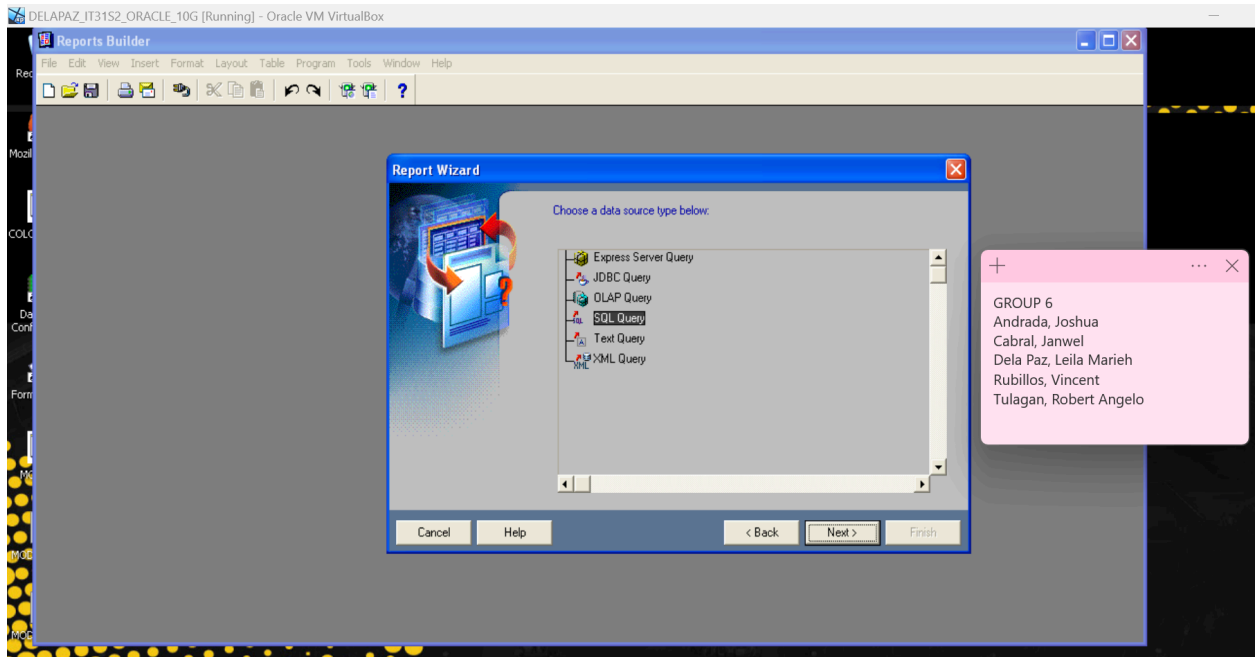
After clicking ok, this will show and just click the Next button.



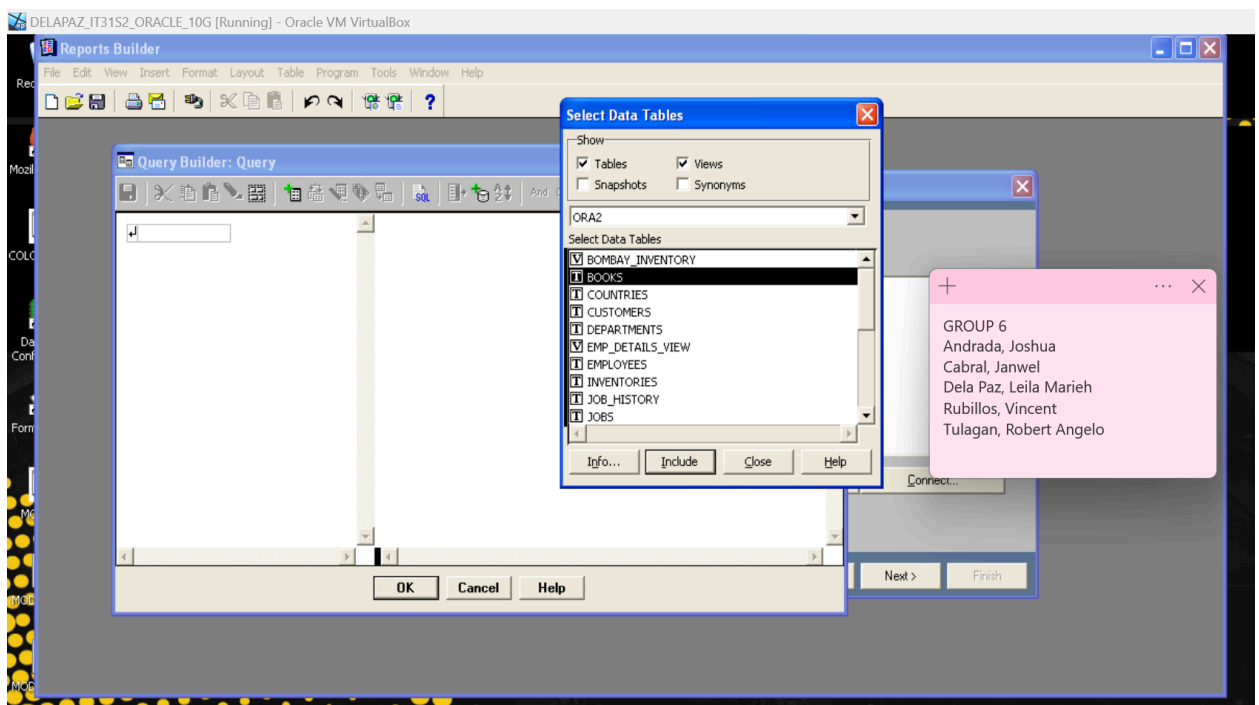
I selected the “Create Paper Layout only” option then clicked the Next button.



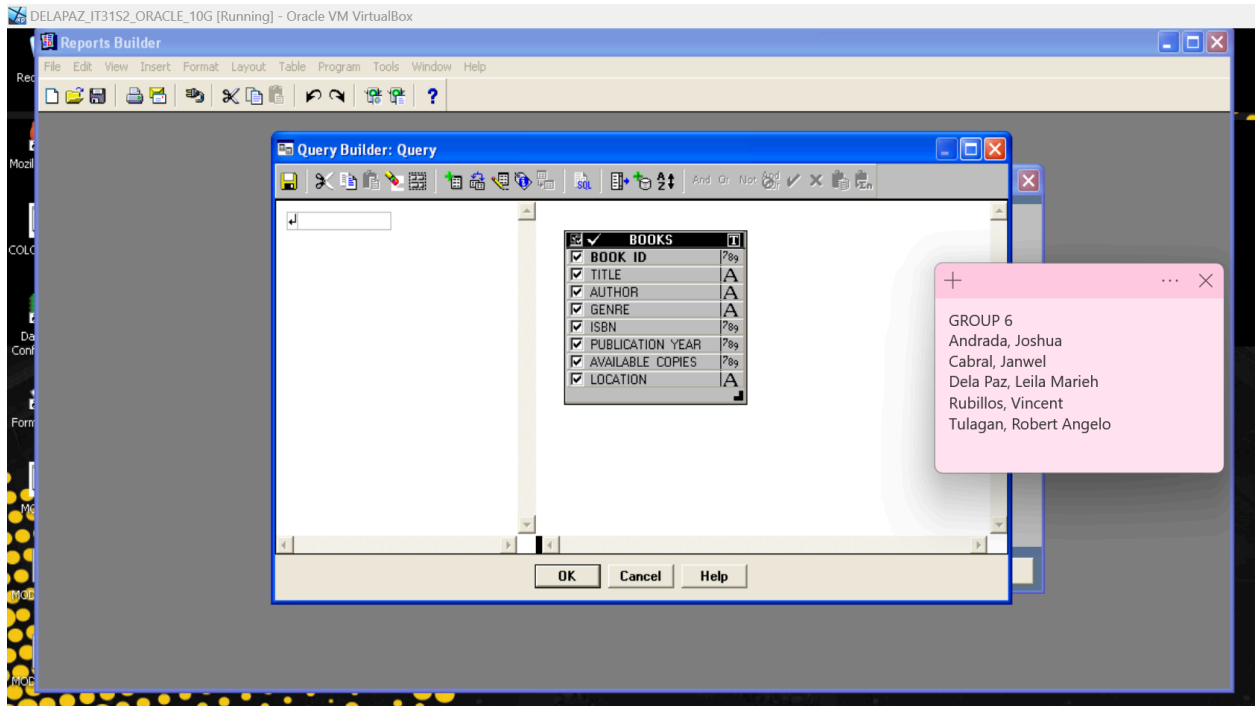
I input the title “List of Books” first before selecting the mailing label layout.



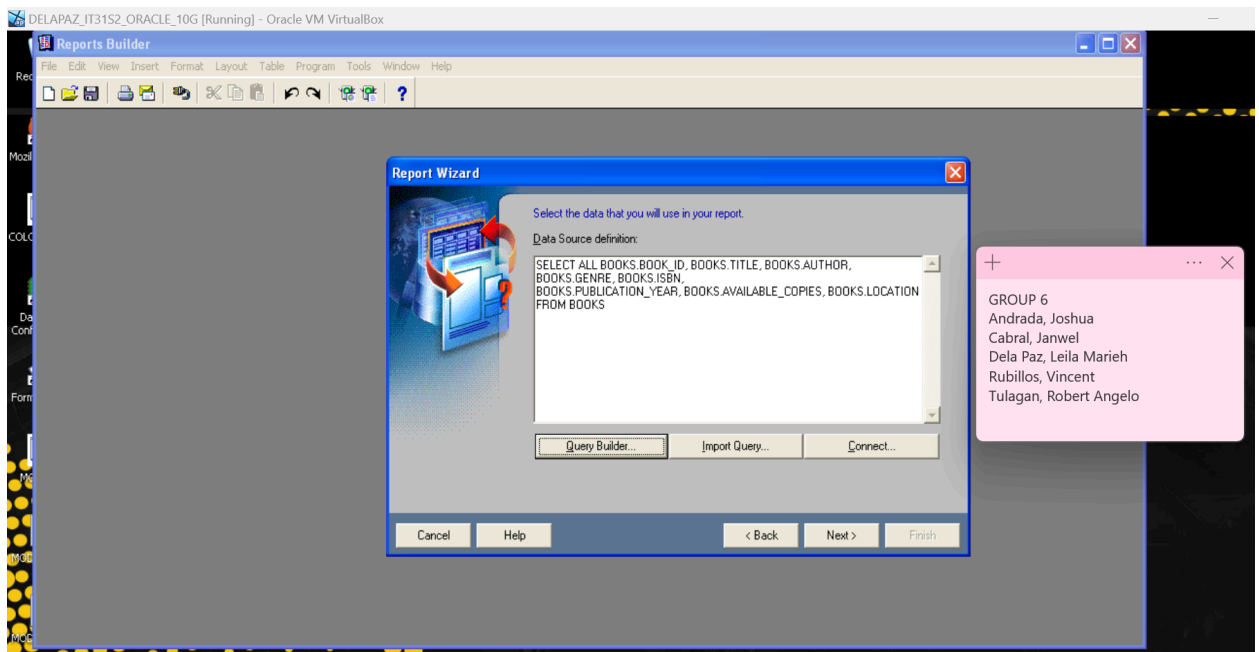
I selected the SQL query.



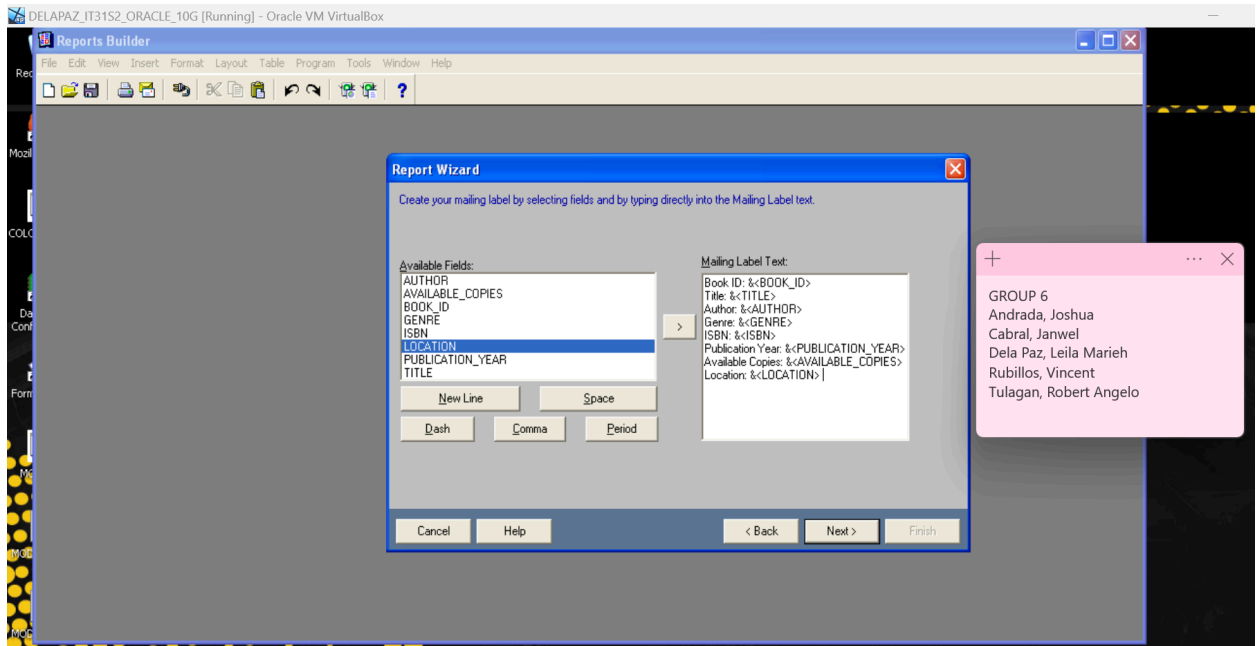
I included the Books table for the mailing label report.



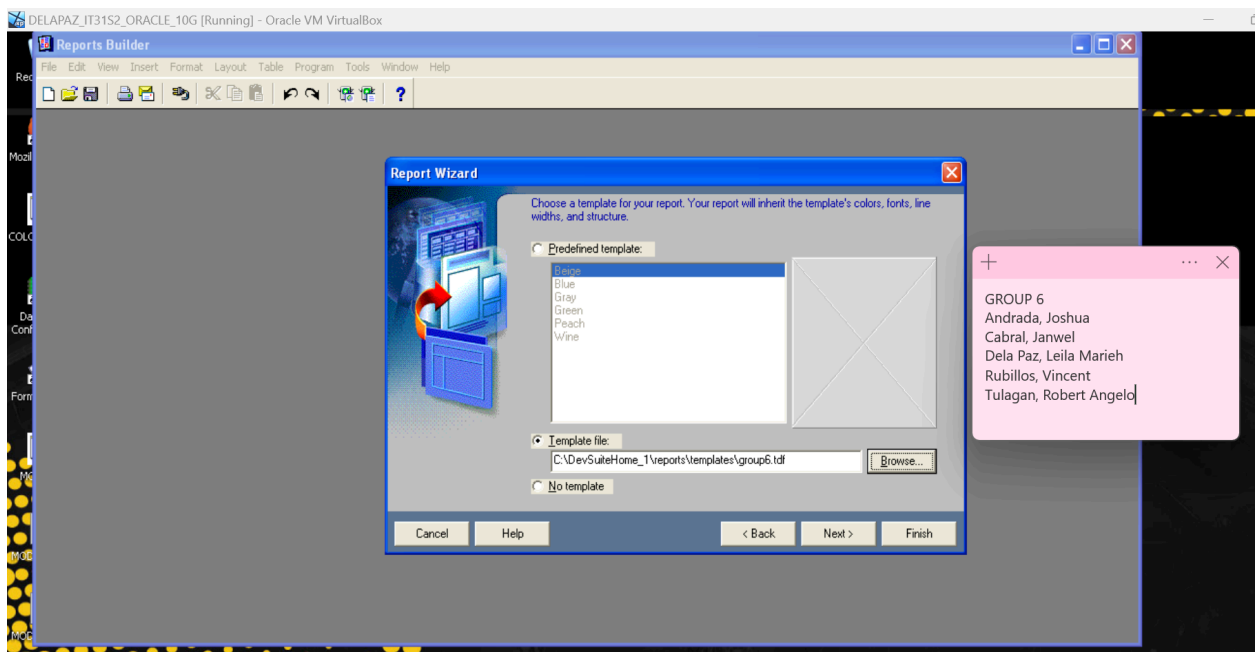
I selected all the checkboxes to include everything in the report.



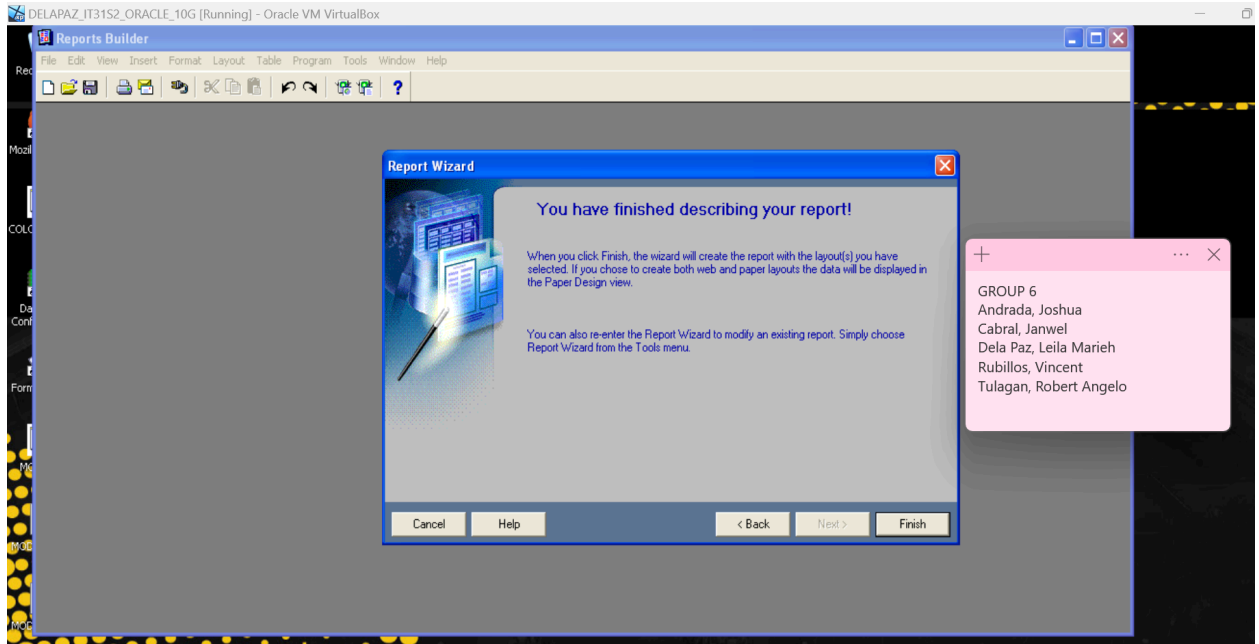
After selecting all the columns, I checked the SQL query if everything is included as it generated itself automatically.



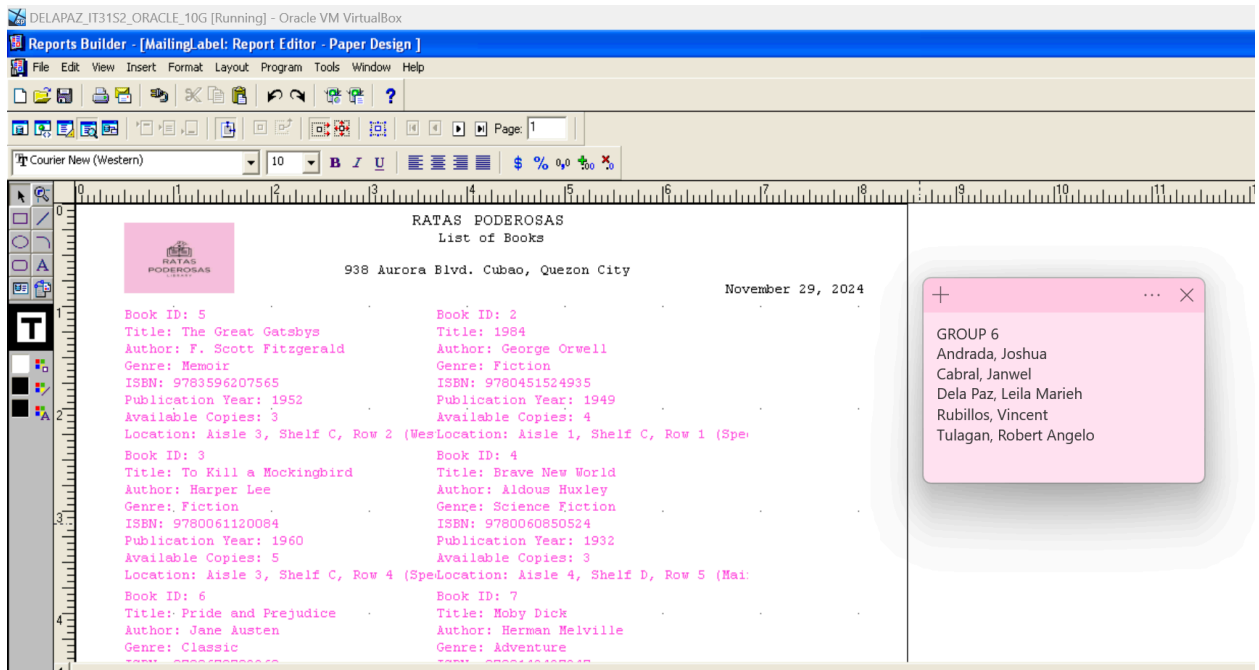
I put some labels/prompts before adding the available fields to add some clarity about the data and avoid confusions.



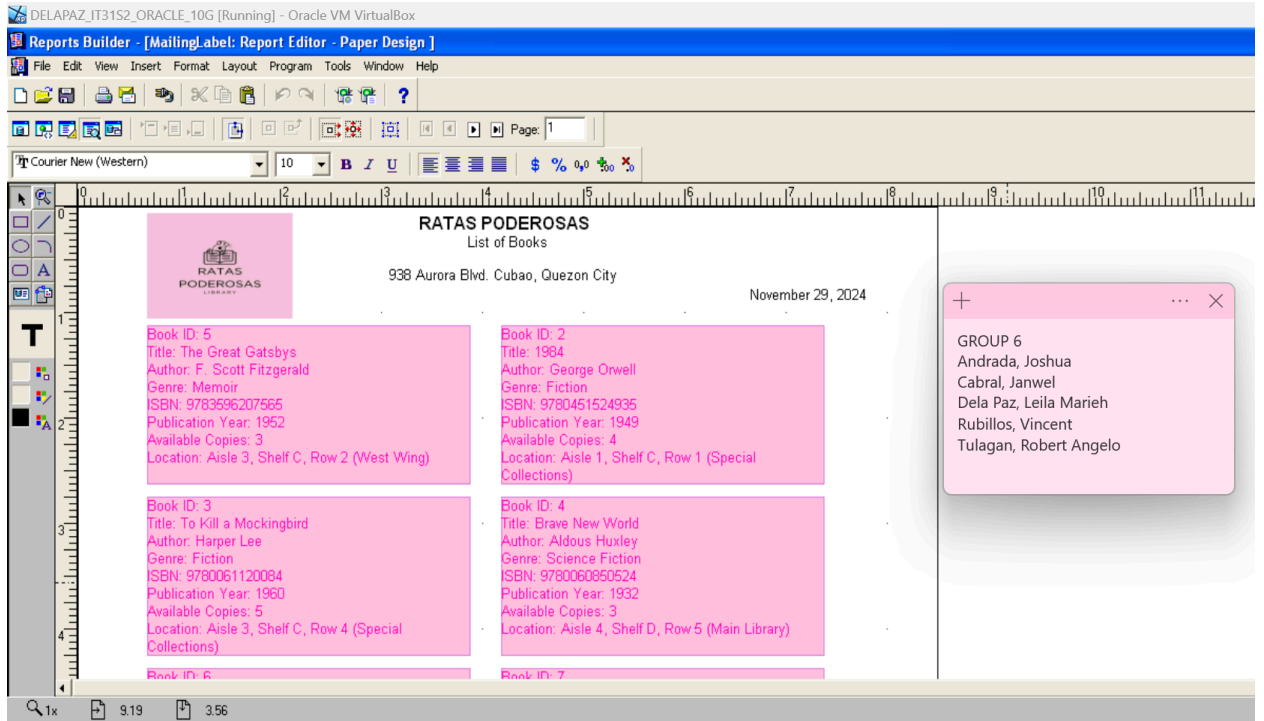
After that, I chose the template for the report and I used the template that I made.



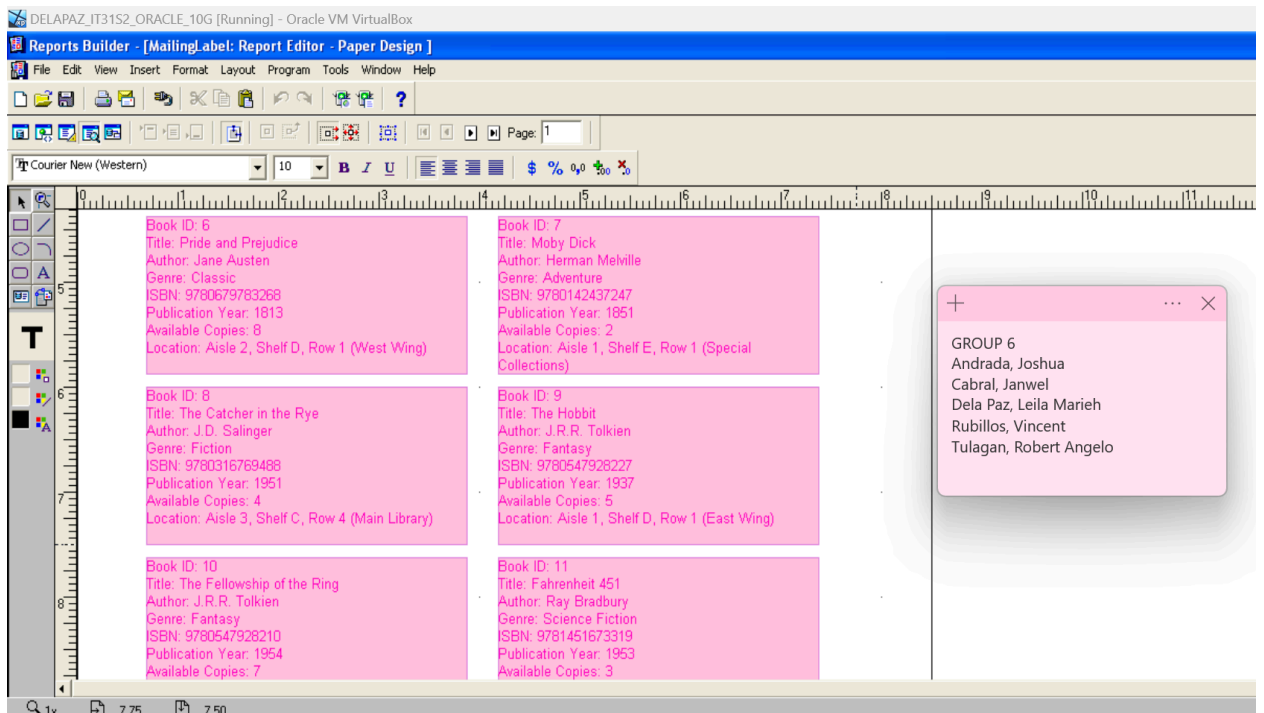
Click the Finish button to generate the mailing label report.



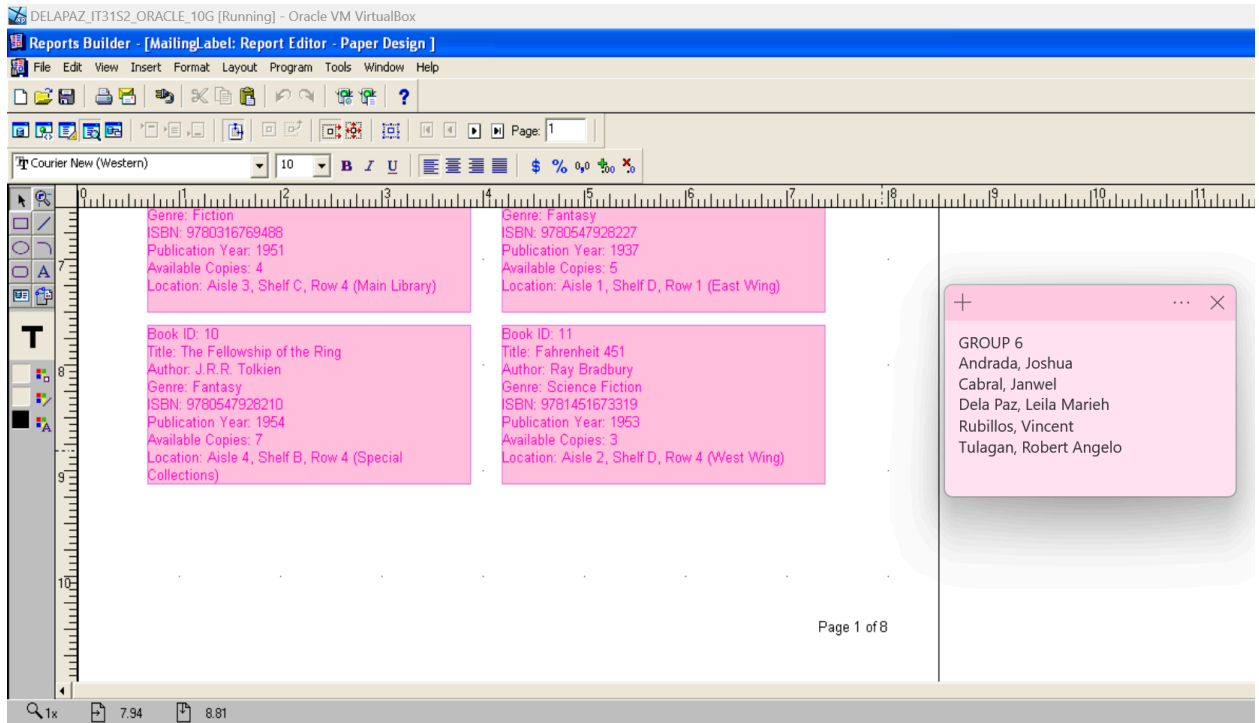
This is the initial Mailing Label report style after clicking the Finish button. This needs more styling.



This is the final Mailing Label report style that I added some colors and changed the font style of the text. I also fixed some positions and added a margin.



This is the second part of the final Mailing Label report style.



This is the last part of the final Mailing Label report style.

g. Report with Bookmark or Hyperlink

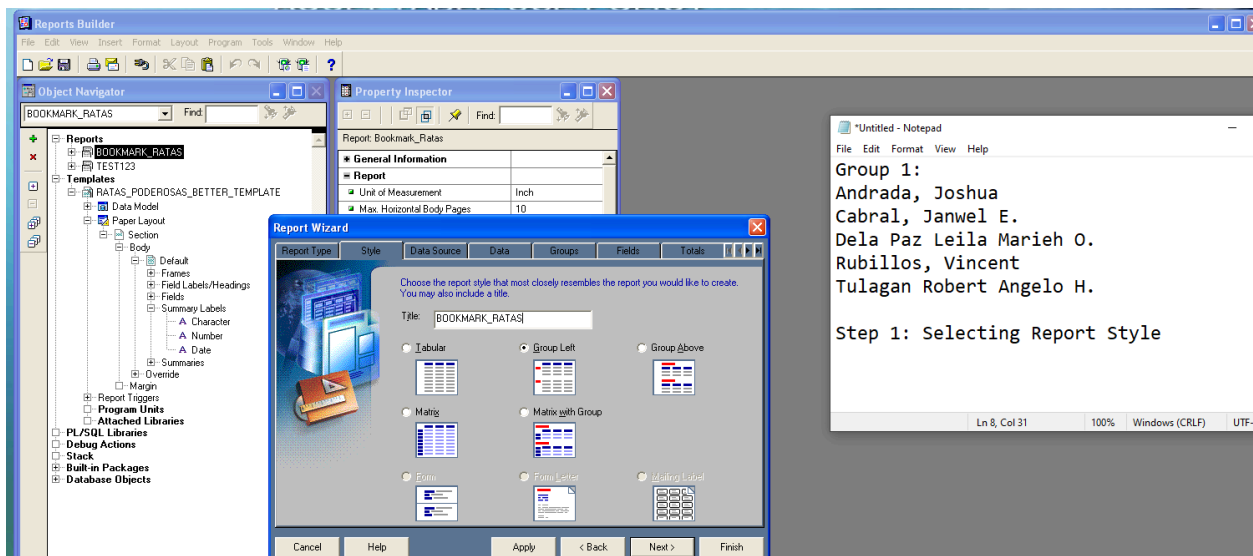


Figure 6,1

Utilizing the report wizard and selecting the Group Left as our report style format

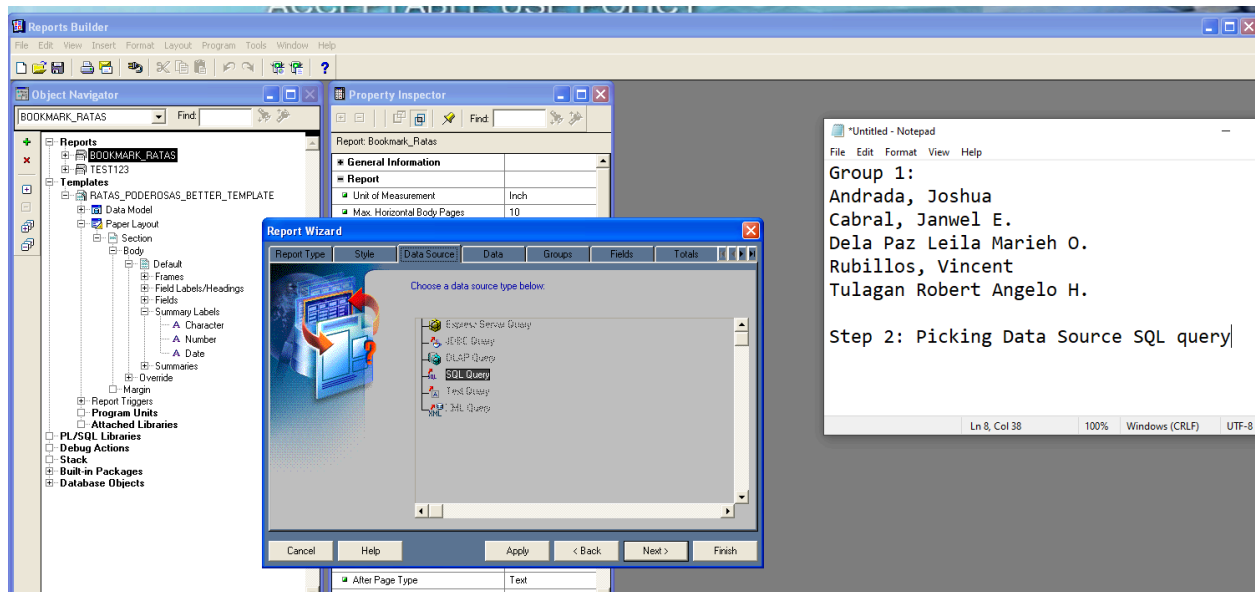


Figure 6.2 Data source selection

Using the SQL query as our data source to manipulate our data in our reports.

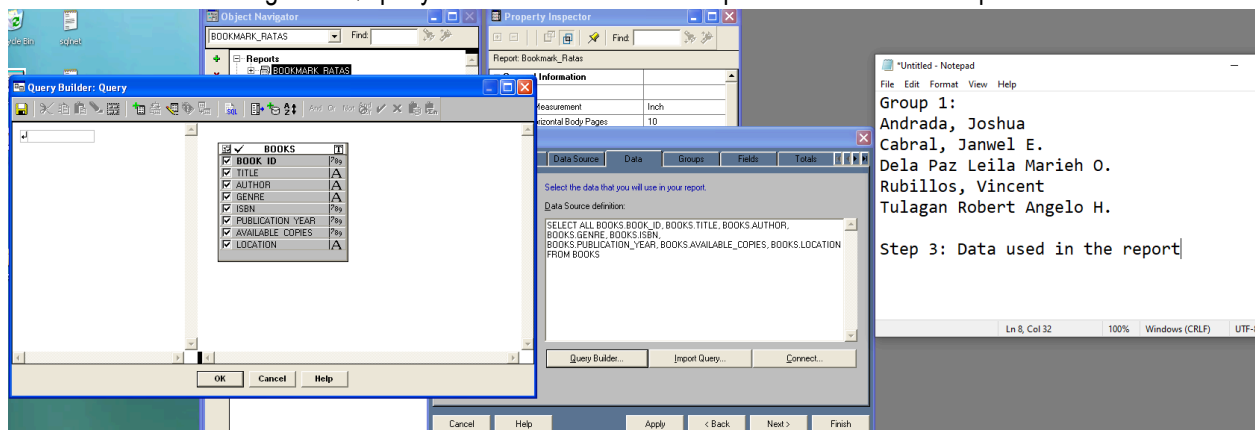


Figure 6.3 Data used in the report

We utilized Query builder and selected our Books table to represent as our data report for this sample

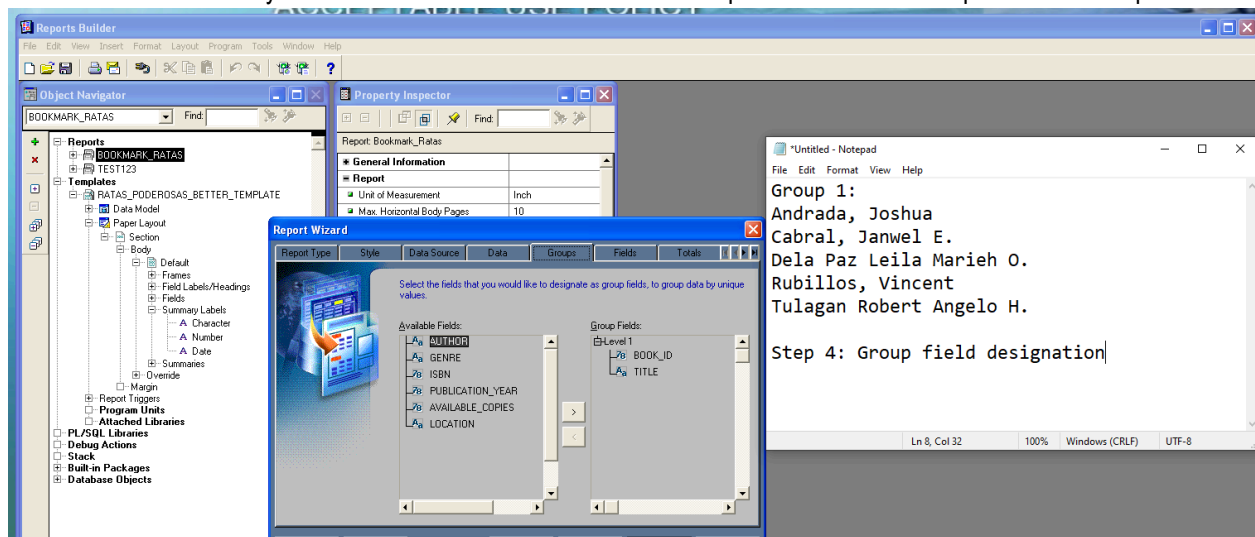


Figure 6.4 Group field designation

We made book_id and title as our group field with unique values

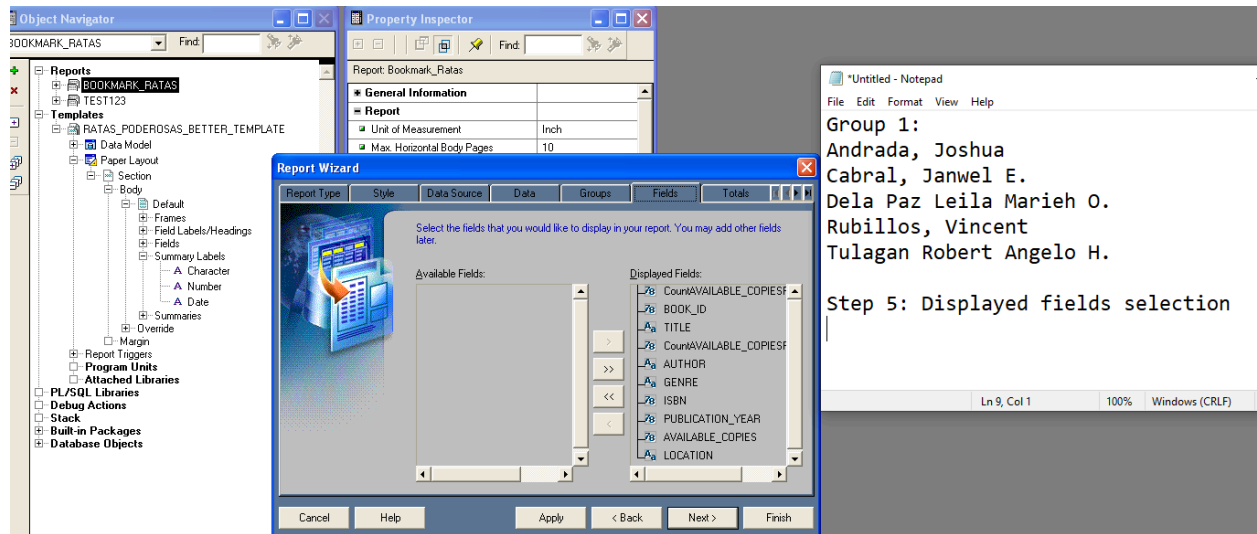


Figure 6.5 Displayed Fields selection

We then selected all our data to be presented and displayed in our report

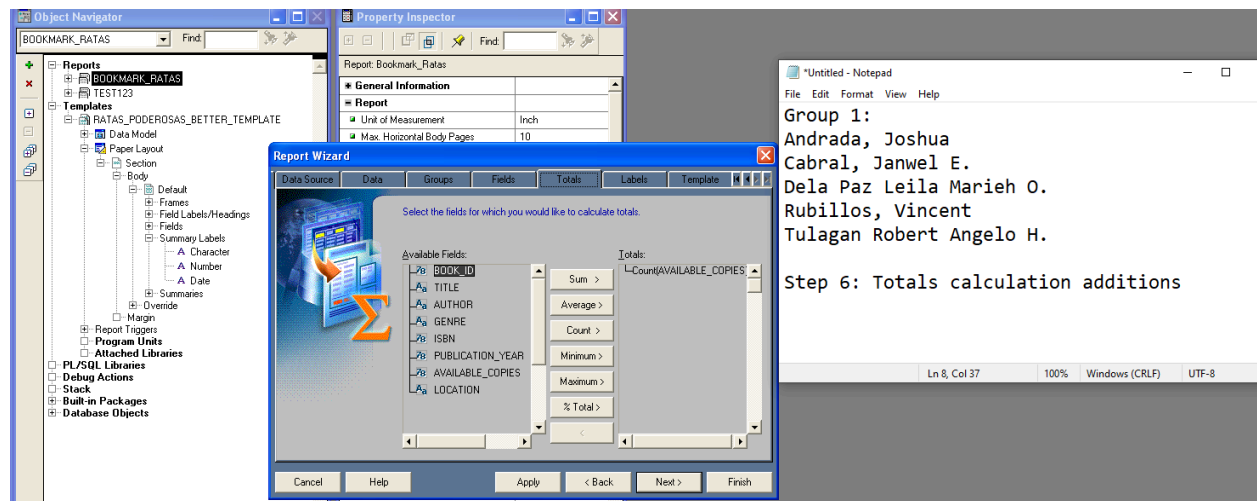


Figure 6.6 Totals Calculation

we decided to use our table column "Available_Copies" as our count operation to count said copies.

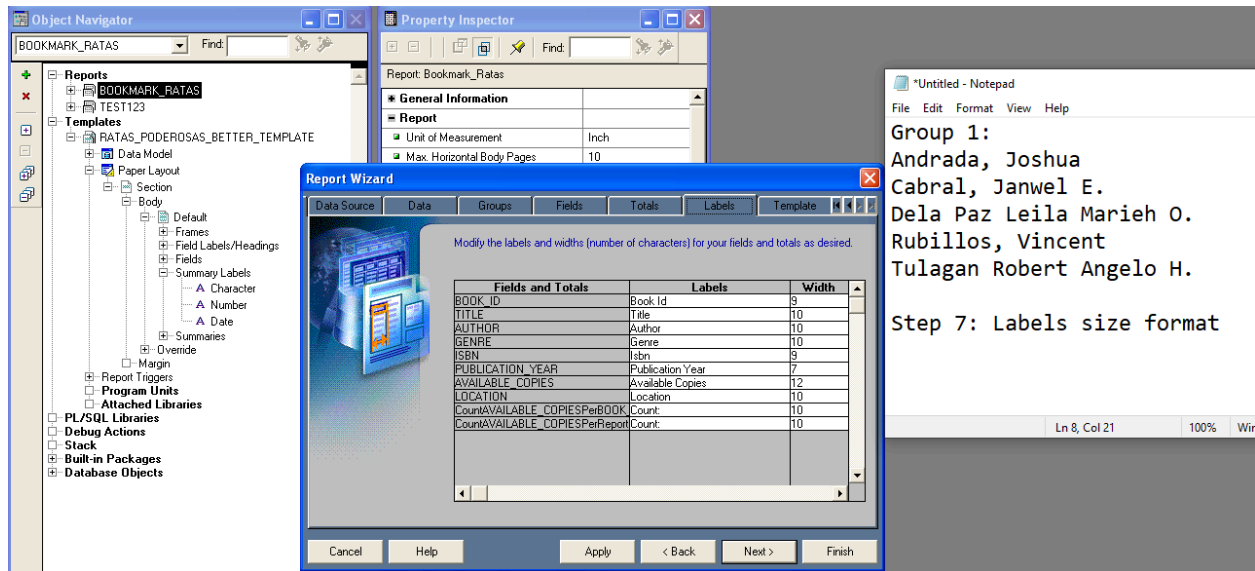


Figure 6.7 Label size format

our data content label format width size and labels remaining the same; default.

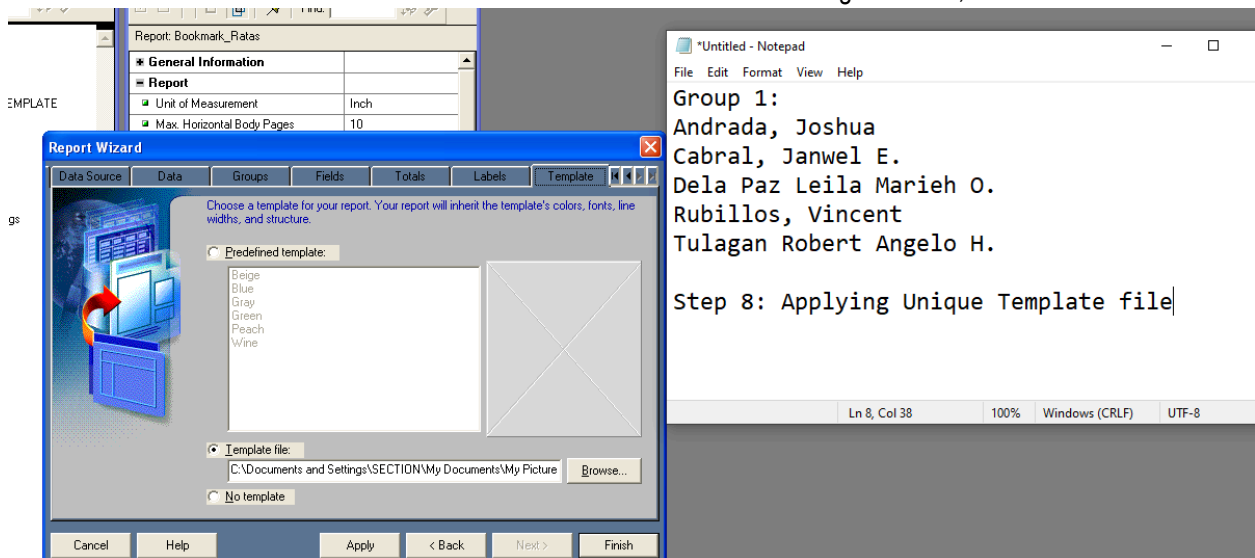


Figure 6.8 Applying the unique template format

We utilized our custom made template that projects our company logo design and color scheme.

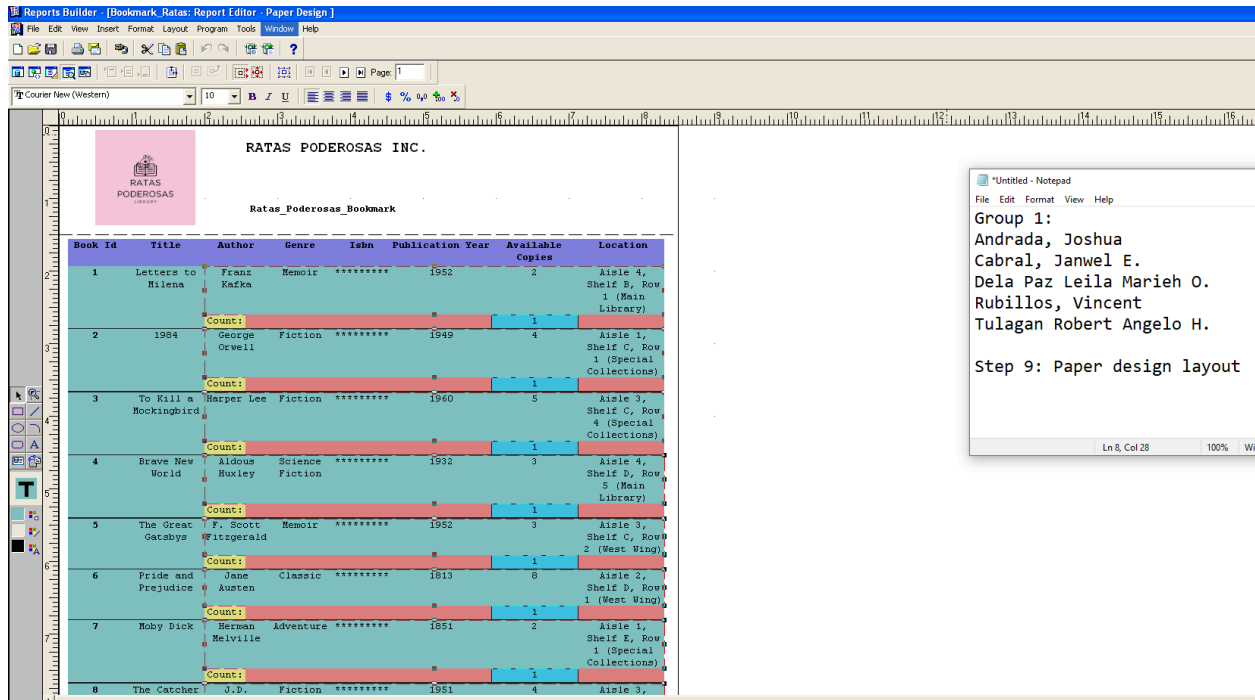


Figure 6.9 Paper design layout
 Once applying and finishing the steps, this is the layout.

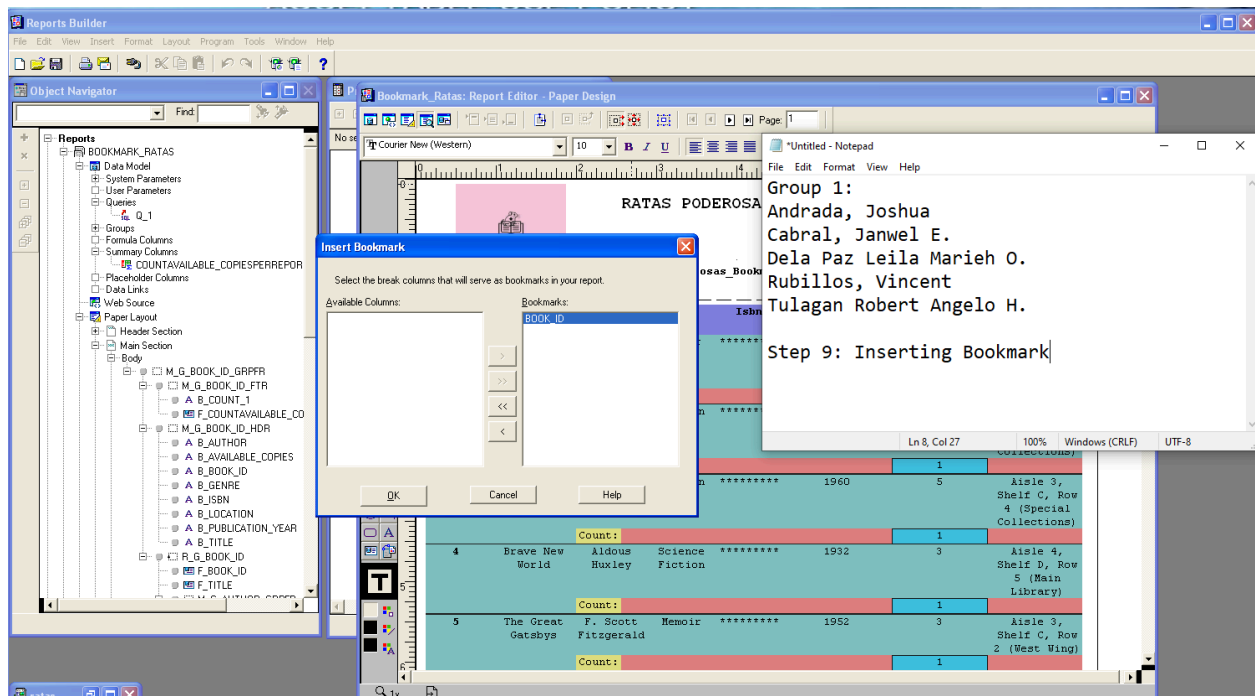


Figure 6.10 Inserting Bookmark
 Utilizing the Book_Id as our bookmark of choice

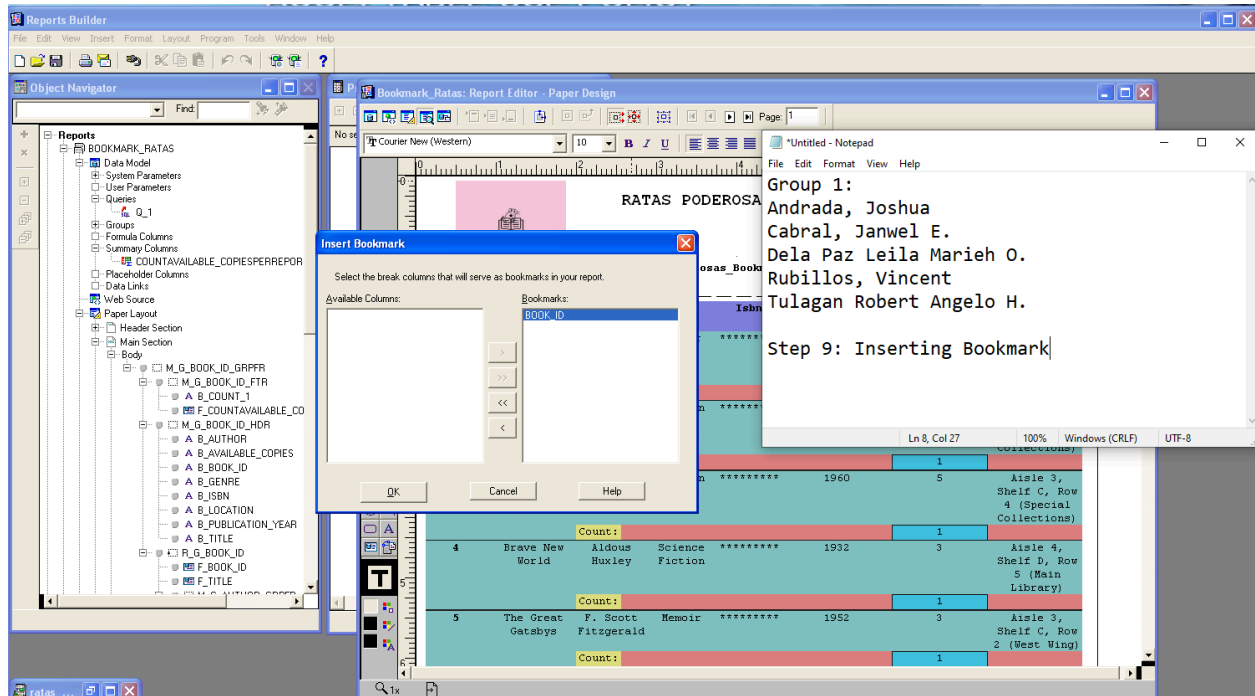


Figure 6.11 Inserting Bookmark
 Utilizing the Book_Id as our bookmark of choice

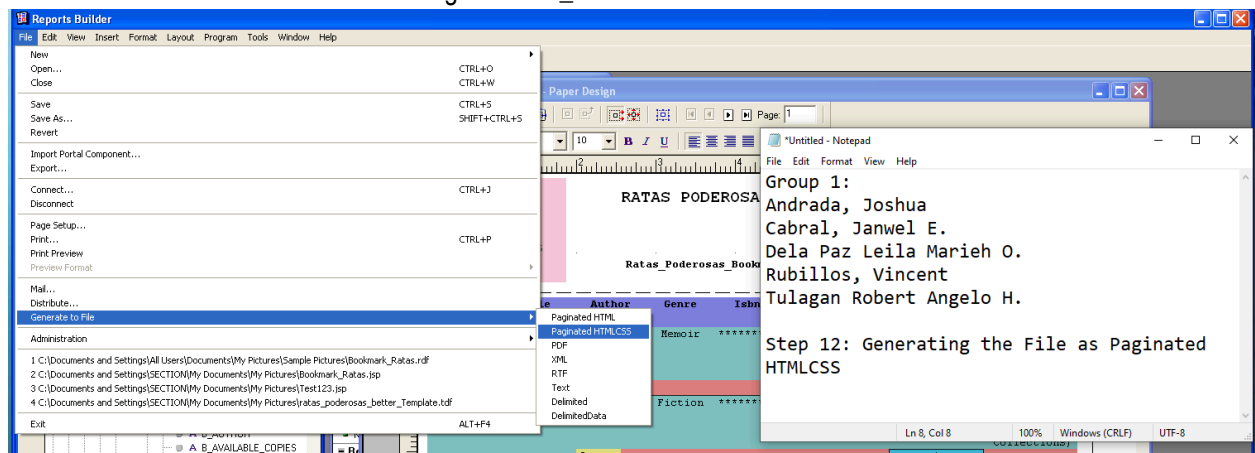


Figure 6.12 Generate File to Pagination HTMLCSS
 After applying the bookmark, we will generate the file and save it to run

The screenshot shows a web browser window displaying an HTML report. The report includes a table of book inventory and a list of names in a Notepad window.

Table 1: Book Inventory

Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location
10	The Fellowship of the Ring	J.R.R. Tolkien	Fantasy	*****	1954	7	Aisle 4, Shelf 2, Row 4 (Special Collections)
11	Fahrenheit 451	Ray Bradbury	Science Fiction	*****	1953	3	Aisle 2, Shelf 2, Row 3
12	Slaughterhouse-Five	Kurt Vonnegut	Fiction	*****	1969	6	Aisle 2, Shelf 2, Row 5 (Special Collections)
13	Crime and Punishment	Fyodor Dostoevsky	Classic	*****	1866	4	Aisle 1, Shelf 2, Row 4 (Special Collections)
14	War and Peace	Leo Tolstoy	Historical Fiction	*****	1869	2	Aisle 4, Shelf 2, Row 2 (West Wing)
15	One Hundred Years of Solitude	Gabriel Garcia Marquez	Fiction	*****	1967	4	Aisle 2, Shelf 2, Row 3 (West Wing)
16	The Stranger	Albert Camus	Philosophy	*****	1942	6	Aisle 5, Shelf 2, Row 1 (West Wing)
17	The Old Man	Ernest	Fiction	*****	1952	8	Aisle 1, Shelf 2, Row 4 (Special Collections)

Table 2: Group 1 Names

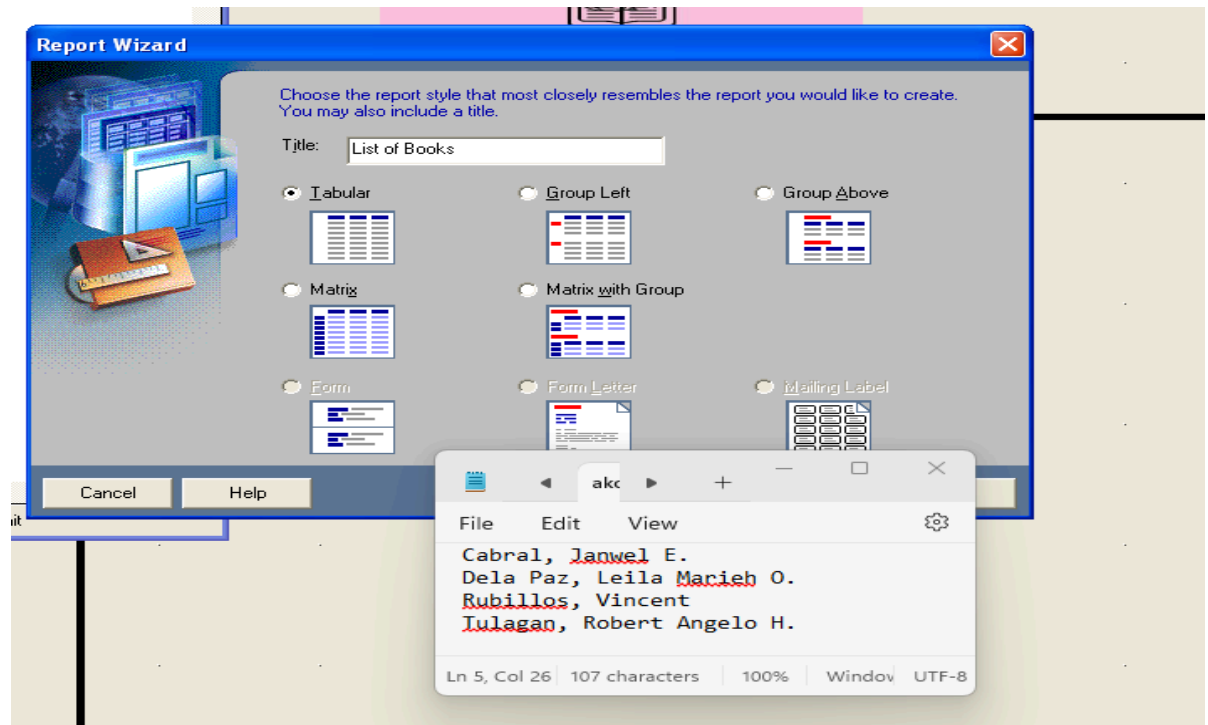
Name
Andrada, Joshua
Cabral, Janwel E.
Dela Paz Leila Marieh O.
Rubillos, Vincent
Tulagan Robert Angelo H.

Table 3: Step 13: Running the generated HTML file

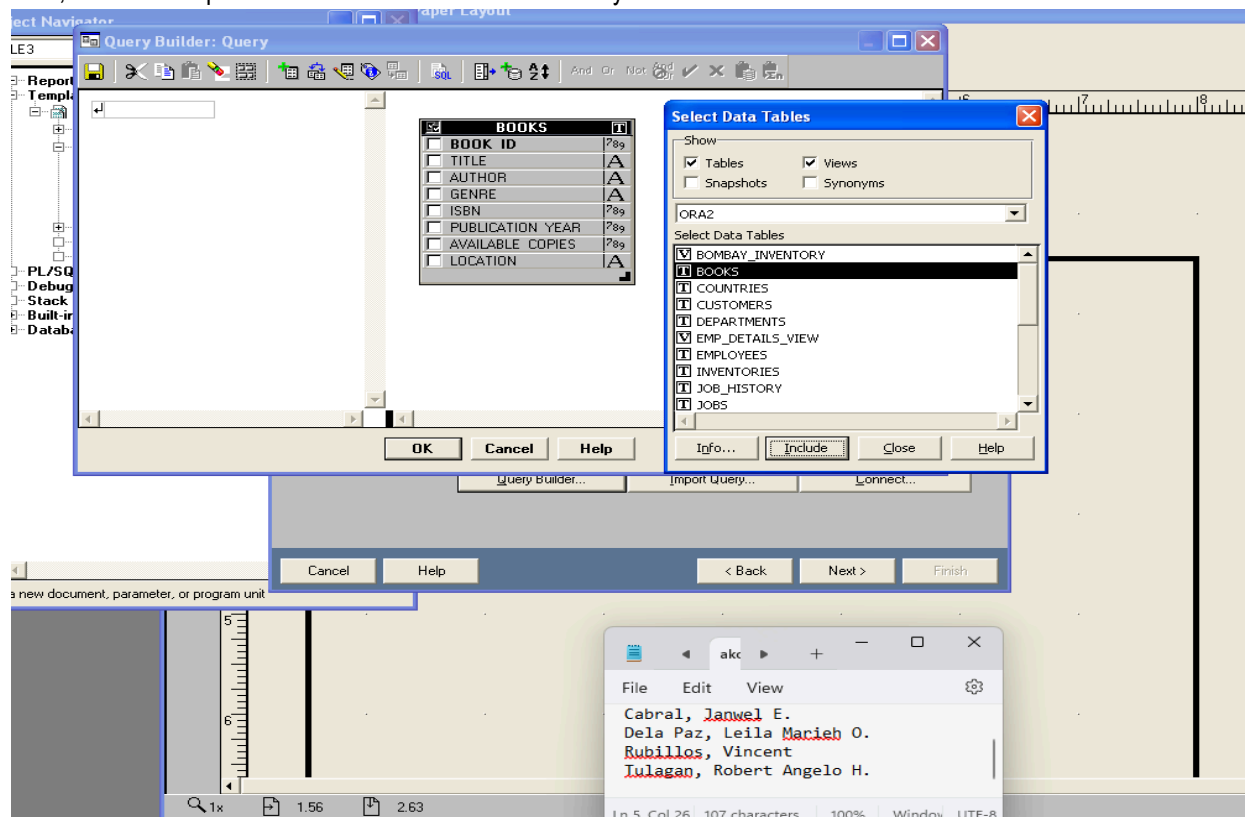
Step	Description
13	Running the generated HTML file

Figure 6.13 Running the HTML output
The final output of the newly generated HTML/CSS Of our report.

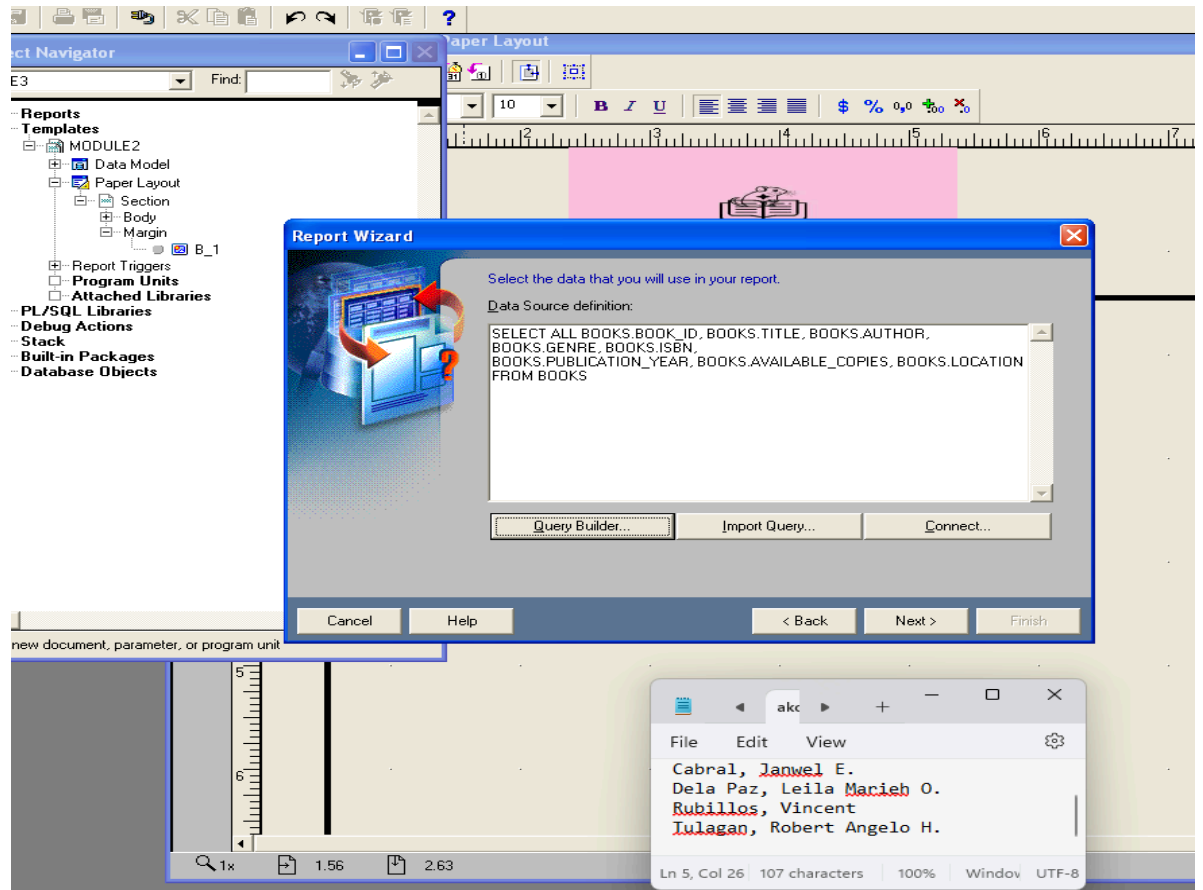
h. Report with Conditional Formatting



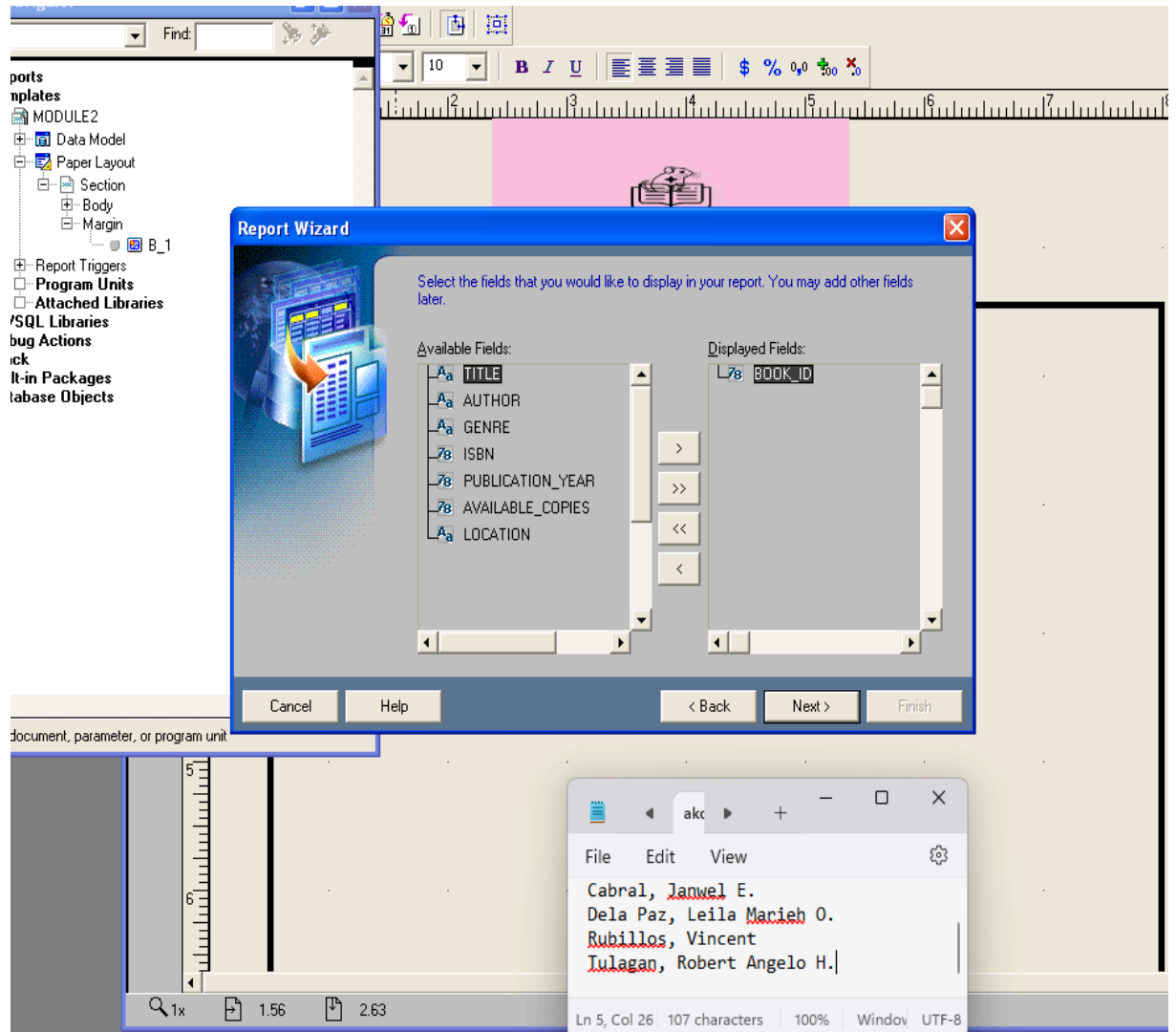
Here, I used the report wizard and selected the tabular layout.



After selecting the layout, I selected the Books table for this report. I also include all the available fields



After selecting the books table, I did not change anything in the PL SQL code so I just proceed to the next part.



Here I include all of the available field in the books table.

Navigator

ports
emplates
MODULE2
Data Model
Paper Layout
Section
Body
Margin
B_1
Report Triggers
Program Units
Attached Libraries
SQL Libraries
Bug Actions
ack
Built-in Packages
Database Objects

document, parameter, or program unit

Report Wizard

Select the fields that you would like to display in your report. You may add other fields later.

Available Fields:

Displayed Fields:

- BOOK_ID
- TITLE
- AUTHOR
- GENRE
- ISBN
- PUBLICATION_YEAR
- AVAILABLE_COPIES
- LOCATION

Cancel Help < Back Next > Finish

File Edit View

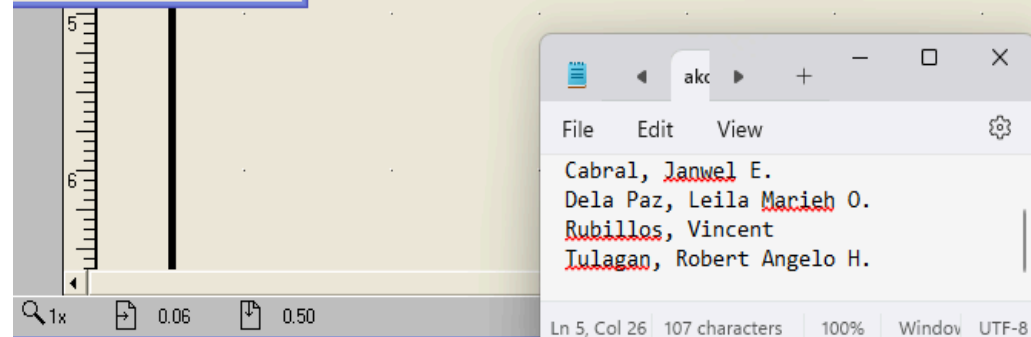
Cabral, Janvel E.
Dela Paz, Leila Marieh O.
Rubillos, Vincent
Tulagan, Robert Angelo H.

Ln 5, Col 26 | 107 characters | 100% | Window UTF-8

B_1
IF_DATE1

ies

or program un



After selecting the fields, I then used the template that we created for our company which consist of our company logo.

MOD015: Report Editor - Paper Design

Q_1

Courier New (Western) 10 B I U \$ % 0,0 ‰ ‰

0 1 2 3 4 5 6 7 8

0 1 2 3 4

List of Books

RATAS PODEROSAS LIBRARY

Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location
109	The Republic	Plato	Philosophy	*****	380	3	Aisle 3, Shelf e, Row 1 (East Wing)
110	Crime and Punishment	Fyodor Dostoevsky	Classic	*****	1866	8	Aisle 1, Shelf e, Row 3 (Special Collection s)
111	The Brothers Karamazov	Fyodor Dostoevsky	Classic	*****	1880	6	Aisle 1, Shelf a, Row 2 (West Wing)
112	War and Peace	Leo Tolstoy	Classic	*****	1869	7	Aisle 1, Shelf b, Row 1 (Main Library)
113	Anna Karenina	Leo Tolstoy	Classic	*****	1877	5	Aisle 3, Shelf c, Row 3 (West Wing)
114	Pride and Prejudice	Jane Austen	Classic	*****	1813	6	Aisle 1, Shelf b, Row 5

File Edit View

Cabral, Janwel E.
Dela Paz, Leila Marieh O.
Rubillos, Vincent
Iulagan, Robert Angelo H.

File Edit View

akc + - □ ×

File Edit View

akc + - □ ×

File Edit View

akc + - □ ×

This is the output/report of the books table. Let's now add some conditional formatting.

The screenshot shows a spreadsheet application with a 'List of Books' table. A 'Conditional Formatting' dialog box is open, showing a blue-green color selection for the 'Book Id' column. The spreadsheet data includes book titles, authors, genres, and publication years.

Book Id	Book Title	Author	Genre	Publication Year
109	Anna Karenina	Leo Tolstoy	Classic	1877
110	Pride and Prejudice	Jane Austen	Classic	1813
111				
112				

The 'Conditional Formatting' dialog box shows a blue-green color selection for the 'Book Id' column. The dialog box also includes a 'Format Exceptions' section and a 'Sample' section.

Here I used the publication year, and also choose a color which is blue green.

Conditional Formatting

You can define exceptions to highlight data for specified conditions with different formatting styles such as font and color.

Format Exceptions

(PUBLICATION_YEAR BETWEEN '1850' and '1900'

Sample

Courier New

Buttons: Apply, OK, Cancel, Help

Book Id	Title	Author	Genre	Isbn	Publication Year	Available	Copies	Location
109	The Republic	Plato	Philosophy	*****380	380	3		Aisle 3, Shelf e, Row 1 (East Wing)
110	Crime and Punishment	Fyodor Dostoevsky	Classic	*****1866	1866	8		Aisle 1, Shelf e, Row 3 (Special Collection s)
111	The Brothers Karamazov	Fyodor Dostoevsky	Classic	*****1880	1880	6		Aisle 1, Shelf a, Row 2 (West Wing)
112	War and Peace	Leo Tolstoy	Classic	*****1869	1869	7		Aisle 1, Shelf b, Row 1 (Main Library)
113	Anna Karenina	Leo Tolstoy	Classic	*****1877	1877	5		Aisle 3, Shelf c, Row 3 (West Wing)
114	Pride and Prejudice	Jane Austen	Classic	*****1813	1813	6		Aisle 1, Shelf b, Row 5

I add a condition which is to highlight the publication year between 1850 and 1900 using the color that I selected earlier.

Book Id	Title	Author	Genre	Isbn	Publication Year	Available	Copies	Location
109	The Republic	Plato	Philosophy	*****380	380	3		Aisle 3, Shelf e, Row 1 (East Wing)
110	Crime and Punishment	Fyodor Dostoevsky	Classic	*****1866	1866	8		Aisle 1, Shelf e, Row 3 (Special Collection s)
111	The Brothers Karamazov	Fyodor Dostoevsky	Classic	*****1880	1880	6		Aisle 1, Shelf a, Row 2 (West Wing)
112	War and Peace	Leo Tolstoy	Classic	*****1869	1869	7		Aisle 1, Shelf b, Row 1 (Main Library)
113	Anna Karenina	Leo Tolstoy	Classic	*****1877	1877	5		Aisle 3, Shelf c, Row 3 (West Wing)
114	Pride and Prejudice	Jane Austen	Classic	*****1813	1813	6		Aisle 1, Shelf b, Row 5

Here, we can see the final output. The publication year that satisfy the condition is not highlighted, while the others that did not satisfy the condition stayed the same.

i. Report with Formula Column or Placeholder

The screenshot displays the Oracle Report Editor interface. In the center, a data model shows a query 'Q_1' connected to a table 'G_BOOK_ID'. The table columns are: TITLE, AUTHOR, GENRE, ISBN, PUBLICATION_YEAR, AVAILABLE_COPIES, CF After Decade, and LOCATION. The 'CF After Decade' column is highlighted.

To the right, the 'Property Inspector' window is open for 'Formula Column: CF_1'. It contains the following settings:

General Information	
Name	CF_After_Decade
Comments	
Column	
Column Type	Formula
Datatype	Number
Width	10
Value if Null	
Break Order	None
Placeholder/Formula	
PL/SQL Formula	...
XML Settings	
XML Tag	CF_AFTER_DECADE
XML Tag Attributes	
Exclude from XML output	No

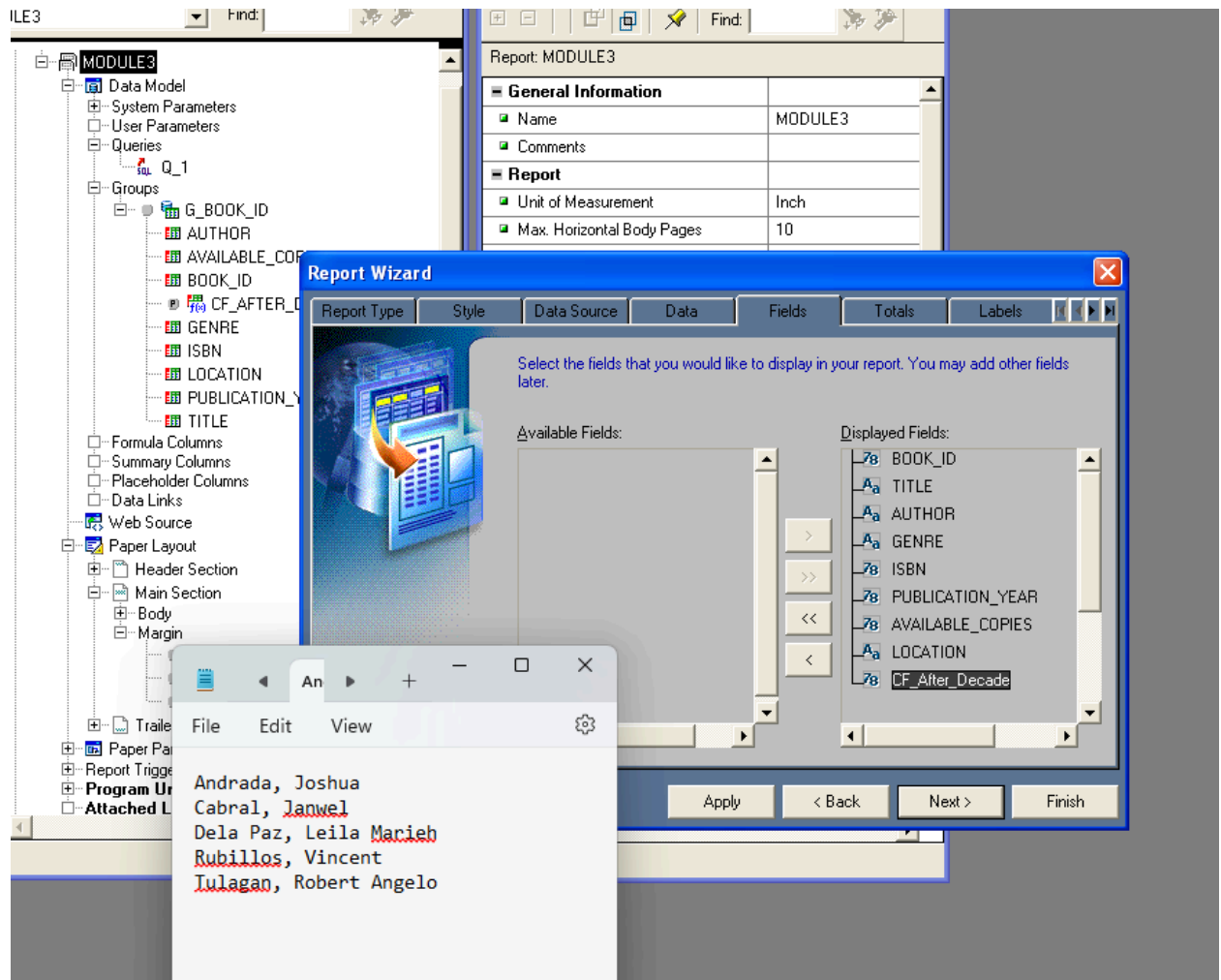
Below the Property Inspector, the 'MODULE3: Program Unit - CF_AFTER_DECADEFORMULA' window is open, showing the following PL/SQL code:

```
function CF_After_DecadeFormula return Number is
begin
    RETURN ((:PUBLICATION_YEAR + 10));
end;
```

On the right side, a preview window shows the report output with the following text:

Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Iulagan, Robert Angelo

Using my previous work in conditional formatting, I also used the books for this report. Here we can see that I add a formula in the publication date which is plus 10 year/decade.



In the fields part of report wizard, I added the newly created formula in displayed field.

MODULE3

Data Model

System Parameters

User Parameters

Queries

Q_1

Groups

G_BOOK_ID

AUTHOR

AVAILABLE_COPIES

BOOK_ID

CF_AFTER_DECAD

GENRE

ISBN

LOCATION

PUBLICATION_YEAR

TITLE

Formula Columns

Summary Columns

Placeholder Columns

Data Links

Web Source

Paper Layout

Header Section

Main Section

Body

Margin

Trailer

Paper Page

Report Trigger

Program User

Attached L

Report: MODULE3

General Information

Name

MODULE3

Comments

Report

Unit of Measurement

Inch

Max. Horizontal Body Pages

10

Report Wizard

Report Type

Style

Data Source

Data

Fields

Totals

Labels

Modify the labels and widths (number of characters) for your fields and totals as desired.

Fields and Totals	Labels	Width
BOOK_ID	Book Id	9
TITLE	Title	10
AUTHOR	Author	10
GENRE	Genre	10
ISBN	Isbn	9
PUBLICATION_YEAR	Publication Year	7
AVAILABLE_COPIES	Available Copies	12
LOCATION	Location	10
*CF_After_Decade	After Decade	12

Apply

< Back

Next >

Finish

File

Edit

View

Andrada, Joshua

Cabral, Janwel

Dela Paz, Leila Mariah

Rubillos, Vincent

Tulagan, Robert Angelo

This formula shows the what would be the publication year if it was published after a decade instead.



List of Books

Monday December 2 2024 3:32 PM

Book Id	Title	Author	Genre	Isbn	Publication Year	Available Copies	Location	After Decade
109	The Republic	Plato	Philosophy	*****	380	3	Aisle 3, 390 Shelf e, Row 1 (East Wing)	
110	Crime and Punishment	Fyodor Dostoevsky	Classic	*****	1866	8	Aisle 1, 1876 Shelf e, Row 3 (Special Collections)	
111	The Brothers	Fyodor Dostoevsky	Classic	*****	1880	6	Aisle 1, 1890 Shelf a, Row 2 (West Wing)	
112	File	Edit	View		1869	7	Aisle 1, 1879 Shelf b, Row 1 (Main Library)	
113	Andrada, Joshua Cabral, Januel Dela Paz, Leila Rubillos, Vincent Tulagan, Robert Angelo	Marieh			1877	5	Aisle 3, 1887 Shelf c, Row 3 (West Wing)	
114					1813	6	Aisle 1, 1823 Shelf b,	

There is now a new column which shows the publication year after a decade, and it is clear that it adds 10 years from the publication year.

The screenshot shows a report design tool interface. On the left is a 'Reports' tree with a 'MODULE3' folder containing a 'Data Model' and various fields like 'M_G_BOOK', 'R_G_BO', 'F_AU', 'F_AV', 'F_BO', 'F_CF', 'F_GE', 'F_ISB', and 'F_LO'. The main area displays a report titled 'List of Books' with a header section containing a logo and the date 'Monday December 2 2024 3:32 PM'. A 'Report Wizard' dialog box is open, showing the 'Totals' tab. The 'Available Fields' list includes 'BOOK_ID', 'TITLE', 'AUTHOR', 'GENRE', 'ISBN', 'PUBLICATION_YEAR', and 'AVAILABLE_COPIES'. The 'Totals' section shows 'L-Max(PUBLICATION_YEAR)' selected. A list of authors is visible in the background, including 'Andrada, Joshua', 'Cabral, Janwel', 'Dela Paz, Leila Marieh', 'Rubillos, Vincent', and 'Iulagan, Robert Angelo'.

Now I created a max value in publication year. This will show the latest publication year, which also mean the latest book.

MODULE3: Report Editor - Paper Design

Page: 12

Courier New (Western) 10 B I U \$ % 0,0 100 X

101	of Tomorrow ThinkingDaniel Non- , Fast KahnemanFiction and Slow	***** 2011	4	Library) Aisle 2, 2021 Shelf b, Row 4 (Main Library)
103	Atomic James Self- Habits Clear help	***** 2018	5	Aisle 2, 2028 Shelf e, Row 1 (East Wing)
104	The 7 Stephen Self- Habits R. Coveyhelp of Highly Effectiv e People	***** 1989	7	Aisle 4, 1999 Shelf c, Row 4 (East Wing)
106	Man's Viktor Self- Search E. help for Frankl Meaning	***** 1946	8	Aisle 2, 1956 Shelf c, Row 4 (West Wing)
107	The Art Sun Tzu Philosop of War hy	***** 500	5	Aisle 2, 510
Maximum:		2018		

Ln 5, Col 14 | 96 characters | 100% | Window | UTF-8

Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Iulagan, Robert Angelo

In this list of books, the latest publication year is 2018. Let's now proceed on creating a placeholder.

The screenshot shows the Reports Builder application interface. The main window is titled "MODULE 3: Report Editor - Data Model". It features a toolbar with various icons for file operations, editing, and viewing. The central workspace displays a data model diagram with the following components:

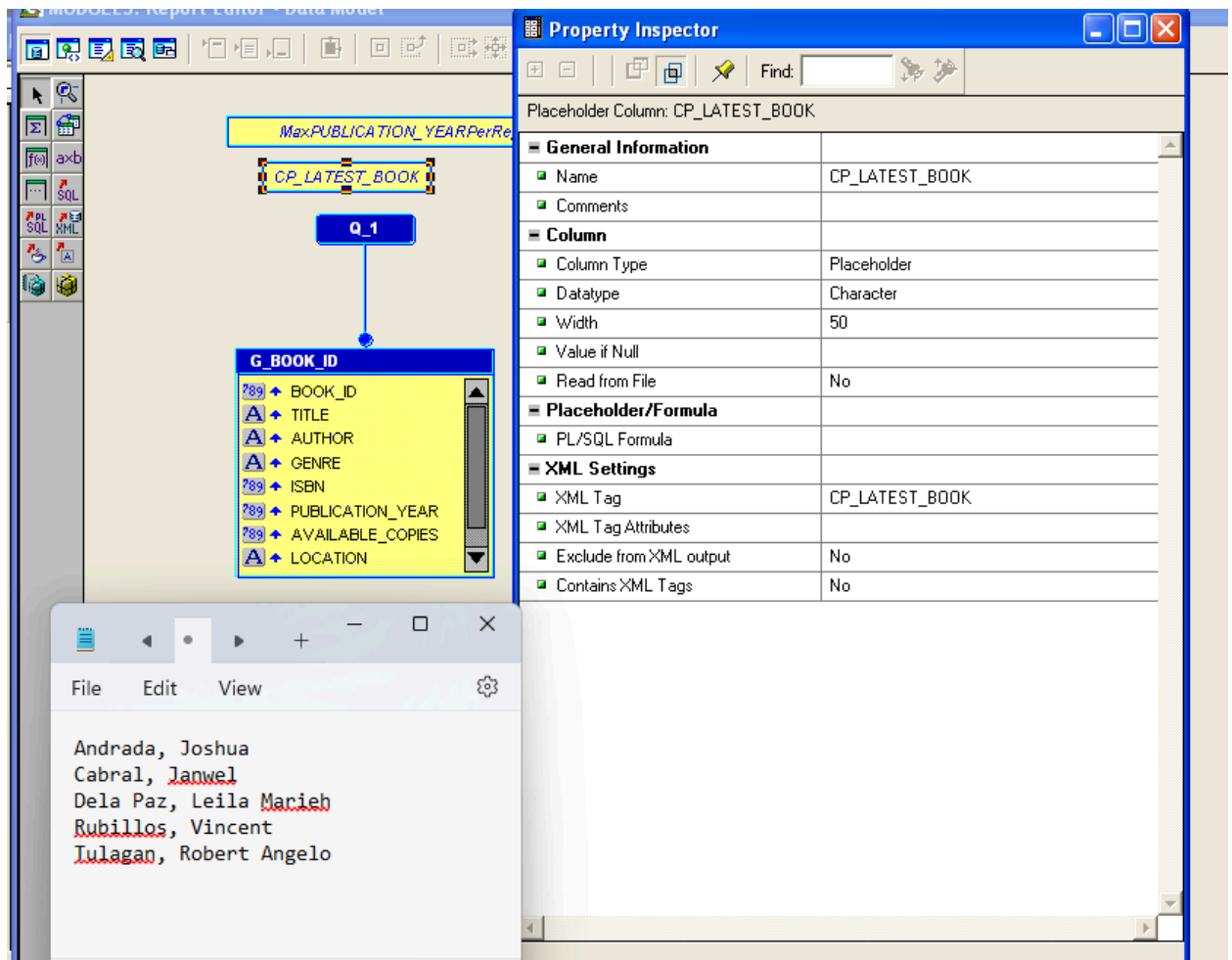
- A yellow box at the top labeled `MaxPUBLICATION_YEARPerReport`.
- A yellow box below it labeled `CP_1`.
- A blue box below that labeled `Q_1`.
- A yellow box at the bottom labeled `G_BOOK_ID`, which is connected to `Q_1` by a blue line. This box contains a list of fields: `BOOK_ID`, `TITLE`, `AUTHOR`, `GENRE`, `ISBN`, `PUBLICATION_YEAR`, `AVAILABLE_COPIES`, and `LOCATION`.

To the right of the main workspace, there is a preview window titled "File Edit View". It displays a list of author names, each on a new line and underlined:

- Andrada, Joshua
- Cabral, Ianuel
- Dela Paz, Leila Marieh
- Rubillos, Vincent
- Tulagan, Robert Angelo

The preview window also shows a status bar at the bottom indicating "Ln 5, Col 14", "96 characters", "100%", "Window", and "UTF-8".

Just above the Q_1, I added a placeholder for the latest book. The title of the book will display in this placeholder



Here, I edited the values of some properties based on the instructions, specifically the data type and name.

The screenshot displays a database reporting tool interface. The main workspace shows a formula editor with the following structure:

- A yellow box labeled `MaxPUBLICATION_YEARPerReport` is connected to a blue box labeled `CP_LATEST_BOOK`.
- The `CP_LATEST_BOOK` box is connected to a blue box labeled `Q.1`.
- The `Q.1` box is connected to a yellow box labeled `G_BOOK_ID`.
- The `G_BOOK_ID` box contains a list of fields: `BOOK_ID`, `TITLE`, `AUTHOR`, `GENRE`, `ISBN`, `PUBLICATION_YEAR`, `AVAILABLE_COPIES`, and `LOCATION`.

The `Property Inspector` window on the right shows the following details for the `MaxPUBLICATION_YEARPerReport` summary column:

Summary Column: MaxPUBLICATION_YEARPerReport	
General Information	
Name	CS_LATEST_PUBLICATION
Comments	
Column	
Column Type	Summary
Datatype	Number
Width	5
Value if Null	
Summary	
Function	Maximum
Source	PUBLICATION_YEAR
Reset At	Report
XML Settings	
XML Tag	MAXPUBLICATION_YEARPERREPORT
XML Tag Attributes	
Exclude from XML output	No

The data preview window at the bottom shows the following data:

File	Edit	View
Andrada, Joshua		
Cabral, Janwel		
Dela Paz, Leila Marieh		
Rubillos, Vincent		
Tulagan, Robert Angelo		

In this screenshot, I created a formula to validate the latest book. This will return the title of the latest book based on the latest publication year.

The screenshot displays a software development environment with several components:

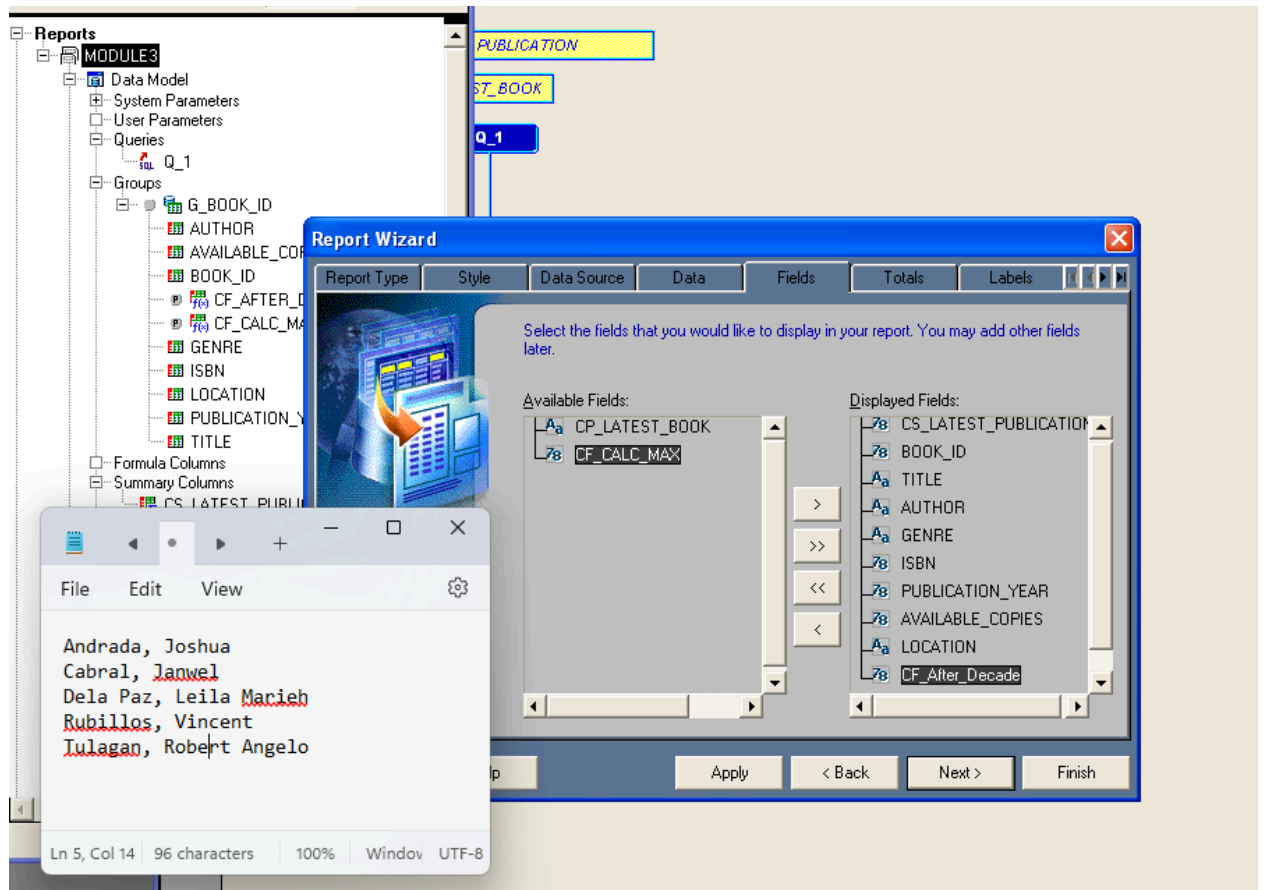
- Query Diagram:** A diagram showing a query `Q_1` connected to a table `G_BOOK_ID`. The table `G_BOOK_ID` has columns: `BOOK_ID`, `TITLE`, `AUTHOR`, `GENRE`, `ISBN`, `PUBLICATION_YEAR`, `AVAILABLE_COPIES`, and `CF_CALC_MAX`. Above the query, there are labels for `CS_LATEST_PUBLICATION` and `CP_LATEST_BOOK`.
- Data Table:** A table window showing the following data:

Author
Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo
- Property Inspector:** A window showing the properties of the column `CF_CALC_MAX`. It includes a "General Information" tab with the following details:

Property	Value
Name	CF_CALC_MAX
Comments	
Column Type	Formula
Datatype	Number
Width	10
Value if Null	
- Code Editor:** A window titled "MODULE3: Program Unit - PU_012" showing the following code:

```
function CF_CALC_MAXFormula return Number is
begin
  IF :PUBLICATION_YEAR = :CS_LATEST_PUBLICATION THEN
    :CP_LATEST_BOOK := (:TITLE);
  END IF;
  RETURN(0);
end;
```

This is the formula that validates the latest book, then displaying the title.



Now, I place the new placeholder on the displayed fields

The screenshot displays a database management interface with a tree view on the left containing categories like System Parameters, User Parameters, Queries, Groups, and Formula Columns. A 'Report Wizard' dialog box is open, showing the 'Labels' tab where fields and their widths are configured. A preview window in the foreground shows a list of authors.

Report Wizard - Labels Tab

Fields and Totals	Labels	Width
BOOK_ID	Book Id	9
TITLE	Title	10
AUTHOR	Author	10
GENRE	Genre	10
ISBN	Isbn	9
PUBLICATION_YEAR	Publication Year	7
AVAILABLE_COPIES	Available Copies	12
LOCATION	Location	10
CF_After_Decade	After Decade	12
CS_LATEST_PUBLICATION	Maximum:	7
*CF_CALC_MAX	Latest Book:	12

Preview Window

Andrada, Joshua
Cabral, Janwel
Dela Paz, Leila Marieh
Rubillos, Vincent
Tulagan, Robert Angelo

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Here, we can see the things we added which are the maximum(latest publication year), and latest book. We can also see the publication year after a decade that I created earlier.

RATAS PODEROSAS LIBRARY									
Book Id	Title	Author	Genre	Isbn	Publication Year	Available	Copies	Location	After Decade Latest Book:
101	Think Daniel Non- ing, Kahnem Ficti Fast an on and Slow	*****	2011		4			Aisle 2, 2021 Shelf b, Row 4 (Main Library)	0
103	Atom James Self- c Clear help Habit s	*****	2018		5			Aisle 2, 2028 Shelf e, Row 1 (East Wing)	0
104	The 7 Stephe Self- Habit R. help s of Covey Highl y Effec tive Peopl e	*****	1989		7			Aisle 4, 1999 Shelf c, Row 4 (East Wing)	0
106	Man's Viktor Self- SearcE. help h for Frankl Meani ng	*****	1946		8			Aisle 2, 1956 Shelf c, Row 4 (West Wing)	0
107	The Sun Philo Art Tzu sophy of War	*****	500		5			Aisle 2, 510 Shelf a, Row 4 (East Wing)	0
Maximum:					2018				
Latest Book					Atomic				

This is the final output. Now we can see that the title of the book with the latest publication year is Atomic. Actually, it is Atomic Habits, which I wasn't able to correct in data modelling, but overall, it validates the latest book and display the title of that book.

J. Report with Header and Trailer section

GROUP_ABOVE_9: Report Editor - Paper Parameter Form

RATAS GROUP_ABOVE

RATAS PODEROSAS INC.

Book Id	1
Title	Letters to Friends
Author	Milena Katka
Genre	Bio
ISBN	*****
Publication Year	1952
Available Copies	2
Count	1

Book Id	2
Title	1984
Author	George Orwell
Genre	Fiction
ISBN	*****
Publication Year	1949
Available Copies	4
Count	1

Book Id	3
Title	To Kill a Mockingbird
Author	Harper Lee
Genre	Fiction
ISBN	*****
Publication Year	1960
Available Copies	5

Step 1:

STEP 1: Create a Group Above report
We'll use this as our pre-existing report to add the header and trailer.

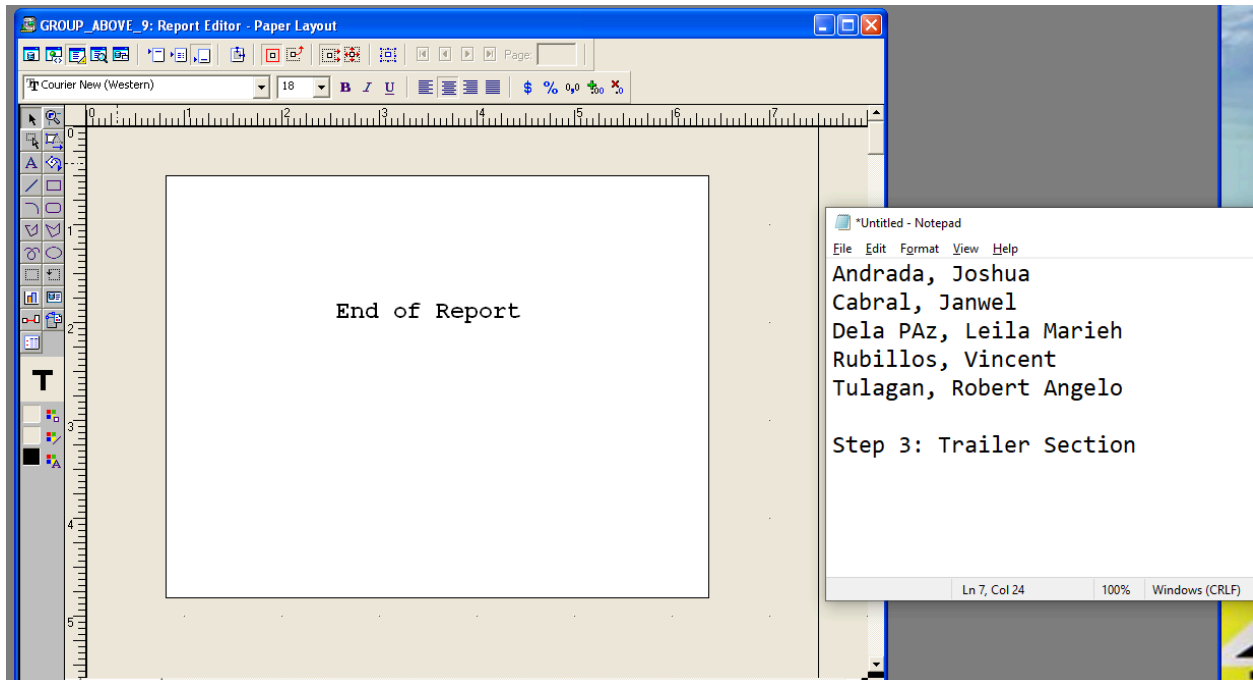
GROUP_ABOVE_9: Report Editor - Paper Layout

Ratas Poderosas
Books Report

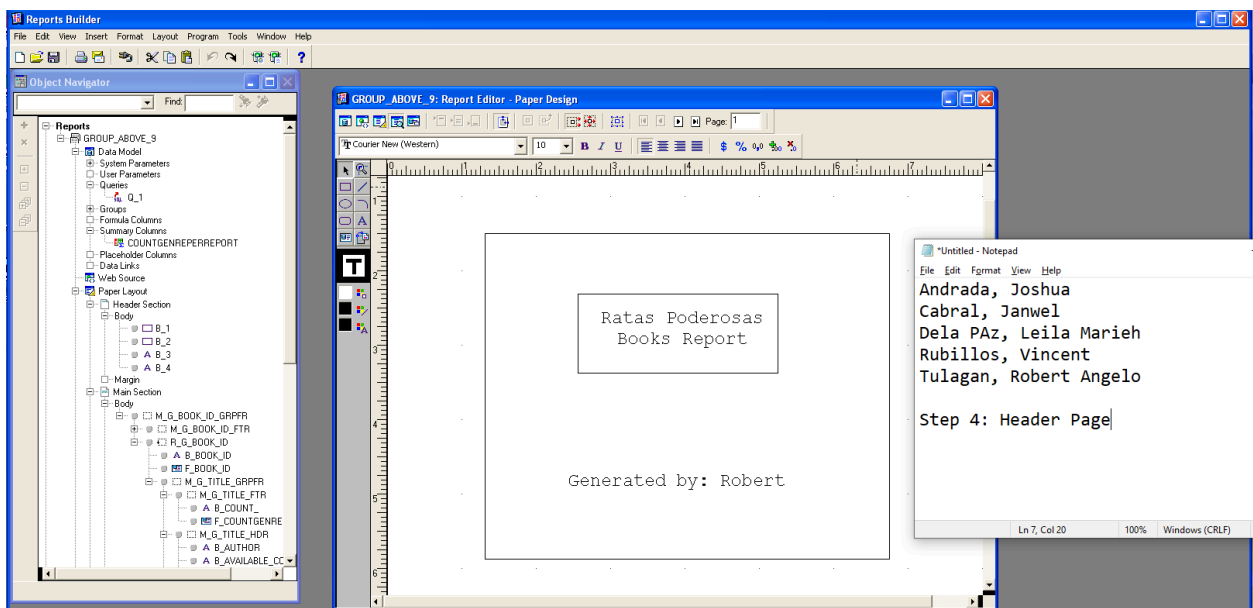
Generated by: Robert

Step 2:

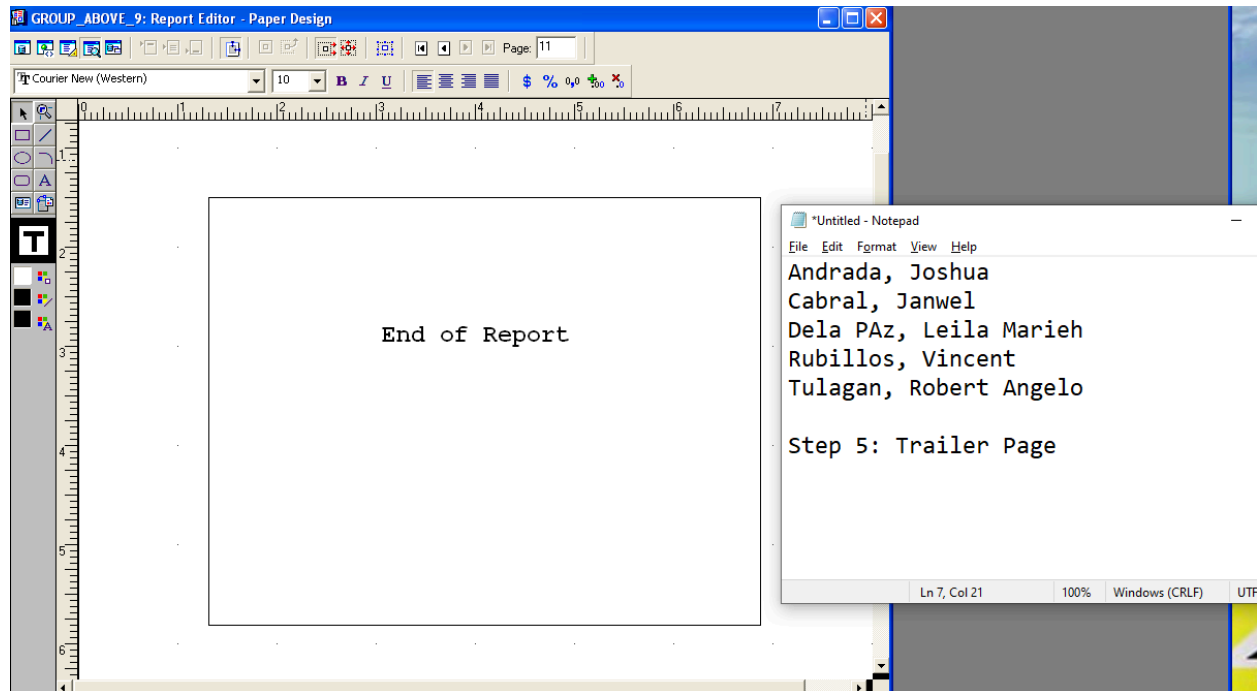
STEP 2: Add Header Section
This will serve as our header Section for our Books Report.



STEP 3: The trailer section
This will appear at the end; last page of our pre-existing report.

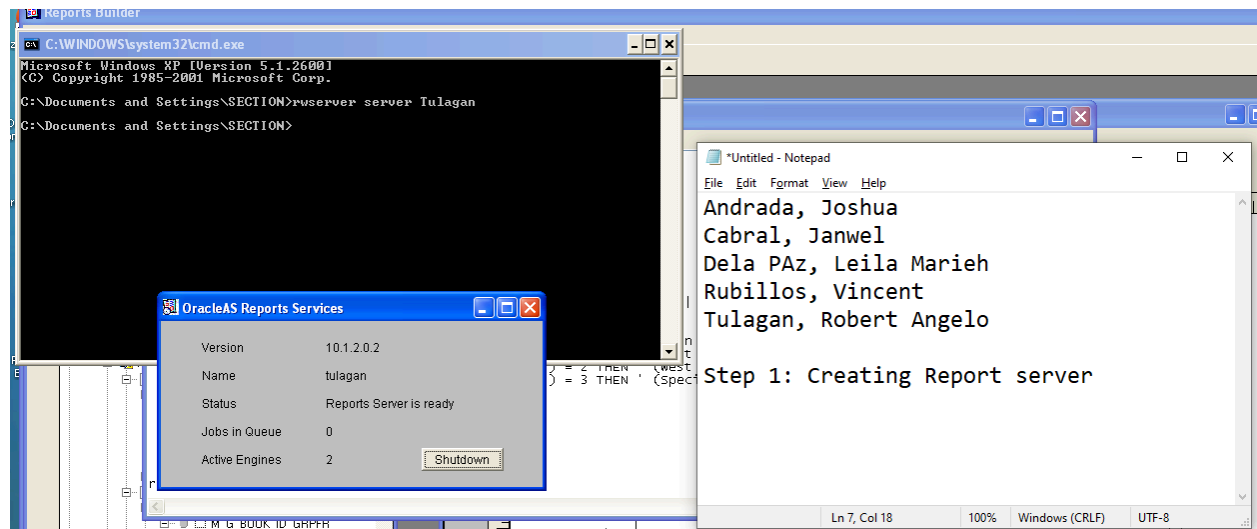


STEP 4: Added Header
This will appear at the first; first page of our pre-existing report.

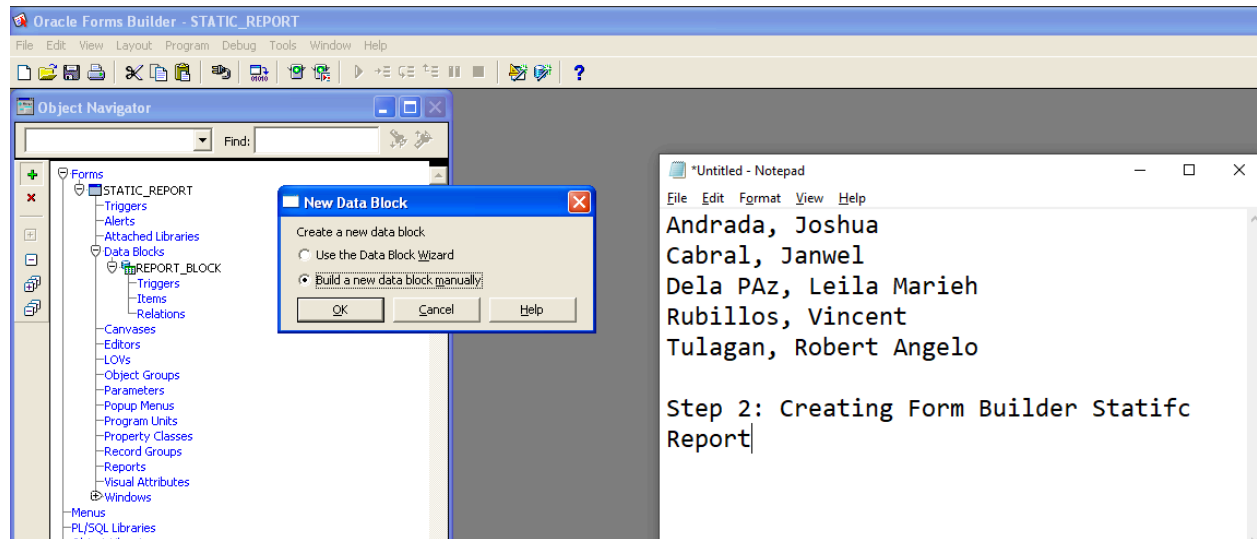


STEP 5: Added Trailer
 This will appear at the end; last page of our pre-existing report.

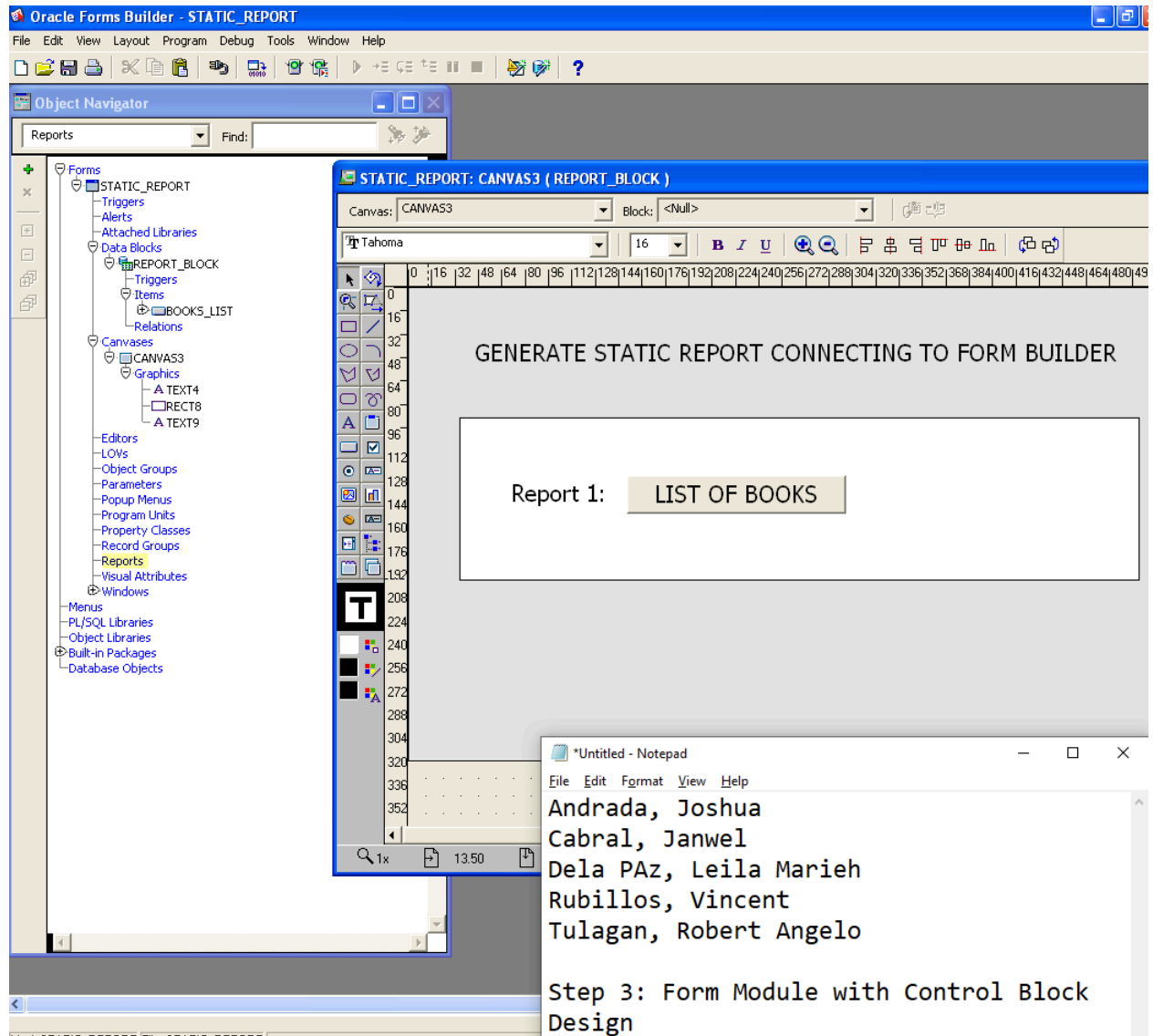
k. Report connected to Forms Builder



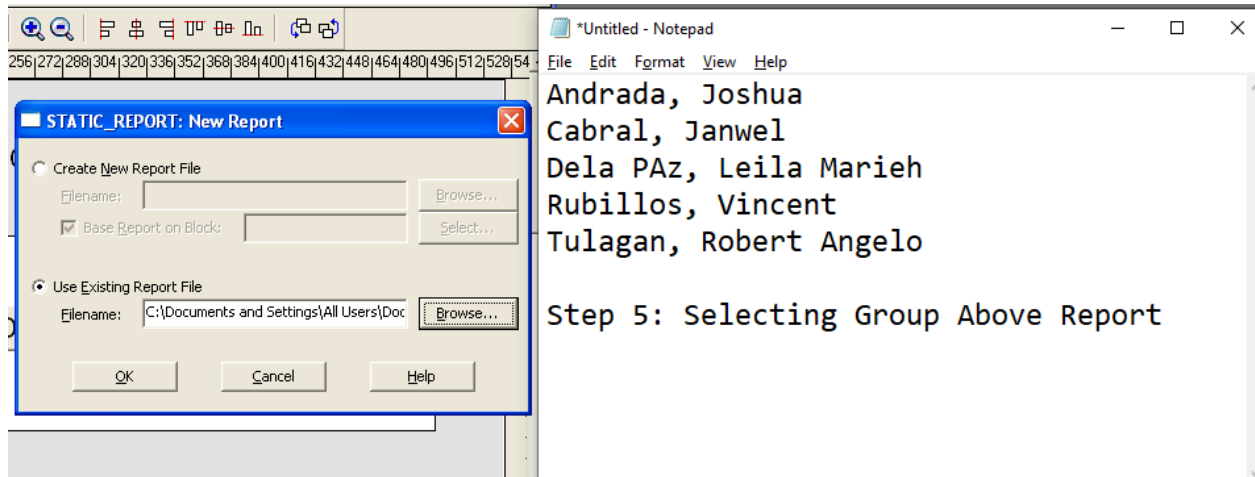
STEP 1: Server Report



STEP 2: Form Builder Creation

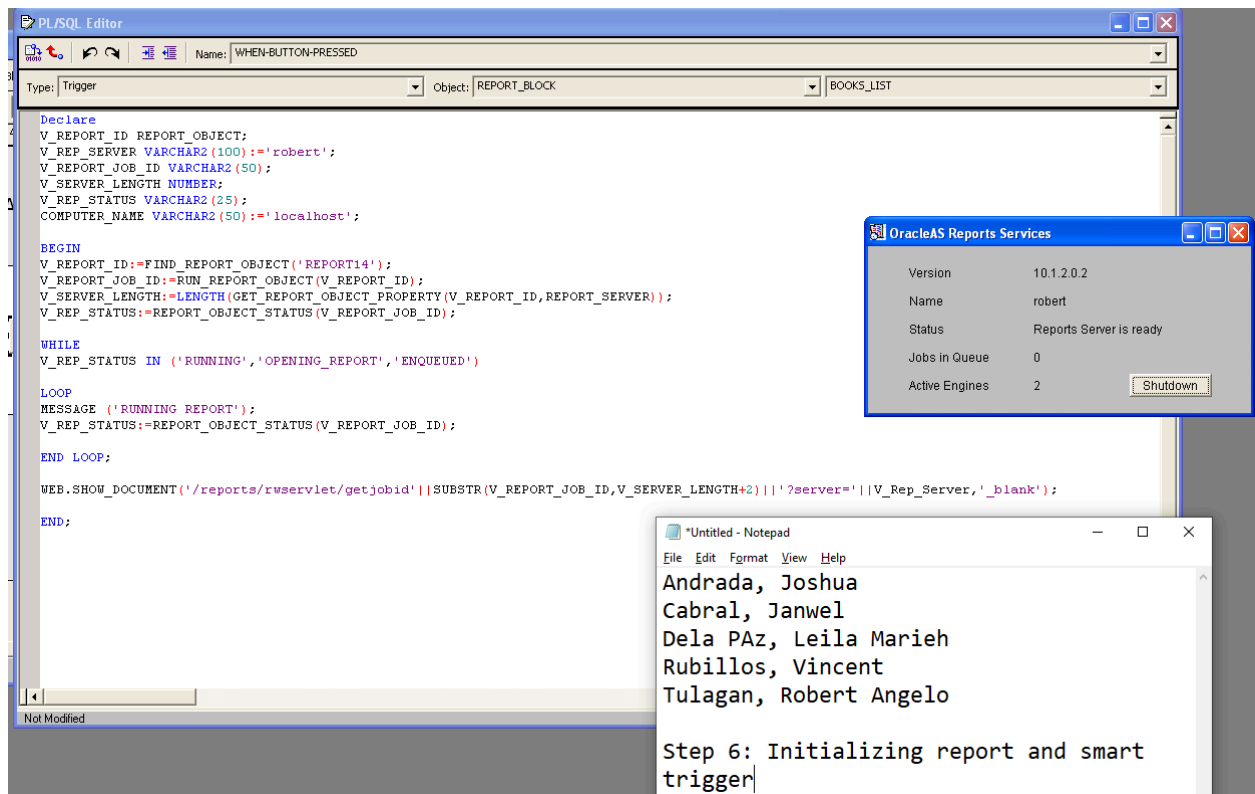


STEP 3: Create Form Module and Control Block With Design



Step 5: Selecting Group Above Report

Step 5: Create Report from the Forms Builder



Step 6: Initializing report and smart trigger

Step 6: Initializing report and smart trigger

The screenshot displays a web application interface with two main windows. The left window, titled "RATAS PODEROSAS INC.", shows a table of books. The right window, titled "Oracle Application Server Forms Services", displays a report titled "REPORT 1:" with a button labeled "LIST OF BOOKS". A Notepad window in the foreground shows the output of the report, listing the names of the authors.

Book Id	Title	Author	Genre	Ishn	Publication Year	Avai Coj
1	Letters to Milena	Frans Kafka	Memoir	*****	1952	
2	1984	Count: George Orwell	Fiction	*****	1949	
3	To Kill a Mockingbird	Count: Harper Lee	Fiction	*****	1960	
4	Brave New World	Count: Aldous Huxley	Science Fiction	*****	1932	
5	The Great Gatsbys	Count: F. Scott Fitzgerald	Memoir	*****	1952	

REPORT 1:

Andrada, Joshua
 Cabral, Janwel
 Dela PAZ, Leila Marieh
 Rubillos, Vincent
 Tulagan, Robert Angelo

Step 7: output

Step 7: output and run.